Mathematical and Computational Methods

Methods

How to solve the standard problems, unit by unit

Most problems on the exercise sheets are instances of a few standard types, and each type has a *method*: a fixed sequence of steps, taken from the notes, that solves every problem of that type. This guide collects them, unit by unit. Each method states the problem it solves, lists its steps and works one example that is not on any sheet. It ends with the mistakes to avoid, where the notes teach it, and a link to a Review set of problems of that kind. The units fall into four parts: foundations and complex numbers, vectors and matrices, differentiation, and integration.

The guide does not say which sheet problem needs which method. Recognising the type is part of what the sheets train, and in an exam nothing tells you. Where a unit's problems look alike but need different methods, its chapter opens with a table for choosing.

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Part I Foundations and complex numbers

Units 1–3. The first unit's methods are working tools for the whole course: solving an inequality, finding a domain, inverting a function, solving an equation. Units 2 and 3 carry them into the complex numbers, where most problems begin by choosing a form: Cartesian for sums, polar or exponential for products, powers and roots.

Unit 1 Foundations: numbers, sets and functions

Six methods cover the unit's computations. Its other problems (sums in Σ notation, hyperbolic identities, reading a piecewise definition, the irrationality proof) are single moves, done directly as the notes do them.

1.1 Solving an inequality by sign analysis

Problem. Solve an inequality in one real variable, and give the solution set as an interval or a union of intervals.

Method

1. Bring everything to one side, $F(x) > 0$ (or $<$, $\leq$, $\geq$). For a quotient, combine into a single fraction.
2. Factor the numerator and the denominator into linear factors. A factor with no real zero, such as $x^2 + 1$, has one sign everywhere: note its sign.
3. Mark every zero of the numerator and of the denominator on a number line. Between consecutive marked points F keeps one sign: find it from a test point, or by counting negative factors. For $\frac{x-a}{x-b}$ this is the notes' rule for $(x-a)(x-b)$: negative exactly between a and b, positive outside.
4. Keep the intervals with the required sign. Include a zero of the numerator only for $\leq$ or $\geq$. Never include a zero of the denominator.
5. Write the answer in interval notation: a square bracket includes the endpoint, a round one excludes it, and $\pm\infty$ always takes a round one.

Example. Solve $\dfrac{6}{x} \leq x - 1$.

Subtract $\frac{6}{x}$ from both sides and combine:

$$x - 1 - \frac{6}{x} = \frac{x^2 - x - 6}{x} = \frac{(x-3)(x+2)}{x} \geq 0.$$

Call the left-hand side $F(x)$. The numerator is zero at $x = -2$ and $x = 3$, the denominator at

$x = 0$, and the signs of the three factors fix the sign of F on each interval between them:

	$x+2$	x	$x-3$	$F(x)$
$x < -2$	$-$	$-$	$-$	$-$
$-2 < x < 0$	$+$	$-$	$-$	$+$
$0 < x < 3$	$+$	$+$	$-$	$-$
$x > 3$	$+$	$+$	$+$	$+$

We need ≥ 0: keep the two positive intervals, include $x = -2$ and $x = 3$ (zeros of the numerator) and exclude $x = 0$:

$$\boxed{[-2, 0) \cup [3, \infty)}$$

Check. One test point in each interval, in the original inequality: $x = -1$ gives $-6 \leq -2$ and $x = 4$ gives $\frac{3}{2} \leq 3$, both true; $x = -3$ gives $-2 \leq -4$ and $x = 1$ gives $6 \leq 0$, both false. Then the endpoints: at $x = -2$ and $x = 3$ the two sides are equal, so both are included; at $x = 0$ the left side is undefined, so it is excluded.

Variants. A *linear* inequality is solved directly, reversing the sign whenever you multiply or divide by a negative number. A *modulus* is removed first, for $a > 0$: $|x| \leq a \iff -a \leq x \leq a$ and $|x| \geq a \iff x \leq -a$ or $x \geq a$.

Pitfalls

- Multiplying both sides by x, whose sign you do not know. Above, it gives $6 \leq x^2 - x$, so $x \leq -2$ or $x \geq 3$: the interval $(-2, 0)$ is lost and $(-\infty, -2)$ wrongly gained.
- Including a zero of the denominator because the inequality is $\geq$.
- Writing $[3, \infty]$: ∞ is not a number, so it cannot be included.

In the notes. Unit 1: Inequalities, and Unit 1: Intervals and infinity.

Practise it. the Review set on this method

1.2 Maximal domain, and range by building up

Problem. A function is given only by a formula. Find its maximal domain, the largest set of real inputs for which the formula makes sense. Find its range, the set of values it actually takes.

Method

1. List every restriction the formula imposes: a denominator $\neq 0$; the expression under an even root ≥ 0; the expression inside a logarithm > 0; the input of arcsin or arccos in $[-1, 1]$.
2. Solve each restriction (an inequality: Method 1.1).
3. Intersect the solution sets, and write the domain in set-builder notation and as an interval or a union.
4. For the range, first rewrite the formula so that x appears only once, as in $\frac{x+1}{x} = 1 + \frac{1}{x}$. Start from the innermost piece and the set of values it takes as x runs over the domain.
5. Push that set through each operation in turn: a shift, a stretch, a reciprocal, a root. Each must be increasing or decreasing on the current set, so that endpoints go to endpoints. A decreasing step (a minus sign, a reciprocal of positive numbers) reverses the order of the

endpoints.
6. Say which endpoint values are attained and which are only approached.

Example. Find the maximal domain of $f(x) = \dfrac{\ln(5-x)}{\sqrt{x+2}}$, and the range of $g(x) = 1 + \dfrac{4}{x^2+2}$.

For f, the logarithm needs $5 - x > 0$, so $x < 5$. The root needs $x + 2 \geq 0$, and since it is in the denominator it also must not vanish: $x + 2 > 0$, so $x > -2$. Intersecting,

$$\boxed{\{\, x \in \mathbb{R} \;:\; -2 < x < 5 \,\} = (-2, 5)}$$

For g the domain is $\mathbb{R}$. Build up from the inside:

$$x^2 \in [0, \infty) \implies x^2 + 2 \in [2, \infty) \implies \frac{4}{x^2+2} \in (0, 2] \implies g(x) \in (1, 3].$$

The reciprocal reverses the order: the smallest denominator, 2, gives the largest value, 2. The value 3 is attained at $x = 0$; the value 1 is approached as $|x| \to \infty$ but never reached. The range is $\boxed{(1, 3]}$.

Check. Test a value just outside each claimed endpoint: $f(-2)$ divides by zero and $f(5)$ takes $\ln 0$; $g(x) = 1$ would need $\frac{4}{x^2+2} = 0$, which has no solution, and $g(x) > 3$ would need $x^2 + 2 < 2$, which has none either.

Pitfalls

- An even root in a denominator needs the expression under it strictly positive, not just non-negative.
- Simplifying before reading off the domain: $\dfrac{x^2-1}{x-1}$ simplifies to $x + 1$, but $x = 1$ is still excluded.
- Giving the codomain as the range. The range is what is attained.

In the notes. Unit 1: What a function is, with its two worked examples, and Unit 1: Power, polynomial and rational functions.

Practise it. the Review set on this method

1.3 Composition, decomposition and the inverse function

Problem. Form a composite $f \circ g$ with its domain; write a given function as a composite of simpler ones; find the inverse of a function, with its domain.

Method

1. Composite: $(f \circ g)(x) = f\big(g(x)\big)$. Substitute $g(x)$ for the variable of f.
2. Its domain: the x in the domain of g for which $g(x)$ lies in the domain of f. Decide it *before* simplifying the formula.
3. Decomposition: name the outer function $F(u)$, the last operation applied, with a fresh letter u, and the inner function $G(x)$, the whole expression it is applied to.
4. Inverse: show f is injective, usually by showing it is strictly monotonic on its domain, and find its range.

5. Write $y = f(x)$ and solve for x in terms of y.
6. Rename the variable to write $f^{-1}(x)$. The domain of f^{-1} is the range of f, and its range is the domain of f.

Example. Let $f(x) = \sqrt{x}$ and $g(x) = 4 - x$. Find $f \circ g$ and $g \circ f$ with their domains, and the inverse of $h = g \circ f$. Then decompose $p(x) = \mathrm{e}^{-x^2/2}$.

$(f \circ g)(x) = \sqrt{4 - x}$, defined when $4 - x \geq 0$: domain $(-\infty, 4]$. And $(g \circ f)(x) = 4 - \sqrt{x}$, defined when x is in the domain of f: domain $[0, \infty)$.

$h(x) = 4 - \sqrt{x}$ is strictly decreasing on $[0, \infty)$, since $\sqrt{x}$ is strictly increasing. Its range is $(-\infty, 4]$, with 4 attained at $x = 0$. Solve $y = 4 - \sqrt{x}$: $\sqrt{x} = 4 - y$, so $x = (4 - y)^2$. Hence

$$\boxed{h^{-1}(x) = (4 - x)^2}$$

with domain $(-\infty, 4]$, the range of h, and range $[0, \infty)$, the domain of h. For p, the last operation is raising e to a power, applied to $-x^2/2$: $G(x) = -\frac{1}{2}x^2$ and $F(u) = \mathrm{e}^u$, so $p = F \circ G$.

Check. $h\big(h^{-1}(1)\big) = 4 - \sqrt{9} = 1$ and $h^{-1}\big(h(9)\big) = h^{-1}(1) = 9$.

Pitfalls

- Reading the domain off the simplified formula. With $g(x) = \sqrt{x}$ and $f(x) = x^2$, $(f \circ g)(x) = x$, but only for $x \geq 0$.
- Dropping the domain of the inverse. $(4 - x)^2$ on all of $\mathbb{R}$ is not injective; it inverts h only on $(-\infty, 4]$.
- Writing f^{-1} for $1/f$. They are different functions.

In the notes. Unit 1: Composition of functions, and Unit 1: Inverse functions and monotonicity.
Practise it. the Review set on this method

1.4 Testing a property: injective, surjective, even, odd, periodic

Problem. Decide whether $f\colon A \to B$ is injective, surjective or bijective; whether it is even, odd or neither; and find its period.

Method

1. Injective: show that f is strictly monotonic on A, or that $f(x_1) = f(x_2)$ forces $x_1 = x_2$. To refute it, give $x_1 \neq x_2$ with $f(x_1) = f(x_2)$.
2. Surjective: solve $y = f(x)$ for $x \in A$, for every $y \in B$; equivalently, show that the range is all of B. To refute it, give one $y \in B$ that is never attained.
3. Even or odd: check that the domain is symmetric about 0. Compute $f(-x)$, simplify, and compare it with $f(x)$ and with $-f(x)$.
4. Neither: give one x with $f(-x) \neq f(x)$ and one with $f(-x) \neq -f(x)$. A single x can do both.
5. Period: for a non-constant f, find the smallest $T > 0$ with $f(x + T) = f(x)$ for all x, if there is one. If f has period T, then $f(bx)$ has period T/b for $b > 0$.

Example. Let $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = \dfrac{x}{1+x^2}$. Is it even or odd, injective, surjective? And what is the period of $q(x) = \cos\dfrac{\pi x}{2}$?

$f(-x) = \dfrac{-x}{1+x^2} = -f(x)$ on the symmetric domain $\mathbb{R}$, so f is *odd*.

Solve $y = f(x)$ for x: $yx^2 - x + y = 0$. For $y \neq 0$ this quadratic has discriminant $1 - 4y^2$, so it has a real root only when $|y| \leq \frac{1}{2}$ (and $y = 0$ gives $x = 0$). The range is $[-\frac{1}{2}, \frac{1}{2}]$, so $y = 1$ is never attained: f is *not surjective*.

For $0 < |y| < \frac{1}{2}$ there are two roots. Taking $y = \frac{2}{5}$ gives $2x^2 - 5x + 2 = 0$, so $x = 2$ or $x = \frac{1}{2}$: $f(2) = \frac{2}{5} = f(\frac{1}{2})$, and f is *not injective*.

cos has period 2π, so q has period $2\pi \,/\, \frac{\pi}{2} = \boxed{4}$.

Check. $q(x+4) = \cos\left(\frac{\pi x}{2} + 2\pi\right) = q(x)$, and no smaller T works: $q(T) = q(0) = 1$ needs $\frac{\pi T}{2}$ to be a multiple of 2π, so T is a multiple of 4. And $f(\pm 1) = \pm\frac{1}{2}$, the ends of the range, are attained.

Pitfalls

- Judging on the wrong domain or codomain. The verdict belongs to the rule *together with* A and B: x^2 is bijective as a map $[0, \infty) \to [0, \infty)$.
- Proving evenness from one value. $f(-x) = f(x)$ must hold for every x; a single value can only refute it.
- Calling a function even or odd when its domain is not symmetric: $\sqrt{x}$ is neither.

In the notes. Unit 1: Injective, surjective and bijective functions, and Unit 1: Symmetry: even, odd and periodic functions.

Practise it. the Review set on this method

1.5 Transforming a graph from a base graph

Problem. Sketch $y = a\,f\big(b(x-c)\big) + d$ from a graph you know, or name the transformations that produce a given formula.

Method

1. Write the target in the form $a\,f\big(b(x-c)\big)+d$: complete the square, or factor the coefficient of x out of the bracket.
2. Every point (u, v) of $y = f(x)$ moves to $\left(c + \frac{u}{b},\ d + av\right)$. Horizontally: scale by $\frac{1}{|b|}$ (and reflect in the y-axis if $b < 0$), then shift by c. Vertically: scale by $|a|$ (and reflect in the x-axis if $a < 0$), then shift by d.
3. Move the key points and the asymptotes by that rule. A moved intercept need not be an intercept of the new graph: find those from its formula, with $x = 0$ and $y = 0$.
4. Read off what is asked: the domain, the range, where the function increases.
5. Sketch, labelling the points you moved.

Example. Obtain $y = 1 + \sqrt{4 - 2x}$ from $y = \sqrt{x}$, and give its domain and range.

Factor the coefficient of x out of the bracket: $4 - 2x = -2(x-2)$, so the target is $f\big(b(x-c)\big)+d$ with $f = \sqrt{}$, $a = 1$, $b = -2$, $c = 2$, $d = 1$. The graph is compressed horizontally by $\frac{1}{2}$ and

reflected in the y-axis, then shifted 2 right and 1 up. The point (u, v) goes to $\left(2 - \frac{u}{2},\ 1 + v\right)$:

$$(0,0) \mapsto (2,1), \qquad (4,2) \mapsto (0,3), \qquad (16,4) \mapsto (-6,5).$$

The graph starts at $(2,1)$ and rises to the *left*, so

$$\boxed{\text{domain } (-\infty, 2], \quad \text{range } [1, \infty)}$$

and the function is decreasing. It meets the y-axis at $(0,3)$ and never meets the x-axis, since $y \geq 1$.

Check. Substitute the moved points: $1 + \sqrt{4-0} = 3$ and $1 + \sqrt{4+12} = 5$. The domain also follows from $4 - 2x \geq 0$.

Pitfalls

- Reading $f(2x-4)$ as a shift by 4. Factor first: $f\big(2(x-2)\big)$ is a shift by 2.
- Moving the wrong way: $f(x-c)$ moves the graph *right* by c when $c > 0$.
- Doing the vertical steps in the wrong order: $af(x)+d$ scales first, then shifts; $a\big(f(x)+d\big)$ is a different graph.

In the notes. Unit 1: Graphs and graph transformations, and its table of transformations.
Practise it. the Review set on this method

1.6 Exponential, logarithmic and trigonometric equations

Problem. Solve an equation in which the unknown sits inside an exponential, a logarithm or a trigonometric function.

Method

1. If a logarithm of an expression in x appears, find where it is defined first.
2. Use the index and log laws to reduce the equation to one exponential or one logarithm on each side, or substitute $u = \mathrm{e}^x$ (so $u > 0$) to get a polynomial in u.
3. Take ln of both sides, or make each side a power of e, and solve. After a substitution, solve the polynomial for u, keep only the roots with $u > 0$, and set $x = \ln u$.
4. Reject any value outside the domain from step 1.
5. For a trigonometric equation, isolate $\sin\theta = c$, $\cos\theta = c$ or $\tan\theta = c$, where θ is the whole argument (such as $2x$), factoring first if there are several terms.
6. Take the principal value α from the table or the inverse function. The second value in one period comes from the symmetry: $\pi - \alpha$ for sine, $-\alpha$ for cosine; tangent has only one value per period.
7. Add whole periods to θ (2π for sine and cosine, π for tangent), then solve for x, and list every solution in the interval asked for.

Example. Solve (a) $\ln(x+4) = 2\ln x$, and (b) $2\sin^2 x = \sin x$ on $[0, 2\pi)$.

(a) Both logarithms need $x > 0$ (then $x + 4 > 0$ too). By the power law $2\ln x = \ln(x^2)$, so $\ln(x+4) = \ln(x^2)$, $x + 4 = x^2$ and $x^2 - x - 4 = 0$, with roots $x = \frac{1 \pm \sqrt{17}}{2}$. Only the plus sign gives $x > 0$: $\boxed{x = \frac{1+\sqrt{17}}{2}}$.

(b) Bring everything to one side and factor: $\sin x\,(2\sin x - 1) = 0$. Then $\sin x = 0$ gives $x = 0,\ \pi$. And $\sin x = \frac{1}{2}$ gives the principal value $\frac{\pi}{6}$ and the second value $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$. So

$$\boxed{x \in \left\{0,\ \tfrac{\pi}{6},\ \tfrac{5\pi}{6},\ \pi\right\}}$$

Check. Substitute: in (a), $x^2 = \frac{9+\sqrt{17}}{2} = x + 4$ and $x > 0$; in (b), $2\sin^2 x = \sin x$ holds at each of the four values ($\frac{1}{2} = \frac{1}{2}$ at $\frac{\pi}{6}$ and $\frac{5\pi}{6}$). Then count: each factor, $\sin x = 0$ and $\sin x = \frac{1}{2}$, has two solutions in $[0, 2\pi)$, so there are four.

Pitfalls

- Dividing by $\sin x$: it loses the solutions $x = 0$ and $x = \pi$. Factor instead, or check the zeros of what you divide by.
- Keeping $x = \frac{1-\sqrt{17}}{2}$ in (a). It satisfies $\ln(x + 4) = \ln(x^2)$, but $\ln x$ is not defined there: $2\ln x = \ln(x^2)$ holds only for $x > 0$.
- Stopping at the principal value. $\arcsin \frac{1}{2}$ is one number; the equation $\sin x = \frac{1}{2}$ has two solutions in every period.

In the notes. Unit 1: Exponential and logarithmic functions, and Unit 1: Trigonometric and inverse-trigonometric functions.

Practise it. the Review set on this method

Unit 2 Complex numbers I: algebra and geometry

Six methods. The powers of i are a step inside the first, and the short proofs of the conjugate and modulus rules are the last step of Method 2.5.

2.1 Cartesian arithmetic, and division by the conjugate

Problem. Write a sum, product, power or quotient of complex numbers in the form $a + b\,\mathrm{i}$; find a conjugate or a modulus.

Method

1. Sum or difference: real parts with real parts, imaginary parts with imaginary parts.
2. Product: expand the brackets, replace i^2 by -1, and group into $(\dots) + (\dots)\,\mathrm{i}$. For i^n, divide n by 4: $\mathrm{i}^n = \mathrm{i}^r$, where r is the remainder, read from $1,\ \mathrm{i},\ -1,\ -\mathrm{i}$. For $n < 0$, first add a multiple of 4: $\mathrm{i}^{-5} = \mathrm{i}^{-5+8} = \mathrm{i}^3 = -\mathrm{i}$.
3. Quotient: multiply the top and the bottom by the conjugate $c - d\,\mathrm{i}$ of the denominator $c + d\,\mathrm{i}$. The denominator becomes the real number $c^2 + d^2$.
4. Expand the numerator, replace i^2 by -1, and split into $\dfrac{p}{c^2+d^2} + \dfrac{q}{c^2+d^2}\,\mathrm{i}$.
5. Conjugate: flip the sign of the imaginary part. Modulus: $|a + b\,\mathrm{i}| = \sqrt{a^2 + b^2}$, and $z\bar{z} = |z|^2$.

Example. Write $\dfrac{7+\mathrm{i}}{1-2\mathrm{i}} - \mathrm{i}^{23}$ in the form $a + b\,\mathrm{i}$, and find its modulus.

Multiply the top and the bottom by $1 + 2\mathrm{i}$, the conjugate of the denominator:

$$\frac{7+\mathrm{i}}{1-2\mathrm{i}} = \frac{(7+\mathrm{i})(1+2\mathrm{i})}{(1-2\mathrm{i})(1+2\mathrm{i})} = \frac{7+14\mathrm{i}+\mathrm{i}+2\mathrm{i}^2}{1^2+2^2} = \frac{5+15\mathrm{i}}{5} = 1+3\mathrm{i}.$$

Since $23 = 4 \cdot 5 + 3$, $\mathrm{i}^{23} = \mathrm{i}^3 = -\mathrm{i}$. So the number is $1 + 3\mathrm{i} - (-\mathrm{i}) = \boxed{1+4\mathrm{i}}$, with modulus $\sqrt{1^2+4^2} = \boxed{\sqrt{17}}$.

Check. Multiply the quotient back by the denominator: $(1+3\mathrm{i})(1-2\mathrm{i}) = 1 - 2\mathrm{i} + 3\mathrm{i} - 6\mathrm{i}^2 = 7 + \mathrm{i}$. Then $\mathrm{i}^{23} = -\mathrm{i}$ because $(-\mathrm{i})\,\mathrm{i} = 1 = \mathrm{i}^{24}$, and $(1+4\mathrm{i})(1-4\mathrm{i}) = 1 - 16\mathrm{i}^2 = 17 = |1+4\mathrm{i}|^2$.

Pitfalls

- Dividing part by part: $\dfrac{a+b\mathrm{i}}{c+d\mathrm{i}} \neq \dfrac{a}{c} + \dfrac{b}{d}\,\mathrm{i}$.
- Losing the sign of i^2: $2\mathrm{i} \cdot \mathrm{i} = -2$, not 2.
- Putting i inside the modulus: $|1 + 4\mathrm{i}| = \sqrt{1+16}$, not $\sqrt{1+(4\mathrm{i})^2}$.

In the notes. Unit 2: The imaginary unit, for the powers of i, Unit 2: Addition, subtraction and multiplication, Unit 2: The modulus, and Unit 2: Division.

Practise it. the Review set on this method

2.2 Real polynomials: complex roots in conjugate pairs

Problem. Solve a real quadratic with a negative discriminant; given one non-real root of a real cubic or quartic, find all its roots.

Method

1. Quadratic: complete the square to $(x+p)^2+q=0$ with $q>0$. Then $(x+p)^2=-q=(\mathrm{i}\sqrt{q})^2$, so $x=-p\pm\mathrm{i}\sqrt{q}$. Or use the formula: $D=b^2-4ac<0$ and $x=\dfrac{-b\pm\mathrm{i}\sqrt{4ac-b^2}}{2a}$.
2. A polynomial with *real* coefficients and a root α also has the root $\overline{\alpha}$.
3. Multiply the pair into a real quadratic: $(x-\alpha)(x-\overline{\alpha})=x^2-2\,\mathrm{Re}(\alpha)\,x+|\alpha|^2$.
4. Find the other factor by comparing coefficients (or by long division). Its roots (by step 1, if it is a quadratic) are the remaining roots.
5. List every root.

Example. (a) Solve $x^2-6x+13=0$. (b) Given that $-1+2\mathrm{i}$ is a root of $p(x)=2x^3+x^2+4x-15$, find the other two.

(a) Complete the square: $(x-3)^2+4=0$, so $(x-3)^2=-4=(2\mathrm{i})^2$ and $\boxed{x=3\pm2\mathrm{i}}$.

(b) The coefficients are real, so $-1-2\mathrm{i}$ is a root too. The pair gives the real factor

$$x^2-2\cdot(-1)\,x+|-1+2\mathrm{i}|^2=x^2+2x+5.$$

The x^3 terms need $2x$ in the other factor, so write $p(x)=(x^2+2x+5)(2x+k)$ and compare the constant terms: $5k=-15$, so $k=-3$. The roots are $\boxed{-1+2\mathrm{i},\ -1-2\mathrm{i},\ \tfrac{3}{2}}$.

Check. Expand: $(x^2+2x+5)(2x-3)=2x^3+x^2+4x-15$. In (a), the pair multiplies back to $x^2-2\cdot3\,x+|3+2\mathrm{i}|^2=x^2-6x+13$.

Pitfalls

- Dropping the i: from $(x-3)^2=-4$ it is $x-3=\pm2\mathrm{i}$, because $(2\mathrm{i})^2=-4$. Do not write $\sqrt{-4}$.
- Using the conjugate-pair rule on a polynomial with non-real coefficients: $x-\mathrm{i}$ has the root i but not $-\mathrm{i}$.
- Stopping at the pair: a cubic has a third root, and it is real.

In the notes. Unit 2: Quadratic equations with complex roots.

Practise it. the Review set on this method

2.3 Cartesian and polar form, with the quadrant rule

Problem. Write a nonzero $z=a+b\,\mathrm{i}$ in polar form $r(\cos\theta+\mathrm{i}\sin\theta)$ with its principal argument, or convert a polar form back to $a+b\,\mathrm{i}$.

Method

1. $r=|z|=\sqrt{a^2+b^2}$.
2. Sketch the point to see its quadrant. A point on an axis has $\theta=0$, $\frac{\pi}{2}$, π or $-\frac{\pi}{2}$ directly.
3. The reference angle is the acute angle $\alpha=\arctan|b/a|$.

4. The principal argument, in $(-\pi, \pi]$, from the quadrant: α (first), $\pi - \alpha$ (second), $-\pi + \alpha$ (third), $-\alpha$ (fourth).
5. Write $z = r(\cos\theta + \mathrm{i}\sin\theta)$.
6. Back to Cartesian: evaluate $\cos\theta$ and $\sin\theta$, and multiply by r.

Example. Write $z = -3 - \sqrt{3}\,\mathrm{i}$ in polar form with its principal argument, and $w = 5\big(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\big)$ in Cartesian form.

$r = \sqrt{9+3} = 2\sqrt{3}$. Both parts are negative: the third quadrant. The reference angle is $\alpha = \arctan\frac{\sqrt{3}}{3} = \frac{\pi}{6}$, so $\theta = -\pi + \frac{\pi}{6} = -\frac{5\pi}{6}$ and

$$\boxed{z = 2\sqrt{3}\Big(\cos\big(-\tfrac{5\pi}{6}\big) + \mathrm{i}\sin\big(-\tfrac{5\pi}{6}\big)\Big)}$$

For w: $\cos\frac{3\pi}{4} = -\frac{\sqrt{2}}{2}$ and $\sin\frac{3\pi}{4} = \frac{\sqrt{2}}{2}$, so $\boxed{w = -\frac{5\sqrt{2}}{2} + \frac{5\sqrt{2}}{2}\,\mathrm{i}}$.

Check. Convert back: $2\sqrt{3}\big(-\frac{\sqrt{3}}{2} - \frac{1}{2}\mathrm{i}\big) = -3 - \sqrt{3}\,\mathrm{i}$.

Pitfalls

- Taking $\arctan(b/a)$ as the argument. Here it gives $\arctan\frac{\sqrt{3}}{3} = \frac{\pi}{6}$, a first-quadrant angle for a third-quadrant point. The sketch fixes the quadrant, not the tangent.
- Giving $\frac{7\pi}{6}$ when the principal argument is asked for. It is an argument of z, but not in $(-\pi, \pi]$.
- Dividing by $a = 0$. A point on the imaginary axis needs no tangent.

In the notes. Unit 2: Modulus and argument, and Unit 2: Converting between Cartesian and polar form.

Practise it. the Review set on this method

2.4 Multiplying and dividing in polar form; rotation

Problem. Multiply or divide complex numbers given in polar form; rotate and scale a point of the plane.

Method

1. Write each factor in polar form (Method 2.3).
2. Product: multiply the moduli and add the arguments. Quotient: divide the moduli and subtract the arguments.
3. If the principal argument is asked for and the result is outside $(-\pi, \pi]$, add or subtract 2π.
4. Convert to $a + b\,\mathrm{i}$ if asked.
5. To turn a point z through α anticlockwise and scale it by $k > 0$, multiply by $k(\cos\alpha + \mathrm{i}\sin\alpha)$. A clockwise turn uses $-\alpha$.

Example. Let $z = 2\big(\cos(-\frac{2\pi}{3}) + \mathrm{i}\sin(-\frac{2\pi}{3})\big)$ and $w = 4\big(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\big)$. Find zw and z/w in Cartesian form. Then turn $2 + 4\mathrm{i}$ a quarter turn clockwise and halve its distance from 0.

zw has modulus 8 and argument $-\frac{2\pi}{3} + \frac{\pi}{2} = -\frac{\pi}{6}$, already in $(-\pi, \pi]$. So

$$zw = 8\Big(\frac{\sqrt{3}}{2} - \frac{1}{2}\,\mathrm{i}\Big) = \boxed{4\sqrt{3} - 4\mathrm{i}}.$$

z/w has modulus $\frac{1}{2}$ and argument $-\frac{2\pi}{3} - \frac{\pi}{2} = -\frac{7\pi}{6}$, which is outside $(-\pi, \pi]$; add 2π to get $\frac{5\pi}{6}$: $z/w = \frac{1}{2}\big(-\frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i}\big) = \boxed{-\frac{\sqrt{3}}{4} + \frac{1}{4}\,\mathrm{i}}$.

The multiplier has $k = \frac{1}{2}$ and $\alpha = -\frac{\pi}{2}$, so it is $\frac{1}{2}(\cos(-\frac{\pi}{2}) + \mathrm{i}\sin(-\frac{\pi}{2})) = -\frac{1}{2}\mathrm{i}$. Then $(2 + 4\mathrm{i})(-\frac{1}{2}\mathrm{i}) = -\mathrm{i} - 2\mathrm{i}^2 = \boxed{2 - \mathrm{i}}$.

Check. Undo the turn and the scaling, multiplying by $2\mathrm{i}$: $(2 - \mathrm{i}) \cdot 2\mathrm{i} = 4\mathrm{i} - 2\mathrm{i}^2 = 2 + 4\mathrm{i}$. And zw is $2 \cdot 4 = 8$ from 0: $\sqrt{48 + 16} = 8$.

Pitfalls

- Adding the moduli. They multiply; only the arguments add.
- Leaving $-\frac{7\pi}{6}$ as the principal argument.
- Turning the wrong way: $\cos\alpha + \mathrm{i}\sin\alpha$ with $\alpha > 0$ turns anticlockwise.

In the notes. Unit 2: Multiplication and division in polar form, and Unit 2: A general product as rotation and scaling.

Practise it. the Review set on this method

2.5 Equating real and imaginary parts

Problem. Find an unknown complex number from an equation it satisfies; prove an identity that holds for all complex numbers.

Method

1. Write the unknown as $z = x + y\,\mathrm{i}$ with x and y real (and $\overline{z} = x - y\,\mathrm{i}$).
2. Expand both sides into (real) + (real) i.
3. Two complex numbers are equal exactly when their real parts are equal and their imaginary parts are equal. That gives two real equations.
4. Solve the two equations for x and y.
5. For an identity in z and w, write $z = a + b\,\mathrm{i}$ and $w = c + d\,\mathrm{i}$, expand both sides and compare the parts. Where moduli appear, it is often shorter to use $|u|^2 = u\,\overline{u}$ with the conjugate rules $\overline{z \pm w} = \overline{z} \pm \overline{w}$ and $\overline{zw} = \overline{z}\,\overline{w}$. A rule for a quotient follows from the one for a product: write $z/w = z \cdot \frac{1}{w}$ and use $w \cdot \frac{1}{w} = 1$.

Example. Find every z with $2z + \mathrm{i}\overline{z} = 3 + 6\mathrm{i}$.

Put $z = x + y\,\mathrm{i}$, so $\overline{z} = x - y\,\mathrm{i}$, and expand:

$$2x + 2y\,\mathrm{i} + \mathrm{i}(x - y\,\mathrm{i}) = (2x + y) + (x + 2y)\,\mathrm{i}.$$

Equate the parts with those of $3 + 6\mathrm{i}$: $2x + y = 3$ and $x + 2y = 6$. The first gives $y = 3 - 2x$;

then $x + 6 - 4x = 6$, so $x = 0$ and $y = 3$. The only solution is $\boxed{z = 3\mathrm{i}}$.

Check. Substitute: $2(3\mathrm{i}) + \mathrm{i}(-3\mathrm{i}) = 6\mathrm{i} + 3$.

Pitfalls

- Treating $\overline{z}$ as z: they have the same real part and opposite imaginary parts.
- Forgetting that x and y must be real. If the two equations have no real solution, no z satisfies the equation.
- Checking an identity at one value of z. That can refute it but not prove it.

In the notes. Unit 2: The imaginary unit, for when two complex numbers are equal, and Unit 2: Why the modulus behaves well, for the $|u|^2 = u\overline{u}$ proofs.

Practise it. the Review set on this method

2.6 Reading a locus on the Argand plane

Problem. Describe and sketch a set of points z given by distances: a circle $|z - c| = r$, a disc such as $|z - c| < r$, or a line $|z - a| = |z - b|$.

Method

1. Read $|z - c|$ as the distance from z to the point c. Rewrite $|z + c|$ as $|z - (-c)|$ first.
2. Name the set. $|z - c| = r$ is the circle with centre c and radius r; $|z - c| < r$ is the inside of that circle. $|z - a| = |z - b|$ is the perpendicular bisector of the segment from a to b.
3. If you are unsure, put $z = x + y\,\mathrm{i}$, square both sides, and simplify to an equation (or inequality) in x and y.
4. Sketch it. For a circle, mark the centre and the radius; for a bisector, the two points. A boundary that belongs to the set is drawn solid; one that does not (a strict inequality) is drawn dashed.

Example. Describe (a) $|z - 1 + 2\mathrm{i}| = 3$ and (b) $|z + 2\mathrm{i}| = |z - 4|$.

(a) $|z - (1 - 2\mathrm{i})| = 3$: the points at distance 3 from $1 - 2\mathrm{i}$, that is, the circle

$$\boxed{\text{centre } (1, -2),\ \text{radius } 3}$$

(b) $|z - (-2\mathrm{i})| = |z - 4|$: the points as far from $-2\mathrm{i}$ as from 4, that is, the perpendicular bisector of the segment from $(0, -2)$ to $(4, 0)$. With $z = x + y\,\mathrm{i}$ and both sides squared,

$$x^2 + (y + 2)^2 = (x - 4)^2 + y^2 \iff 4y + 4 = -8x + 16 \iff \boxed{y = 3 - 2x}.$$

Check. The midpoint $(2, -1)$ of the segment lies on $y = 3 - 2x$, and the slope -2 is perpendicular to the segment's slope $\frac{1}{2}$.

Pitfalls

- Reading $|z + 2\mathrm{i}|$ as the distance from $2\mathrm{i}$. It is the distance from $-2\mathrm{i}$.
- Getting the centre's sign wrong: $|z - 1 + 2\mathrm{i}|$ is centred at $1 - 2\mathrm{i}$, not at $1 + 2\mathrm{i}$.
- Drawing the boundary of $|z - c| < r$ solid.

In the notes. Unit 2: Addition and subtraction as vectors, where $|z - w|$ is the distance between z and w.

Practise it. the Review set on this method

Unit 3 Complex numbers II: exponential form, De Moivre, roots

Five methods, and the problems look alike: most start by writing a number as $re^{i\theta}$. What decides the method is what the question asks for. The fundamental theorem of algebra is a fact to quote; given one non-real root, a real polynomial is factored as in Method 2.2.

Which method?

The question asks for	Method
a product, a quotient or a rotation	3.1, exponential form and index laws
an integer power z^n	3.2, De Moivre (not the binomial theorem)
every w with $w^n = z$; the roots of unity	3.3, nth roots
a square root of z when $\arg z$ is not a standard angle	3.3, variant: equate parts
$\cos n\theta$ or $\sin n\theta$ in powers of $\cos\theta$ and $\sin\theta$	3.4, expand $(\cos\theta + i\sin\theta)^n$
$\cos^n\theta$ or $\sin^n\theta$ as a sum of multiple angles	3.4, the inverse Euler formulas
cos, sin, cosh or sinh of an imaginary number	3.4, the inverse Euler formulas at ix
one sinusoid equal to a sum of sinusoids of one frequency	3.5, phasors

3.1 Exponential form and the index laws

Problem. Write a nonzero complex number as $re^{i\theta}$ or back as $a + bi$; multiply, divide, invert or rotate in exponential form.

Method

1. To exponential form: find r and the argument θ exactly as for the polar form (Method 2.3, quadrant first), and write $z = re^{i\theta}$.
2. Back to Cartesian: $re^{i\theta} = r\cos\theta + ir\sin\theta$.
3. Product and quotient: multiply or divide the moduli, add or subtract the exponents. The reciprocal is $\frac{1}{r}e^{-i\theta}$ and the conjugate is $re^{-i\theta}$.
4. Rotation: multiplying by $e^{i\alpha}$ turns a point through α anticlockwise about 0 ($e^{-i\alpha}$ turns it clockwise), and $se^{i\alpha}$ also scales it by s.
5. If the principal argument is asked for, bring the exponent into $(-\pi, \pi]$ by whole turns.

Example. Let $z = -2 + 2i$ and $w = \sqrt{2} - \sqrt{2}\,i$. Find zw and z/w through the exponential form.

z is in the second quadrant with $|z| = 2\sqrt{2}$ and reference angle $\frac{\pi}{4}$, so $z = 2\sqrt{2}\,e^{i3\pi/4}$. w is in the fourth with $|w| = 2$, so $w = 2e^{-i\pi/4}$. Then

$$zw = 4\sqrt{2}\,e^{i(3\pi/4-\pi/4)} = 4\sqrt{2}\,e^{i\pi/2} = \boxed{4\sqrt{2}\,i}, \qquad \frac{z}{w} = \sqrt{2}\,e^{i(3\pi/4+\pi/4)} = \sqrt{2}\,e^{i\pi} = \boxed{-\sqrt{2}}.$$

Check. In Cartesian form, $(-2 + 2i)(\sqrt{2} - \sqrt{2}i) = -2\sqrt{2} + 2\sqrt{2}i + 2\sqrt{2}i + 2\sqrt{2} = 4\sqrt{2}\,i$, and $(-\sqrt{2})(\sqrt{2} - \sqrt{2}i) = -2 + 2i = z$.

Pitfalls

- Subtracting the exponents the wrong way round in a quotient: it is the argument of the top minus the argument of the bottom.
- Writing $e^{i\theta}$ with θ in degrees. The exponent is in radians.
- Taking the argument from $\arctan(b/a)$ without the quadrant, as in Method 2.3.

In the notes. Unit 3: The exponential form of a complex number, and Unit 3: $e^{i\theta}$ as a rotation.
Practise it. the Review set on this method

3.2 Integer powers by De Moivre

Problem. Compute z^n for an integer n, positive or negative.

Method

1. Write $z = re^{i\theta}$ (Method 3.1).
2. Then $z^n = r^n e^{in\theta}$: raise the modulus to the power and multiply the argument by n. This works for negative n too.
3. Remove whole turns (2π each) from $n\theta$ to bring it into $(-\pi, \pi]$.
4. Evaluate cos and sin of it and write $a + b\,\mathrm{i}$.

Why it works. De Moivre's theorem, $(\cos\theta + \mathrm{i}\sin\theta)^n = \cos n\theta + \mathrm{i}\sin n\theta$. The proof is by induction: check $n = 1$; assume it for $n = k$; split $(\cos\theta + \mathrm{i}\sin\theta)^{k+1}$ into the kth power times one more factor, use the assumption, and add the arguments.

Example. Compute (a) $(-\sqrt{3} - \mathrm{i})^7$ and (b) $(1 - \sqrt{3}\,\mathrm{i})^{-5}$.

(a) $-\sqrt{3} - \mathrm{i}$ is in the third quadrant with modulus 2 and reference angle $\frac{\pi}{6}$, so $-\sqrt{3} - \mathrm{i} = 2e^{-\mathrm{i}5\pi/6}$ and $(-\sqrt{3} - \mathrm{i})^7 = 2^7 e^{-\mathrm{i}35\pi/6}$. Add three whole turns: $-\frac{35\pi}{6} + 6\pi = \frac{\pi}{6}$. So the power is $128e^{\mathrm{i}\pi/6} = \boxed{64\sqrt{3} + 64\mathrm{i}}$.

(b) $1 - \sqrt{3}\,\mathrm{i} = 2e^{-\mathrm{i}\pi/3}$, so

$$(1 - \sqrt{3}\,\mathrm{i})^{-5} = 2^{-5}e^{\mathrm{i}5\pi/3} = \tfrac{1}{32}e^{-\mathrm{i}\pi/3} = \tfrac{1}{32}\left(\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\mathrm{i}\right) = \boxed{\tfrac{1}{64} - \tfrac{\sqrt{3}}{64}\,\mathrm{i}}.$$

Check. Multiply out a low power. $(-\sqrt{3}-\mathrm{i})^3 = -8\mathrm{i}$, so $(-\sqrt{3}-\mathrm{i})^7 = (-8\mathrm{i})^2(-\sqrt{3}-\mathrm{i}) = 64\sqrt{3}+64\mathrm{i}$. And $(1 - \sqrt{3}\,\mathrm{i})^3 = -8$, so $(1 - \sqrt{3}\,\mathrm{i})^{-5} = (1 - \sqrt{3}\,\mathrm{i})/(1 - \sqrt{3}\,\mathrm{i})^6 = \frac{1}{64}(1 - \sqrt{3}\,\mathrm{i})$.

Pitfalls

- Raising the modulus but not multiplying the argument, or the reverse.
- Expanding by the binomial theorem: it gives the same answer in eight terms instead of one line.
- Losing track of the turns: $-\frac{35\pi}{6}$ is $\frac{\pi}{6}$ less three whole turns, not $-\frac{5\pi}{6}$ less some.

In the notes. Unit 3: Statement and proof, and Unit 3: Integer powers of complex numbers.
Practise it. the Review set on this method

3.3 nth roots, and the roots of unity

Problem. Find every w with $w^n = z$, in exponential or Cartesian form.

Method

1. Write $z = re^{i\theta}$, with the quadrant checked. (If $z = 0$, the only root is $w = 0$.)
2. The roots all have modulus $\rho = r^{1/n}$, the positive real root.
3. Their arguments are $\phi_k = \dfrac{\theta + 2\pi k}{n}$ for $k = 0, 1, \ldots, n-1$: exactly n of them.
4. Write $w_k = \rho\, e^{i\phi_k}$, and convert to Cartesian form if asked. Each root is the previous one turned through $\frac{2\pi}{n}$: $w_k = w_0\, e^{2\pi i k/n}$.
5. The roots are the corners of a regular n-gon on the circle of radius ρ.

Variants. The *roots of unity* are the case $z = 1$: the powers of $\omega = e^{2\pi i/n}$, which add up to 0 for $n \geq 2$. A *square root* of z when $\arg z$ is not a standard angle: put $(a+bi)^2 = z$, equate the parts, and add $a^2 + b^2 = |z|$ from the moduli. Solve for a^2 and b^2, and choose the signs so that $2ab = \operatorname{Im} z$: the roots are $\pm(a+bi)$. An equation such as $z^n = c\,\bar{z}$: treat $z = 0$ separately, then match the moduli and the arguments.

Example. Solve $w^4 = -8 + 8\sqrt{3}\,i$ in Cartesian form.

$|z| = \sqrt{64 + 192} = 16$, and z is in the second quadrant with reference angle $\arctan\sqrt{3} = \frac{\pi}{3}$, so $\theta = \frac{2\pi}{3}$. Then $\rho = 16^{1/4} = 2$ and

$$\phi_k = \frac{2\pi/3 + 2\pi k}{4} = \frac{\pi}{6} + \frac{k\pi}{2}, \qquad k = 0, 1, 2, 3: \quad \tfrac{\pi}{6},\ \tfrac{2\pi}{3},\ \tfrac{7\pi}{6},\ \tfrac{5\pi}{3}.$$

The roots, the corners of a square on the circle of radius 2, are

$$\boxed{\sqrt{3} + i,\quad -1 + \sqrt{3}\,i,\quad -\sqrt{3} - i,\quad 1 - \sqrt{3}\,i}$$

Check. Raise one root to the fourth power: $(\sqrt{3}+i)^4 = (2e^{i\pi/6})^4 = 16e^{i2\pi/3} = -8 + 8\sqrt{3}\,i$. The four roots add up to 0, as the corners of a regular polygon centred at 0 must.

Pitfalls

- Forgetting the $2\pi k$: it gives one root instead of n.
- Running k up to n: $k = n$ repeats the root for $k = 0$.
- Taking $\rho = r$ instead of $r^{1/n}$.

In the notes. Unit 3: The nth roots of a complex number, Unit 3: Square roots the algebraic way, and Unit 3: The roots of unity.

Practise it. the Review set on this method

3.4 Identities from De Moivre and the inverse Euler formulas

Problem. Write $\cos n\theta$ or $\sin n\theta$ in powers of $\cos\theta$ and $\sin\theta$; write a power of $\cos\theta$ or $\sin\theta$ as a sum of multiple angles; evaluate a trigonometric or hyperbolic function at an imaginary number.

Method

1. Multiple angle: expand $(\cos\theta + \mathrm{i}\sin\theta)^n$ by the binomial theorem. By De Moivre it equals $\cos n\theta + \mathrm{i}\sin n\theta$. Compare the real parts and the imaginary parts. Use $\sin^2\theta = 1 - \cos^2\theta$ if one function is asked for.
2. Power: substitute $\cos\theta = \frac{1}{2}(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta})$ or $\sin\theta = \frac{1}{2\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta})$, take the factor $(\frac{1}{2})^n$ or $(\frac{1}{2\mathrm{i}})^n$ outside, and expand.
3. Pair each $\mathrm{e}^{\mathrm{i}k\theta}$ with $\mathrm{e}^{-\mathrm{i}k\theta}$: their sum is $2\cos k\theta$, their difference $2\mathrm{i}\sin k\theta$. Simplify.
4. Imaginary argument: put $\mathrm{i}x$ into the inverse Euler formula, use $\mathrm{i}^2 = -1$, and recognise $\cosh x$ or $\sinh x$. So $\cos(\mathrm{i}x) = \cosh x$ and $\sin(\mathrm{i}x) = \mathrm{i}\sinh x$. Putting $\mathrm{i}x$ into $\cosh z = \frac{1}{2}(\mathrm{e}^z + \mathrm{e}^{-z})$ and $\sinh z = \frac{1}{2}(\mathrm{e}^z - \mathrm{e}^{-z})$ gives $\cosh(\mathrm{i}x) = \cos x$ and $\sinh(\mathrm{i}x) = \mathrm{i}\sin x$.

Example. (a) Write $\sin 5\theta$ in powers of $s = \sin\theta$. (b) Write $\cos^3\theta$ as a sum of cosines of multiples of θ. (c) Evaluate $\sin(\mathrm{i}\ln 3)$.

(a) With $c = \cos\theta$, the imaginary terms of $(c + \mathrm{i}s)^5$ are $5c^4(\mathrm{i}s) + 10c^2(\mathrm{i}s)^3 + (\mathrm{i}s)^5 = \mathrm{i}(5c^4s - 10c^2s^3 + s^5)$. Comparing imaginary parts with $\cos 5\theta + \mathrm{i}\sin 5\theta$, and using $c^2 = 1 - s^2$,

$$\begin{aligned}\sin 5\theta &= 5(1-s^2)^2 s - 10(1-s^2)s^3 + s^5 \\ &= \boxed{16s^5 - 20s^3 + 5s}.\end{aligned}$$

(b) Put $u = \mathrm{e}^{\mathrm{i}\theta}$. Then

$$\cos^3\theta = \tfrac{1}{8}\left(u + u^{-1}\right)^3 = \tfrac{1}{8}\left(u^3 + u^{-3} + 3(u + u^{-1})\right) = \tfrac{1}{8}\left(2\cos 3\theta + 6\cos\theta\right) = \boxed{\tfrac{1}{4}\cos 3\theta + \tfrac{3}{4}\cos\theta}.$$

(c) $\sin(\mathrm{i}\ln 3) = \mathrm{i}\sinh(\ln 3) = \mathrm{i}\cdot\frac{1}{2}\left(3 - \frac{1}{3}\right) = \boxed{\frac{4}{3}\mathrm{i}}$.

Check. Test at an angle where nothing vanishes. At $\theta = \frac{\pi}{6}$, where $s = \frac{1}{2}$: $\sin\frac{5\pi}{6} = \frac{1}{2} = \frac{16}{32} - \frac{20}{8} + \frac{5}{2}$. At $\theta = \frac{\pi}{3}$: $\cos^3\frac{\pi}{3} = \frac{1}{8} = \frac{1}{4}(-1) + \frac{3}{4}\cdot\frac{1}{2}$. For (c), from the inverse Euler formula directly: $\sin(\mathrm{i}\ln 3) = \frac{1}{2\mathrm{i}}(\mathrm{e}^{-\ln 3} - \mathrm{e}^{\ln 3}) = \frac{1}{2\mathrm{i}}(\frac{1}{3} - 3) = \frac{4}{3}\mathrm{i}$.

Pitfalls

- Getting $(2\mathrm{i})^n$ wrong: $(2\mathrm{i})^2 = -4$ and $(2\mathrm{i})^3 = -8\mathrm{i}$.
- Comparing parts before the expansion is complete, or putting a term with i^2 in the imaginary part.
- Writing $\cos(\mathrm{i}x) = \mathrm{i}\cosh x$. The cosine of an imaginary number is real: $\cos(\mathrm{i}x) = \cosh x$.

In the notes. Unit 3: Multiple-angle identities, Unit 3: Sine and cosine as exponentials, and Unit 3: A link to the hyperbolic functions.

Practise it. the Review set on this method

3.5 Adding sinusoids with phasors

Problem. Write a sum of sinusoids of one angular frequency ω (not the root of unity of Method 3.3) as a single sinusoid $R\cos(\omega t + \alpha)$.

Method

1. Write every term as $A\cos(\omega t+\varphi)$ with $A>0$. A sine becomes a cosine by $\sin x=\cos(x-\frac{\pi}{2})$, and a minus sign by $-\cos x=\cos(x+\pi)$.
2. Each term's phasor is $P=Ae^{i\varphi}$.
3. Add the phasors in Cartesian form. If they add up to 0, the signal is identically 0 and has no phase.
4. Otherwise find the modulus R and the argument α of the sum, quadrant first.
5. The sum is $R\cos(\omega t+\alpha)$. If the question asks for $R\cos(\omega t-\beta)$, then $\beta=-\alpha$.

Example. Write $4\cos(\omega t+\frac{5\pi}{6})-4\sin\omega t$ as a single sinusoid.

By both rules of step 1, $-\sin x=-\cos(x-\frac{\pi}{2})=\cos(x+\frac{\pi}{2})$, so the phasors are $P_1=4e^{i5\pi/6}=-2\sqrt{3}+2i$ and $P_2=4e^{i\pi/2}=4i$. Their sum is $-2\sqrt{3}+6i$, in the second quadrant, with reference angle $\arctan\frac{6}{2\sqrt{3}}=\arctan\sqrt{3}=\frac{\pi}{3}$. So

$$R=\sqrt{12+36}=4\sqrt{3},\qquad \alpha=\pi-\tfrac{\pi}{3}=\tfrac{2\pi}{3},$$

and the sum is $\boxed{4\sqrt{3}\cos(\omega t+\frac{2\pi}{3})}$, with amplitude about 6.93.

Check. At $\omega t=0$: $4\cos\frac{5\pi}{6}=-2\sqrt{3}=4\sqrt{3}\cos\frac{2\pi}{3}$. At $\omega t=\frac{\pi}{2}$, where the sine term counts in full: $4\cos\frac{4\pi}{3}-4=-6=4\sqrt{3}\cos\frac{7\pi}{6}$.

Pitfalls

- Adding the amplitudes: $4+4=8$ is too big unless the phases agree.
- Turning a sine the wrong way: $\sin x=\cos(x-\frac{\pi}{2})$, not $\cos(x+\frac{\pi}{2})$.
- Reading α off arctan when the phasor sum is in the second or third quadrant: here $\arctan\frac{6}{-2\sqrt{3}}=-\frac{\pi}{3}$ points the wrong way.

In the notes. Unit 3: Application: phasors and AC analysis.

Practise it. the Review set on this method

Part II Vectors and matrices

Units 4–6: vectors, matrices, and the eigenvalues and eigenvectors of a matrix. The methods build on one another. The cross product gives a plane its normal, row reduction gives both an inverse and an eigenvector, and the powers of a matrix come from its eigenvectors.

Unit 4 Vectors

Seven methods: the algebra of components first, then three on the two products and three on lines and planes. Proofs from the dot-product rules are a variant of Method 4.2.

4.1 Working in components: length, unit vectors, combinations

Problem. Add and scale vectors; find the vector between two points and the distance between them; normalise a vector; write a vector as a linear combination of two others, and say whether those two are independent.

Method

1. For the vector between two points, take end minus start: $\overrightarrow{PQ} = (q_1 - p_1,\ q_2 - p_2,\ q_3 - p_3)$.
2. Add, subtract and scale componentwise. Scale each vector first, then combine them component by component.
3. The length is $\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$. The distance from P to Q is $\left\|\overrightarrow{PQ}\right\|$.
4. To normalise $\mathbf{v} \neq \mathbf{0}$, divide by its length: $\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$. Check that $\|\hat{\mathbf{v}}\| = 1$.
5. To write $\mathbf{c} = \alpha\mathbf{a} + \beta\mathbf{b}$, match the components. That gives two equations in α and β. Solve them by substitution, and check by recombining.
6. Two vectors are independent exactly when neither is a multiple of the other. Try $\mathbf{b} = \lambda\mathbf{a}$: if the components need different values of λ, the two vectors are independent. Two independent vectors in $\mathbb{R}^2$ form a basis: every vector is a combination of them in exactly one way.

Example. (a) For $P(1, 4, -2)$ and $Q(3, -2, 1)$, find $\overrightarrow{PQ}$, the distance PQ, and the unit vector in the direction from P to Q. (b) Write $\mathbf{c} = (8, -3)$ as a combination of $\mathbf{a} = (2, 1)$ and $\mathbf{b} = (1, -3)$. Are $\mathbf{a}$ and $\mathbf{b}$ independent?

(a) End minus start, $\overrightarrow{PQ} = (3 - 1,\ -2 - 4,\ 1 - (-2))$, so

$$\boxed{\overrightarrow{PQ} = (2, -6, 3)}$$

Then $\left\|\overrightarrow{PQ}\right\| = \sqrt{4 + 36 + 9} = \sqrt{49}$, so $\boxed{PQ = 7}$. With $\mathbf{v} = \overrightarrow{PQ}$, dividing by 7 gives

$$\boxed{\hat{\mathbf{v}} = \left(\tfrac{2}{7}, -\tfrac{6}{7}, \tfrac{3}{7}\right)}$$

(b) Matching components gives $2\alpha + \beta = 8$ and $\alpha - 3\beta = -3$. The first equation gives $\beta = 8 - 2\alpha$. Substituting, $\alpha - 3(8 - 2\alpha) = -3$, so $7\alpha = 21$. Hence $\alpha = 3$ and $\beta = 2$:

$$\boxed{\mathbf{c} = 3\mathbf{a} + 2\mathbf{b}}$$

For independence, suppose $\mathbf{b} = \lambda\mathbf{a}$. The first components need $\lambda = \frac{1}{2}$, and the second need $\lambda = -3$. (Likewise $\mathbf{a} = \mu\mathbf{b}$ needs $\mu = 2$ and $\mu = -\frac{1}{3}$.) So neither vector is a multiple of the other, and $\mathbf{a}$ and $\mathbf{b}$ are $\boxed{\text{independent}}$.

Check. $\|\hat{\mathbf{v}}\| = \frac{1}{7}\sqrt{4 + 36 + 9} = 1$. Recombining, $3\mathbf{a} + 2\mathbf{b} = (6, 3) + (2, -6) = (8, -3) = \mathbf{c}$. This check also exercises step 2.

Pitfalls

- Subtracting the wrong way round. $P - Q = (-2, 6, -3)$ has the same length, 7, so the distance survives, but the direction and the unit vector are reversed.
- Dividing by the wrong number. The component sum here is $2 - 6 + 3 = -1$, and dividing by it gives $(-2, 6, -3)$, of length 7. Dividing by $\left\|\overrightarrow{PQ}\right\|^2 = 49$ gives a vector of length $\frac{1}{7}$. The check $\|\hat{\mathbf{v}}\| = 1$ catches both.
- Taking "no two are parallel" to mean independent. No two of $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are parallel, yet $\mathbf{c} = 3\mathbf{a} + 2\mathbf{b}$ is a combination of the other two. So the three together are not independent; in $\mathbb{R}^2$ three vectors never are.

In the notes. Unit 4: Components, Unit 4: Addition, subtraction and scalar multiplication, Unit 4: Magnitude and unit vectors, and Unit 4: Linear combinations, independence and basis (intuition).

Practise it. the Review set on this method

4.2 The angle between vectors, and between planes

Problem. Find the angle between two vectors in $\mathbb{R}^2$, $\mathbb{R}^3$ or $\mathbb{R}^n$, where its cosine is the cosine similarity; find the angle between two planes.

Method

1. Collect the ingredients: $\mathbf{u} \cdot \mathbf{v}$, $\|\mathbf{u}\|$ and $\|\mathbf{v}\|$. If the vectors join points, form them first by subtracting coordinates, $\overrightarrow{PQ} = (q_1 - p_1,\ q_2 - p_2,\ q_3 - p_3)$.
2. Divide: $\cos\theta = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\ \|\mathbf{v}\|}$, simplified. The sign of $\mathbf{u} \cdot \mathbf{v}$ alone gives the type of angle: positive acute, negative obtuse, zero a right angle, except that parallel vectors give $\theta = 0$ (same way) or π (opposite ways).
3. Give θ in $[0, \pi]$: exactly if $\cos\theta$ is a standard value, otherwise the exact $\cos\theta$ and then $\theta \approx \ldots^\circ$.
4. Two planes: read the normals off the coefficients. The plane $ax + by + cz = d$ has normal (a, b, c).
5. Take steps 1–3 on the normals, and report the angle between the planes in $[0, \frac{\pi}{2}]$, the smaller of θ and $\pi - \theta$, from $|\cos\theta|$. Perpendicular planes give $\frac{\pi}{2}$.

Variants. Orthogonality is the case $\mathbf{u} \cdot \mathbf{v} = 0$. A *proof* from the dot-product rules writes each squared length as a dot product, $\|\mathbf{w}\|^2 = \mathbf{w} \cdot \mathbf{w}$, expands, naming each rule it uses (distributive, scalar, commutative), and simplifies to the claim.

Example. Find the angle between $\mathbf{u} = (1, -2, 1)$ and $\mathbf{v} = (-2, 1, 1)$, and the angle between the planes $x + 2y - 2z = 1$ and $-4x + 3z = 2$.

The ingredients are $\mathbf{u} \cdot \mathbf{v} = -2 - 2 + 1 = -3$, $\|\mathbf{u}\| = \sqrt{1+4+1} = \sqrt{6}$ and $\|\mathbf{v}\| = \sqrt{4+1+1} = \sqrt{6}$. So

$$\cos\theta = \frac{-3}{\sqrt{6}\,\sqrt{6}} = -\frac{1}{2},$$

a standard value, and $\boxed{\theta = \frac{2\pi}{3} = 120^\circ}$: obtuse, as the negative dot product said.

The normals are $\mathbf{n}_1 = (1, 2, -2)$ and $\mathbf{n}_2 = (-4, 0, 3)$, with $\mathbf{n}_1 \cdot \mathbf{n}_2 = -4 + 0 - 6 = -10$, $\|\mathbf{n}_1\| = 3$ and $\|\mathbf{n}_2\| = 5$. So

$$\cos\theta = \frac{-10}{3 \cdot 5} = -\frac{2}{3},$$

not a standard value. The angle $\varphi \in [0, \frac{\pi}{2}]$ between the planes has $\cos\varphi = |\cos\theta| = \frac{2}{3}$:

$$\boxed{\cos\varphi = \tfrac{2}{3}}, \qquad \boxed{\varphi \approx 48.2^\circ}$$

Check. The law of cosines, from which the formula came: $\mathbf{u} - \mathbf{v} = (3, -3, 0)$ has $\|\mathbf{u} - \mathbf{v}\|^2 = 18$, and $\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\,\|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta = 6 + 6 + 6 = 18$. For the planes, the reversed normal $(4, 0, -3)$ describes the same second plane, as $4x - 3z = -2$, and gives $\cos\varphi = \frac{10}{15} = \frac{2}{3}$ directly.

Pitfalls

- Stopping at the dot product. $\mathbf{u} \cdot \mathbf{v} = -3$ is not the cosine, which always lies in $[-1, 1]$: divide by the lengths.
- Reporting 131.8° for the planes. That is the angle between the chosen normals; the planes meet at the angle in $[0, \frac{\pi}{2}]$, from $|\cos\theta|$.
- Building a normal from the right-hand side. Only the coefficients of x, y and z make it, and a missing variable gives a 0, as in $(-4, 0, 3)$.

In the notes. Unit 4: The scalar (dot) product, Unit 4: Beyond three dimensions: $\mathbb{R}^n$ and a data-science note, for cosine similarity, and Unit 4: Where a line meets a plane; the angle between two planes.

Practise it. the Review set on this method

4.3 Projection, resolving, work

Problem. Project one vector onto another; resolve a vector into a part along a direction and a part across it; find the work done by a force and its component along the motion.

Method

1. Compute $\mathbf{u} \cdot \mathbf{v}$ and $\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}$.
2. The projection of $\mathbf{u}$ onto $\mathbf{v}$ is the multiple of $\mathbf{v}$

$$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2}\,\mathbf{v}.$$

3. To resolve $\mathbf{u}$: the parallel part is $\mathbf{u}_{\parallel} = \operatorname{proj}_{\mathbf{v}} \mathbf{u}$, and the perpendicular part is the remainder, $\mathbf{u}_{\perp} = \mathbf{u} - \mathbf{u}_{\parallel}$.
4. Check that $\mathbf{u}_{\perp} \cdot \mathbf{v} = 0$ and $\mathbf{u}_{\parallel} + \mathbf{u}_{\perp} = \mathbf{u}$.

5. The scalar projection, the signed length of the shadow, is $\dfrac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|} = \mathbf{u}\cdot\hat{\mathbf{v}}$.
6. Work: a constant force $\mathbf{F}$ moving through a displacement $\mathbf{d}$ does work $W = \mathbf{F}\cdot\mathbf{d}$. Its scalar component along the motion is $\mathbf{F}\cdot\hat{\mathbf{d}} = W/\|\mathbf{d}\|$.

Example. A force $\mathbf{F} = (5, 2, 4)$ (newtons) moves its point of application through $\mathbf{d} = (2, 2, 1)$ (metres). Find the work done and the scalar component of $\mathbf{F}$ along the motion, and resolve $\mathbf{F}$ into parts along and across the motion.

$W = \mathbf{F}\cdot\mathbf{d} = 10 + 4 + 4$, so $\boxed{W = 18\ \text{J}}$. With $\|\mathbf{d}\| = \sqrt{4+4+1} = 3$, the scalar component along the motion is $W/\|\mathbf{d}\| = \boxed{6\ \text{N}}$.

For the parts, $\mathbf{F}\cdot\mathbf{d} = 18$ and $\|\mathbf{d}\|^2 = 9$, so $\mathbf{F}_{\parallel} = \frac{18}{9}(2, 2, 1) = (4, 4, 2)$, and the remainder is $\mathbf{F}_{\perp} = (5, 2, 4) - (4, 4, 2)$:

$$\boxed{\mathbf{F}_{\parallel} = (4, 4, 2)\ \text{N}}, \qquad \boxed{\mathbf{F}_{\perp} = (1, -2, 2)\ \text{N}}$$

Check. $\mathbf{F}_{\perp}\cdot\mathbf{d} = 2 - 4 + 2 = 0$ and $\mathbf{F}_{\parallel} + \mathbf{F}_{\perp} = (5, 2, 4) = \mathbf{F}$. The parallel part has length $\|(4, 4, 2)\| = 6$, the scalar component, and Pythagoras holds: $\|\mathbf{F}\|^2 = 45 = 36 + 9$.

Pitfalls

- Dividing by $\|\mathbf{d}\|$ instead of $\|\mathbf{d}\|^2$. It gives $6(2, 2, 1) = (12, 12, 6)$, three times too long. The vector projection divides by the length squared, the scalar projection by the length.
- Projecting the wrong way round. $\text{proj}_{\mathbf{d}}\,\mathbf{F}$ lies along $\mathbf{d}$; $\text{proj}_{\mathbf{F}}\,\mathbf{d}$ lies along $\mathbf{F}$, and is a different vector.
- Mixing up the two components. The scalar component is the number 6 N, the parallel part the vector $(4, 4, 2)$ N; and work is a scalar, in joules.

In the notes. Unit 4: The scalar (dot) product, with its projection example and the remark on work.

Practise it. the Review set on this method

4.4 The cross product: normal, area, volume, torque

Problem. In $\mathbb{R}^3$, find a vector perpendicular to two given vectors, the area of a parallelogram or a triangle, the volume of a parallelepiped, or a torque.

Method

1. Compute $\mathbf{u}\times\mathbf{v}$ from its components $u_2v_3 - u_3v_2$, $u_3v_1 - u_1v_3$ and $u_1v_2 - u_2v_1$, showing every product. The determinant with $\mathbf{i}, \mathbf{j}, \mathbf{k}$ in the top row is a way to remember them.
2. Check: the result dotted with $\mathbf{u}$, and with $\mathbf{v}$, gives 0.
3. Normal: $\mathbf{u}\times\mathbf{v}$ is perpendicular to both factors. Divide out any common factor; for a unit normal, divide by its length.
4. Area: $\|\mathbf{u}\times\mathbf{v}\|$ for the parallelogram on $\mathbf{u}$ and $\mathbf{v}$, and half of it for the triangle. For a triangle ABC, take $\mathbf{u} = \overrightarrow{AB}$ and $\mathbf{v} = \overrightarrow{AC}$.
5. Volume (optional enrichment, as in the notes): compute $\mathbf{v}\times\mathbf{w}$ first, then $\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})$. The volume of the parallelepiped is its absolute value, and 0 means the three vectors are coplanar.
6. Torque: $\boldsymbol{\tau} = \mathbf{r}\times\mathbf{F}$, by step 1, with $\mathbf{r}$ the position of the point where $\mathbf{F}$ acts, measured from

the pivot.

Example. Let $A(0,1,1)$, $B(2,4,-1)$, $C(1,3,-1)$ and $D(1,2,4)$. (a) Find a unit vector perpendicular to the plane of A, B and C, and the area of the triangle ABC. (b) Find the volume of the parallelepiped on $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{AD}$. (c) Find the torque about a pivot of the force $\mathbf{F} = (3,0,-1)$ (newtons) acting at $\mathbf{r} = (0,2,1)$ (metres) from it.

(a) $\overrightarrow{AB} = (2,3,-2)$ and $\overrightarrow{AC} = (1,2,-2)$. The components of their cross product are $3\cdot(-2) - (-2)\cdot 2 = -2$, $(-2)\cdot 1 - 2\cdot(-2) = 2$ and $2\cdot 2 - 3\cdot 1 = 1$:

$$\boxed{\overrightarrow{AB} \times \overrightarrow{AC} = (-2,2,1)}$$

Check: $\overrightarrow{AB}\cdot(-2,2,1) = -4+6-2 = 0$ and $\overrightarrow{AC}\cdot(-2,2,1) = -2+4-2 = 0$. Its length is $\sqrt{4+4+1} = 3$, so

$$\boxed{\hat{\mathbf{n}} = \pm\tfrac{1}{3}(-2,2,1)}, \qquad \boxed{\text{area} = \tfrac{3}{2}}$$

(b) The box on $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{AD}$ is the same whichever is called $\mathbf{u}$, so take $\mathbf{u} = \overrightarrow{AD} = (1,1,3)$ and reuse the cross product: $\overrightarrow{AD}\cdot(\overrightarrow{AB}\times\overrightarrow{AC}) = -2+2+3 = 3$, so the volume is $\boxed{3}$, and the four points are not coplanar.

(c) $\boldsymbol{\tau} = \mathbf{r}\times\mathbf{F}$ has components $2\cdot(-1) - 1\cdot 0 = -2$, $1\cdot 3 - 0\cdot(-1) = 3$ and $0\cdot 0 - 2\cdot 3 = -6$:

$$\boxed{\boldsymbol{\tau} = (-2,3,-6)\ \text{N m}}$$

of magnitude 7 N m. Check: $\boldsymbol{\tau}\cdot\mathbf{r} = 0+6-6 = 0$ and $\boldsymbol{\tau}\cdot\mathbf{F} = -6+0+6 = 0$.

Check. The area also follows from $\|\mathbf{u}\|\,\|\mathbf{v}\|\sin\theta$. Here $\left\|\overrightarrow{AB}\right\| = \sqrt{17}$, $\left\|\overrightarrow{AC}\right\| = 3$ and $\overrightarrow{AB}\cdot\overrightarrow{AC} = 12$, so $\cos\theta = \frac{4}{\sqrt{17}}$ and $\sin\theta = \frac{1}{\sqrt{17}}$. The parallelogram's area is $\sqrt{17}\cdot 3\cdot\frac{1}{\sqrt{17}} = 3$, twice the triangle's.

Pitfalls

- Getting the middle component's sign wrong. It is $u_3v_1 - u_1v_3$, the minus sign on $\mathbf{j}$ in the determinant: above, $(-2)\cdot 1 - 2\cdot(-2) = 2$, not -2.
- Forgetting the half: 3 is the area of the parallelogram, and the triangle's is $\frac{3}{2}$.
- Reversing the order. $\overrightarrow{AC}\times\overrightarrow{AB} = (2,-2,-1)$ is still a normal, but $\mathbf{F}\times\mathbf{r}$ is minus the torque. A negative triple product still gives a positive volume: take the absolute value.

In the notes. Unit 4: The vector (cross) product, and Unit 4: Optional: the scalar triple product.
Practise it. the Review set on this method

4.5 A line through two points; is a point on it?

Problem. Find the line through two points in vector, parametric or symmetric form, and decide whether a given point lies on it.

Method

1. The direction is $\mathbf{d} = \overrightarrow{AB}$: subtract the coordinates of A from those of B. A tidier multiple serves as well. The length $\left\|\overrightarrow{AB}\right\|$ is the distance from A to B.
2. Take A as the base point, with position vector $\mathbf{r}_0$. The vector form is $\mathbf{r} = \mathbf{r}_0 + t\,\mathbf{d}$, and in

coordinates the parametric form is $x = x_0 + t\,d_1$, $y = y_0 + t\,d_2$, $z = z_0 + t\,d_3$.

3. If asked, eliminate t for the symmetric form $\dfrac{x - x_0}{d_1} = \dfrac{y - y_0}{d_2} = \dfrac{z - z_0}{d_3}$, possible when no d_i is 0.
4. Check: some value of t returns B.
5. Point test: solve for t from one coordinate whose $d_i \neq 0$, then put that t into the others. The point is on the line only if every coordinate matches.

Example. Find the line through $A(3, 1, -2)$ and $B(-1, 3, 4)$ in vector, parametric and symmetric form, and the distance AB. Do $C(7, -1, -8)$ and $D(1, 2, 2)$ lie on it?

$\overrightarrow{AB} = (-4, 2, 6) = 2(-2, 1, 3)$, so $AB = \sqrt{16 + 4 + 36} = \sqrt{56} = \boxed{2\sqrt{14}}$, and the tidier direction is $\mathbf{d} = (-2, 1, 3)$. With base point A,

$$\boxed{\mathbf{r} = (3, 1, -2) + t(-2, 1, 3)}$$

that is, $x = 3 - 2t$, $y = 1 + t$, $z = -2 + 3t$. Eliminating t,

$$\boxed{\frac{x - 3}{-2} = y - 1 = \frac{z + 2}{3}}$$

Putting $t = 2$ gives $(3 - 4,\ 1 + 2,\ -2 + 6) = (-1, 3, 4) = B$.

For C: $3 - 2t = 7$ gives $t = -2$, and then $y = 1 - 2 = -1$ and $z = -2 - 6 = -8$ both match, so $\boxed{C \text{ is on the line}}$, at $t = -2$. For D: $3 - 2t = 1$ gives $t = 1$; then $y = 2$ matches, but $z = -2 + 3 = 1 \neq 2$, so $\boxed{D \text{ is not on the line}}$.

Check. $\overrightarrow{AC} = (4, -2, -6) = -2\mathbf{d}$ is parallel to the direction, as it must be for a point of the line; $\overrightarrow{AD} = (-2, 1, 4)$ is not a multiple of $\mathbf{d}$.

Pitfalls

- Testing too few coordinates. D matches in x and in y, and fails only in z.
- Using $\overrightarrow{AB}$ as the base point. $\mathbf{r} = (-4, 2, 6) + t(-2, 1, 3)$ is a parallel line that misses A and B.
- Writing a symmetric form with a 0 in a denominator. When some $d_i = 0$, that coordinate is constant along the line; state it separately.

In the notes. Unit 4: Components, for the vector between two points, and Unit 4: The equation of a line.

Practise it. the Review set on this method

4.6 A plane through three points

Problem. Find the plane through three points, in point-normal, Cartesian or parametric form.

Method

1. *Two vectors in the plane.* With P as the base point, form $\overrightarrow{PQ}$ and $\overrightarrow{PR}$.
2. *A normal, by the cross product*: $\mathbf{n} = \overrightarrow{PQ} \times \overrightarrow{PR}$ (Method 4.4). Any nonzero multiple serves, so divide out a common factor.

3. *The equation.* Write the point-normal form $\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$ at P, that is $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$, and multiply it out to $ax + by + cz = d$, with $d = \mathbf{n} \cdot \mathbf{r}_0$.
4. Check: all three points satisfy it.
5. Parametric form, if asked: $\mathbf{r} = \mathbf{r}_0 + s\,\overrightarrow{PQ} + t\,\overrightarrow{PR}$, with $s, t \in \mathbb{R}$.

Why it works. A point $\mathbf{r}$ lies in the plane exactly when $\mathbf{r} - \mathbf{r}_0$ lies flat in it, that is, when it is orthogonal to $\mathbf{n}$. If $\overrightarrow{PQ} \times \overrightarrow{PR} = \mathbf{0}$, the three points are on one line and do not fix a plane.

Example. Find the plane through $P(1, -1, 2)$, $Q(-1, 0, 1)$ and $R(1, 2, 3)$ in Cartesian and in parametric form.

Two vectors in the plane: $\overrightarrow{PQ} = (-2, 1, -1)$ and $\overrightarrow{PR} = (0, 3, 1)$.

A normal: the components of $\overrightarrow{PQ} \times \overrightarrow{PR}$ are $1 \cdot 1 - (-1) \cdot 3 = 4$, $(-1) \cdot 0 - (-2) \cdot 1 = 2$ and $(-2) \cdot 3 - 1 \cdot 0 = -6$. So it is $(4, 2, -6) = 2(2, 1, -3)$; take $\mathbf{n} = (2, 1, -3)$.

The equation: point-normal form at P,

$$2(x - 1) + 1(y + 1) - 3(z - 2) = 0,$$

which multiplies out to $2x + y - 3z + 5 = 0$:

$$\boxed{2x + y - 3z = -5}$$

Check: P gives $2 - 1 - 6 = -5$, Q gives $-2 + 0 - 3 = -5$ and R gives $2 + 2 - 9 = -5$.

In parametric form,

$$\mathbf{r} = (1, -1, 2) + s(-2, 1, -1) + t(0, 3, 1),$$

that is, $\boxed{\mathbf{r} = (1 - 2s,\ -1 + s + 3t,\ 2 - s + t)}$.

Check. The normal is perpendicular to both vectors in the plane: $(2, 1, -3) \cdot (-2, 1, -1) = -4 + 1 + 3 = 0$ and $(2, 1, -3) \cdot (0, 3, 1) = 0 + 3 - 3 = 0$. The parametric point at $s = 1$, $t = -1$ is $(-1, -3, 0)$, and $2(-1) + (-3) - 3 \cdot 0 = -5$.

Pitfalls

- Crossing the position vectors instead of the vectors in the plane: $\overrightarrow{OQ} \times \overrightarrow{OR} = (-2, 4, -2)$ is not perpendicular to $\overrightarrow{PQ}$ (their dot product is 10).
- Stopping at $2x + y - 3z = 0$. The right-hand side is $d = \mathbf{n} \cdot \mathbf{r}_0 = -5$; it is 0 only for a plane through the origin.
- Checking only the base point. P satisfies the equation by construction; Q and R are what test the normal.

In the notes. Unit 4: The equation of a plane, with its worked example.

Practise it. the Review set on this method

4.7 Where a line meets a plane; distance from a point to a plane

Problem. Find where a line meets a plane, or show that it misses the plane or lies in it; find the foot of the perpendicular from a point to a plane, and the distance.

Method

1. Write the coordinates of a point of the line in terms of t: $x = x_0 + t\,d_1$, $y = y_0 + t\,d_2$, $z = z_0 + t\,d_3$.
2. Substitute them into the plane's equation, and solve for t.
3. Put that t back into the line: it gives the crossing point. Check that the point satisfies the plane's equation.
4. If t cancels, then $\mathbf{d}\cdot\mathbf{n} = 0$ and the direction lies flat. The base point decides: on the plane, the line lies in it (every t works); off it, the line is parallel (no t works).
5. Distance from a point $\mathbf{p}$: step from $\mathbf{p}$ along the normal, $\mathbf{p} + t\,\mathbf{n}$. Substitute into the plane's equation and solve for t; that point is the foot of the perpendicular.
6. The distance is $|t|\ \|\mathbf{n}\|$. Check it by the formula $\dfrac{|ap_1 + bp_2 + cp_3 - d|}{\|\mathbf{n}\|}$.

Example. Take the plane $2x - 3y + 6z = 5$. (a) Where does the line $\mathbf{r} = (3,1,0) + t(1,1,0)$ meet it? (b) And the line $\mathbf{r} = (1,1,1) + t(3,0,-1)$? (c) Find the foot of the perpendicular from $P(2,-1,2)$ to the plane, and the distance from P to it.

(a) A point of the line is $(3+t,\ 1+t,\ 0)$. Substituting,

$$2(3+t) - 3(1+t) + 6\cdot 0 = 5 \iff 3 - t = 5,$$

so $t = -2$, and the crossing point is $\boxed{(1,-1,0)}$. It satisfies $2 + 3 + 0 = 5$.

(b) A point of the line is $(1+3t,\ 1,\ 1-t)$. Substituting,

$$2(1+3t) - 3 + 6(1-t) = 5 \iff 5 = 5,$$

which holds for every t. The t cancelled because $\mathbf{d}\cdot\mathbf{n} = 6 + 0 - 6 = 0$, and the base point is on the plane $(2 - 3 + 6 = 5)$, so $\boxed{\text{the line lies in the plane}}$.

(c) The normal is $\mathbf{n} = (2,-3,6)$, with $\|\mathbf{n}\| = \sqrt{4+9+36} = 7$. Step from P: $\mathbf{p} + t\,\mathbf{n} = (2+2t,\ -1-3t,\ 2+6t)$ lies on the plane when

$$2(2+2t) - 3(-1-3t) + 6(2+6t) = 5,$$

that is $19 + 49t = 5$, so $t = -\frac{14}{49} = -\frac{2}{7}$. The foot is $(2,-1,2) - \frac{2}{7}(2,-3,6)$:

$$\boxed{F = \left(\tfrac{10}{7}, -\tfrac{1}{7}, \tfrac{2}{7}\right)}$$

and the distance is $|t|\ \|\mathbf{n}\| = \frac{2}{7}\cdot 7 = \boxed{2}$.

Check. The formula gives $\dfrac{|2\cdot 2 - 3\cdot(-1) + 6\cdot 2 - 5|}{7} = \dfrac{14}{7} = 2$, and the foot is on the plane: $\frac{20}{7} + \frac{3}{7} + \frac{12}{7} = 5$. The foot is also 2 from P: $\overrightarrow{FP} = \frac{1}{7}(4,-6,12)$ has length $\frac{1}{7}\sqrt{196} = 2$.

Pitfalls

- Stopping at t. In (a), $t = -2$ is a parameter value, not the point: put it back into the line.
- Calling the line in (b) parallel because $\mathbf{d}\cdot\mathbf{n} = 0$. That says only that the direction lies flat; the base point decides between parallel and contained.
- Giving $|t| = \frac{2}{7}$ as the distance. The step was $t\,\mathbf{n}$, so the distance is $|t|\ \|\mathbf{n}\| = 2$, unless $\mathbf{n}$ is a unit vector.

In the notes. Unit 4: Where a line meets a plane; the angle between two planes, with its note on when a line crosses a plane and its distance example.

Practise it. the Review set on this method

Unit 5 Matrices

Five methods, and each starts from the sizes and the shape of the matrix. A short proof about matrices (a property of the inverse or of the transpose) follows one pattern: state the goal, rewrite it with the named rules (associativity, the transpose rules, uniqueness of the inverse) and end with a concluding sentence. For a trace identity, write out the entries of both sides instead.

Which method?

The question asks for	Method
whether a sum or a product is defined, and its size	5.1, compare the sizes first
a product, a transpose or a trace	5.1, row times column
a 3×3 determinant with zeros in it	5.2, expand along the row or column with most zeros
the value of a parameter that makes a matrix singular	5.2, solve det = 0
the inverse of a 2×2 matrix	5.3, swap, negate, divide by det
the inverse of a 3×3 matrix that is not triangular	5.3, cofactors and the adjugate
the inverse of a triangular or larger matrix	5.3, row-reduce $(\,\mathbf{A} \mid \mathbf{I}\,)$
the solution of a system by the inverse method, or with $\mathbf{A}^{-1}$ known	5.4, variant: $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ ($\mathbf{A}^{-1}$ by 5.3)
the solution of a system, otherwise	5.4, Gaussian elimination
whether a system has one, no or infinitely many solutions	5.4, read the echelon form, $0 = c$ first
the matrix of a linear map of the plane, or of two maps in turn	5.5, columns are the images of $\mathbf{i}, \mathbf{j}$
the effect of a map on area and orientation	5.5, the sign and size of det

5.1 Products, transpose and trace: check the sizes first

Problem. Decide whether a sum or a product of matrices is defined, and compute it; find a transpose or a trace.

Method

1. Write down the size of every matrix, $m \times n$ (m rows, n columns).
2. Sum or difference: the sizes must be equal. Then, as for a scalar multiple, work entry by entry.
3. Product $\mathbf{AB}$: write the sizes as $(m \times n)(n' \times p)$. If $n \neq n'$ the product is undefined: stop. Otherwise it is $m \times p$.
4. The (i, j) entry of $\mathbf{AB}$ is row i of $\mathbf{A}$ dotted with column j of $\mathbf{B}$.
5. Transpose: swap rows and columns, so the (i, j) entry of $\mathbf{A}^{\mathsf{T}}$ is a_{ji}. For a product, $(\mathbf{AB})^{\mathsf{T}} = \mathbf{B}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}$.
6. Trace, of a square matrix only: the sum of the diagonal entries.

Example. Let $\mathbf{A} = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 3 & -2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 4 \\ 0 & -1 \\ 3 & 2 \end{pmatrix}$. Which of $\mathbf{A}+\mathbf{B}$, $\mathbf{A}+\mathbf{B}^{\mathsf{T}}$, $\mathbf{AB}$ and $\mathbf{BA}$ are defined? Compute $\mathbf{A}+\mathbf{B}^{\mathsf{T}}$, $\mathbf{AB}$ and $\operatorname{tr}(\mathbf{AB})$.

$\mathbf{A}$ is 2×3 and $\mathbf{B}$ is 3×2, so $\boxed{\mathbf{A}+\mathbf{B} \text{ is undefined}}$. $\mathbf{B}^{\mathsf{T}}$ is 2×3, so $\mathbf{A}+\mathbf{B}^{\mathsf{T}}$ is defined and 2×3. The product $(2\times 3)(3\times 2)$ has matching inner sizes and is 2×2; $\mathbf{BA}$ is $(3\times 2)(2\times 3)$, so it is 3×3. Entrywise,

$$\mathbf{A}+\mathbf{B}^{\mathsf{T}} = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 3 & -2 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 3 \\ 4 & -1 & 2 \end{pmatrix} = \boxed{\begin{pmatrix} 3 & -1 & 3 \\ 5 & 2 & 0 \end{pmatrix}}.$$

Row times column,

$$\mathbf{AB} = \begin{pmatrix} 2+0+0 & 8+1+0 \\ 1+0-6 & 4-3-4 \end{pmatrix} = \boxed{\begin{pmatrix} 2 & 9 \\ -5 & -3 \end{pmatrix}},$$

and the trace is the diagonal sum, $2+(-3)$: $\boxed{\operatorname{tr}(\mathbf{AB}) = -1}$.

Check. The cyclic property says $\operatorname{tr}(\mathbf{BA}) = \operatorname{tr}(\mathbf{AB})$, although $\mathbf{BA}$ is 3×3. Its diagonal entries are $1\cdot 2+4\cdot 1 = 6$, $0\cdot(-1)+(-1)\cdot 3 = -3$ and $3\cdot 0+2\cdot(-2) = -4$, and $6-3-4 = -1$. That tests only the diagonal. For every entry use associativity, $(\mathbf{AB})\mathbf{v} = \mathbf{A}(\mathbf{Bv})$, with $\mathbf{v} = (1,1)^{\mathsf{T}}$: $\mathbf{Bv} = (5,-1,5)^{\mathsf{T}}$ and $\mathbf{A}(\mathbf{Bv}) = (11,-8)^{\mathsf{T}}$, the row sums $2+9$ and $-5-3$ of $\mathbf{AB}$. A wrong entry changes a row sum.

Pitfalls

- Adding $\mathbf{A}$ and $\mathbf{B}$ because both have six entries. The sizes 2×3 and 3×2 differ, so the sum does not exist.
- Multiplying entry by entry, or giving a product the inner sizes: the size of $\mathbf{BA}$ comes from the outer ones, 3×3.
- Transposing a product without reversing it. Here $\mathbf{A}^{\mathsf{T}}\mathbf{B}^{\mathsf{T}}$ is 3×3, so it cannot be $(\mathbf{AB})^{\mathsf{T}}$, which is 2×2.

In the notes. Unit 5: Addition, subtraction and scalar multiplication, Unit 5: Matrix multiplication, and Unit 5: Transpose and trace.

Practise it. the Review set on this method

5.2 Determinants by cofactor expansion; the parameter that makes a matrix singular

Problem. Evaluate the determinant of a 2×2 or 3×3 matrix; decide whether the matrix is invertible; find the values of a parameter for which it is singular.

Method

1. For a 2×2 matrix, $\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$.
2. For a 3×3 matrix, choose the row or column with the most zeros (the first row if none stands out).
3. Write each entry of that row or column times its cofactor $C_{ij} = (-1)^{i+j}M_{ij}$. The sign follows the checkerboard $\begin{smallmatrix} + & - & + \\ - & + & - \\ + & - & + \end{smallmatrix}$, and the minor M_{ij} is the 2×2 determinant left when row i and column j are deleted. A zero entry drops out.

4. Evaluate the 2×2 minors and add up.
5. The matrix is invertible exactly when $\det \neq 0$; $\det = 0$ means singular.
6. For a parameter k: find det as an expression in k, set it equal to 0 and solve. Those values, and only those, make the matrix singular.

Variants. A *triangular* matrix has determinant equal to the product of its diagonal entries, with no expansion needed. A larger matrix expands the same way: for a 4×4 the minors are 3×3 determinants.

Example. (a) Evaluate $\det \mathbf{A}$ for $\mathbf{A} = \begin{pmatrix} 2 & 1 & -3 \\ 4 & 0 & 5 \\ -1 & 0 & 2 \end{pmatrix}$. (b) For which k is $\mathbf{B} = \begin{pmatrix} k & 1 & 0 \\ 3 & k & 2 \\ 0 & 1 & 1 \end{pmatrix}$ singular?

(a) Column 2 has two zeros, so expand along it. Only the entry $a_{12} = 1$ survives, and position $(1, 2)$ carries the sign $-$:

$$\det \mathbf{A} = -1 \begin{vmatrix} 4 & 5 \\ -1 & 2 \end{vmatrix} = -(8 + 5) = \boxed{-13}.$$

(b) Expand along the first row, whose third entry is 0:

$$\det \mathbf{B} = k \begin{vmatrix} k & 2 \\ 1 & 1 \end{vmatrix} - 1 \begin{vmatrix} 3 & 2 \\ 0 & 1 \end{vmatrix} = k(k - 2) - 3.$$

So $\det \mathbf{B} = k^2 - 2k - 3 = (k - 3)(k + 1)$, which is 0 for $\boxed{k = -1 \text{ or } k = 3}$.

Check. (a) Expanding along the first row instead gives $2 \cdot 0 - 1 \cdot 13 + (-3) \cdot 0 = -13$. (b) At $k = 3$ the rows of $\mathbf{B}$ are $(3, 1, 0)$, $(3, 3, 2)$, $(0, 1, 1)$, and the second is the first plus twice the third. At $k = -1$ they are $(-1, 1, 0)$, $(3, -1, 2)$, $(0, 1, 1)$, and the second is twice the third minus three times the first. Either way the three rows lie in one plane, so the determinant must be 0: both roots check.

Pitfalls

- Dropping the checkerboard sign. In (a) the cofactor of a_{12} is -13, so $\det \mathbf{A} = -13$, not 13.
- Using the diagonal product on a matrix that is not triangular.
- Losing a root: $k^2 - 2k - 3 = 0$ has two solutions, and both make $\mathbf{B}$ singular.

In the notes. Unit 5: The determinant, and Unit 5: The inverse, for invertible exactly when $\det \neq 0$.

Practise it. the Review set on this method

5.3 The inverse: the 2×2 formula, cofactors, or row reduction

Problem. Find the inverse of a square matrix, or show that it has none.

Method

1. Find $\det \mathbf{A}$ (Method 5.2). If it is 0 there is no inverse: stop.
2. A 2×2 matrix: swap the diagonal entries, negate the off-diagonal ones, and divide by the determinant, $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \dfrac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

3. A 3×3 matrix by cofactors: find all nine cofactors $C_{ij} = (-1)^{i+j} M_{ij}$ and write the cofactor matrix $\mathbf{C} = (C_{ij})$.
4. Transpose it to get the adjugate, $\operatorname{adj}(\mathbf{A}) = \mathbf{C}^{\mathsf{T}}$, and divide by the determinant: $\mathbf{A}^{-1} = \dfrac{1}{\det \mathbf{A}} \operatorname{adj}(\mathbf{A})$.
5. Or by row reduction: write $(\,\mathbf{A} \mid \mathbf{I}\,)$ and apply row operations until the left block is $\mathbf{I}$ (echelon form, then scale each leading entry to 1 and clear above it). The right block is then $\mathbf{A}^{-1}$.
6. Check one product: $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$.

Example. Invert (a) $\mathbf{P} = \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix}$ and (b) $\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 3 & -1 \\ 2 & 1 & -1 \end{pmatrix}$.

(a) $\det \mathbf{P} = 3(-1) - (-2)(4) = 5 \neq 0$. Swap, negate and divide by 5:

$$\boxed{\mathbf{P}^{-1} = \frac{1}{5} \begin{pmatrix} -1 & 2 \\ -4 & 3 \end{pmatrix}}$$

(b) Expand along the first row: $\det \mathbf{A} = 1 \cdot \begin{vmatrix} 3 & -1 \\ 1 & -1 \end{vmatrix} = -2$. Deleting each row and column in turn and attaching the checkerboard sign gives

$$\mathbf{C} = \begin{pmatrix} -2 & 1 & -3 \\ 0 & -1 & -1 \\ 0 & 1 & 3 \end{pmatrix},$$

for instance $C_{12} = -\begin{vmatrix} 3 & -1 \\ 2 & -1 \end{vmatrix} = 1$ and $C_{23} = -\begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = -1$. Transpose, then divide by -2:

$$\operatorname{adj}(\mathbf{A}) = \mathbf{C}^{\mathsf{T}} = \begin{pmatrix} -2 & 0 & 0 \\ 1 & -1 & 1 \\ -3 & -1 & 3 \end{pmatrix},$$

$$\boxed{\mathbf{A}^{-1} = \frac{1}{2} \begin{pmatrix} 2 & 0 & 0 \\ -1 & 1 & -1 \\ 3 & 1 & -3 \end{pmatrix}}$$

Check. $\mathbf{P} \begin{pmatrix} -1 & 2 \\ -4 & 3 \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$, so $\mathbf{P}\mathbf{P}^{-1} = \mathbf{I}$. And

$$\mathbf{A} \begin{pmatrix} 2 & 0 & 0 \\ -1 & 1 & -1 \\ 3 & 1 & -3 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix},$$

so $\mathbf{A}\mathbf{A}^{-1} = \frac{1}{2} \cdot 2\mathbf{I} = \mathbf{I}$.

Pitfalls

- Forgetting the transpose. In (b) $C_{12} = 1$, but the $(1,2)$ entry of the adjugate is $C_{21} = 0$.
- Dividing by $|\det \mathbf{A}| = 2$ instead of by -2: every entry of $\mathbf{A}^{-1}$ changes sign.
- Inverting entry by entry. $\mathbf{P}^{-1}$ is not the matrix of reciprocals $\frac{1}{3}$, $-\frac{1}{2}$, $\frac{1}{4}$, -1.

In the notes. Unit 5: The inverse, and Unit 5: Gaussian elimination, for the inverse by row reduction.

Practise it. the Review set on this method

5.4 Gaussian elimination and reading the echelon form

Problem. Solve a linear system $\mathbf{Ax} = \mathbf{b}$, or decide whether it has one solution, none or infinitely many.

Method

1. Write the augmented matrix $(\,\mathbf{A} \mid \mathbf{b}\,)$.
2. Reduce it to echelon form with labelled row operations $R_i \to R_i + kR_j$, stacked when done at once. Keep the entries integers (a step such as $R_2 \to 2R_2 + 3R_1$ is allowed). If the entry that should be the next leading entry is 0, swap in a lower row where it is not.
3. Look *first* for a row $(\,0 \;\cdots\; 0 \mid c\,)$ with $c \neq 0$. It reads $0 = c$: there is no solution. Stop.
4. Otherwise count the leading entries. With one for each unknown the solution is unique: back-substitute, from the last row up.
5. With fewer leading entries than unknowns, an unknown whose column has no leading entry is free. Set it equal to t, back-substitute, and write the solution as $\mathbf{x} = \mathbf{p} + t\,\mathbf{d}$, $t \in \mathbb{R}$: a line. With two free unknowns, set them equal to s and t: the solution $\mathbf{x} = \mathbf{p} + s\,\mathbf{u} + t\,\mathbf{v}$ is a plane.
6. Check the answer in every original equation (for a line, at $t = 0$ and $t = 1$).

Variants. *Gauss–Jordan* continues from the echelon form: scale each leading entry to 1 and clear above it. When the solution is unique this reaches $(\,\mathbf{I} \mid \mathbf{x}\,)$, and the solution is read off. The *inverse method* solves $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$, with $\mathbf{A}^{-1}$ found by Method 5.3 if it is not given. With $\mathbf{P}^{-1}$ from Method 5.3, the system $3x - 2y = 4$, $4x - y = 7$ gives

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{5}\begin{pmatrix} -1 & 2 \\ -4 & 3 \end{pmatrix}\begin{pmatrix} 4 \\ 7 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} -4+14 \\ -16+21 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

For a square system, $\det \mathbf{A} \neq 0$ guarantees exactly one solution; $\det = 0$ leaves none or infinitely many, and only the echelon form says which.

Example. For which values of a and b does the system

$$x + 2y - z = 2, \qquad 2x + 5y + 3z = 7, \qquad 3x + 7y + az = b$$

have exactly one solution, none, or infinitely many? Solve it in the last case.

Reduce the augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 2 & 5 & 3 & 7 \\ 3 & 7 & a & b \end{array}\right)$$

$$\xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & 1 & 5 & 3 \\ 0 & 1 & a+3 & b-6 \end{array}\right)$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & 1 & 5 & 3 \\ 0 & 0 & a-2 & b-9 \end{array}\right).$$

If $a \neq 2$ there are three leading entries for three unknowns: $\boxed{a \neq 2:\ \text{exactly one solution}}$. If $a = 2$ the last row is $(\,0\ 0\ 0 \mid b-9\,)$, which reads $0 = b - 9$. So $\boxed{a = 2,\ b \neq 9:\ \text{no solution}}$.

If $a = 2$ and $b = 9$ the last row is all zeros, and z is free. Set $z = t$. Row 2 gives $y = 3 - 5t$, and row 1 gives $x = 2 - 2(3 - 5t) + t = -4 + 11t$:

$$\boxed{\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -4 \\ 3 \\ 0 \end{pmatrix} + t \begin{pmatrix} 11 \\ -5 \\ 1 \end{pmatrix},\ t \in \mathbb{R}}$$

a line on which the three planes meet.

Check. With $a = 2$, $b = 9$: $t = 0$ gives $(-4, 3, 0)$ and $t = 1$ gives $(7, -2, 1)$, and $7 - 4 - 1 = 2$, $14 - 10 + 3 = 7$, $21 - 14 + 2 = 9$ (the first point works the same way). Also $\det \mathbf{A} = a - 2$ by cofactors, so $a \neq 2$ is exactly $\det \mathbf{A} \neq 0$.

Pitfalls

- Reading a zero row of coefficients as "infinitely many" before looking at the right-hand side. With $a = 2$, $b = 10$ the row is $(\,0\ 0\ 0 \mid 1\,)$: no solution.
- Changing the coefficients and forgetting the right-hand column, or using a row in the same step as it is being changed.
- Deciding between none and infinitely many from $\det \mathbf{A} = 0$. The determinant only says the solution is not unique.

In the notes. Unit 5: The inverse-matrix method, Unit 5: Gaussian elimination, and Unit 5: What the solutions look like.

Practise it. the Review set on this method

5.5 A matrix as a transformation of the plane

Problem. Find the matrix of a linear map of the plane, or of two maps done in turn; find the image of a point or a shape; describe the effect on area and orientation.

Method

1. Only a linear map has a matrix: a translation moves the origin, and no 2×2 matrix does. The columns of the matrix are the images of $\mathbf{i} = (1, 0)$ and $\mathbf{j} = (0, 1)$. Or take it from the standard list: rotation by θ, $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$; reflection in $y = x$; enlargement; stretch; shear.
2. "Do **S**, then **R**" is the single matrix **RS**: the map done first stands on the right.
3. Apply the matrix to the vector, or to each corner of a shape. The edges between corners go to the edges between their images.

4. Read the determinant: |det| is the factor by which areas scale, a negative det flips the orientation, and det = 0 collapses the plane onto a line, or onto the origin for the zero matrix (the map is not invertible).
5. To name a map, see where it sends **i** and **j**.

Example. Let $\mathbf{S} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, the shear $(x, y) \mapsto (x + 2y, y)$, and let $\mathbf{T}$ be the map that sends **i** to $(1, 1)$ and **j** to $(2, -1)$. (a) Find $\mathbf{T}$. (b) Find the matrix $\mathbf{M}$ of "first $\mathbf{S}$, then $\mathbf{T}$". (c) Find the image of the triangle with corners $(0, 0)$, $(3, 0)$, $(1, 2)$ under $\mathbf{M}$, and the effect on area and orientation.

(a) The images of **i** and **j** are the columns: $\boxed{\mathbf{T} = \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix}}$.

(b) The shear is done first, so it stands on the right:

$$\mathbf{M} = \mathbf{TS} = \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \boxed{\begin{pmatrix} 1 & 4 \\ 1 & 1 \end{pmatrix}}$$

(c) $\mathbf{M}$ fixes $(0, 0)$ and sends $(3, 0)$ to $(3, 3)$ and $(1, 2)$ to $(1+8,\ 1+2)$. The image is the triangle with corners $\boxed{(0, 0),\ (3, 3),\ (9, 3)}$. Since $\boxed{\det \mathbf{M} = 1 - 4 = -3}$, areas are multiplied by 3 and the orientation is flipped.

Check. Half the 2×2 determinant of the two edges from $(0, 0)$ gives the signed area. Before: $\frac{1}{2}(3 \cdot 2 - 0 \cdot 1) = 3$. After: $\frac{1}{2}(3 \cdot 3 - 3 \cdot 9) = -9$: the area is $9 = 3 \times 3$, and the sign has reversed.

Pitfalls

- Multiplying in the order of the words. $\mathbf{ST} = \begin{pmatrix} 3 & 0 \\ 1 & -1 \end{pmatrix}$ applies $\mathbf{T}$ first, and sends $(1, 2)$ to $(3, -1)$, not $(9, 3)$.
- Writing the images of **i** and **j** as the rows instead of the columns.
- Reading $\det \mathbf{M} = -3$ as a negative area. The area scales by $|\det| = 3$; the sign is the orientation.

In the notes. Unit 5: Matrices as transformations, and Unit 5: The determinant, for the determinant as a signed area.

Practise it. the Review set on this method

Unit 6 Eigensystems

Six methods, and the first is a step in all the others. True-or-false questions and short proofs are not a method: answer each with a one-line reason that cites a result from the notes, or with one concrete counterexample, and set a proof out as set-up, derivation, conclusion.

6.1 The eigensystem: 2×2, 3×3, complex eigenvalues

Problem. Find the eigenvalues of a square matrix and an eigenvector for each, real or complex.

Method

1. Write $\det(\mathbf{A} - \lambda\mathbf{I})$ and expand it: a 2×2 directly, a 3×3 along the row or column with the most zeros.
2. Factor the characteristic polynomial and name its roots $\lambda_1, \lambda_2, \ldots$: these are the eigenvalues. A real 2×2 with a negative discriminant gives a complex pair $a \pm b\,\mathrm{i}$.
3. For each eigenvalue in turn, write $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$. For a 2×2, one row is enough: read off how v_1 and v_2 are related and choose a simple $\mathbf{v}$.
4. For a 3×3, row-reduce $\mathbf{A} - \lambda\mathbf{I}$, or read the equations off directly when the zeros make them immediate. Set the free unknown to 1 and back-substitute.
5. For a complex pair, find the eigenvector of $a + b\,\mathrm{i}$ from row 1; the eigenvector of $a - b\,\mathrm{i}$ is its conjugate. Only if asked: $|\lambda|$ sets the growth and $\arg\lambda$ the turning, on average per step over many steps; exactly at every step only for $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$.
6. Check one pair with $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$, and the eigenvalues with $\sum \lambda_i = \operatorname{tr}\mathbf{A}$ and $\prod \lambda_i = \det\mathbf{A}$.

Example. Find the eigenvalues and eigenvectors of (a) $\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 4 & -2 \\ 1 & 1 & 1 \end{pmatrix}$ and (b) $\mathbf{B} = \begin{pmatrix} 4 & -5 \\ 2 & -2 \end{pmatrix}$.

(a) Row 1 has two zeros, so expand along it:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (1 - \lambda)\big[(4 - \lambda)(1 - \lambda) + 2\big] = (1 - \lambda)(\lambda - 2)(\lambda - 3),$$

since $(4 - \lambda)(1 - \lambda) + 2 = \lambda^2 - 5\lambda + 6$. So $\lambda_1 = 1$, $\lambda_2 = 2$, $\lambda_3 = 3$. For $\lambda_1 = 1$, swap R_1 and R_3 of $\mathbf{A} - \mathbf{I}$, then take $R_2 \to R_2 - R_1$:

$$\mathbf{A} - \mathbf{I} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 3 & -2 \\ 1 & 1 & 0 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & -2 \\ 0 & 0 & 0 \end{pmatrix}.$$

With $v_3 = 1$, back-substitution gives $v_2 = 1$ and $v_1 = -1$. For $\lambda_2 = 2$ and $\lambda_3 = 3$ the first row of $\mathbf{A} - \lambda\mathbf{I}$ is $(1 - \lambda, 0, 0)$, so $v_1 = 0$ at once. Then row 2 reads $2v_2 - 2v_3 = 0$ for $\lambda = 2$ and $v_2 - 2v_3 = 0$ for $\lambda = 3$:

$$\boxed{\lambda = 1 : \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}}, \qquad \boxed{\lambda = 2 : \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}},$$

$$\boxed{\lambda = 3 : \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}}$$

(b) $\det(\mathbf{B} - \lambda\mathbf{I}) = (4-\lambda)(-2-\lambda) + 10 = \lambda^2 - 2\lambda + 2 = (\lambda-1)^2 + 1$, so $\lambda = 1 \pm \mathrm{i}$. For $\lambda = 1+\mathrm{i}$, row 1 of $\mathbf{B} - \lambda\mathbf{I}$ reads $(3-\mathrm{i})v_1 - 5v_2 = 0$; take $v_1 = 5$, $v_2 = 3 - \mathrm{i}$. The eigenvector of $1 - \mathrm{i}$ is the conjugate:

$$\boxed{\lambda = 1 \pm \mathrm{i}, \quad \mathbf{v} = (5,\ 3 \mp \mathrm{i})^{\mathsf{T}}}$$

If the geometry is asked for: $|1+\mathrm{i}| = \sqrt{2}$ and $\arg(1+\mathrm{i}) = \frac{\pi}{4}$, so on average each step stretches the iterates of $\mathbf{B}$ by $\sqrt{2}$ and turns them through $\frac{\pi}{4}$: they spiral outwards. After four steps they are back on their own line, reversed: $(1+\mathrm{i})^4 = -4$, and indeed $\mathbf{B}^4 = -4\mathbf{I}$.

Check. $\mathbf{A}(0,2,1)^{\mathsf{T}} = (0,6,3)^{\mathsf{T}} = 3(0,2,1)^{\mathsf{T}}$; $\operatorname{tr}\mathbf{A} = 6 = 1+2+3$ and $\det\mathbf{A} = 6 = 1\cdot 2\cdot 3$. In (b), $\mathbf{B}(5, 3-\mathrm{i})^{\mathsf{T}} = (5+5\mathrm{i},\ 4+2\mathrm{i})^{\mathsf{T}} = (1+\mathrm{i})(5, 3-\mathrm{i})^{\mathsf{T}}$; $\operatorname{tr}\mathbf{B} = 2$ and $\det\mathbf{B} = 2 = (1+\mathrm{i})(1-\mathrm{i})$.

Pitfalls

- Subtracting λ from one diagonal entry only, or from every entry. It comes off each diagonal entry and nowhere else.
- Row-reducing $\mathbf{A} - \lambda\mathbf{I}$ to a matrix with no zero row. Then the only solution is $\mathbf{0}$, and λ is not an eigenvalue: recheck the characteristic polynomial.
- Keeping the same eigenvector for both halves of a complex pair. The eigenvector of $1 - \mathrm{i}$ is $(5, 3+\mathrm{i})^{\mathsf{T}}$, not $(5, 3-\mathrm{i})^{\mathsf{T}}$.

In the notes. Unit 6: Finding eigenvalues and eigenvectors, with its trace and determinant checks, and Unit 6: Complex eigenvalues.

Practise it. the Review set on this method

6.2 Is it diagonalisable? Multiplicities

Problem. Decide whether a matrix is diagonalisable, giving the algebraic and geometric multiplicity of each eigenvalue.

Method

1. Find the eigenvalues (Method 6.1), and the *algebraic multiplicity* of each: how many times it repeats as a root. A triangular matrix has its diagonal entries as eigenvalues.
2. If the n eigenvalues are distinct, the matrix is diagonalisable. Stop.
3. For each repeated eigenvalue, solve $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ and count the free unknowns: that is its *geometric multiplicity*, the number of independent eigenvectors.
4. Compare. If geometric = algebraic for every eigenvalue, the matrix is diagonalisable. If any eigenvalue falls short, it is defective and cannot be diagonalised.

Why it works. An $n \times n$ matrix is diagonalisable exactly when it has n independent eigenvectors. Eigenvectors of distinct eigenvalues are independent, and the geometric multiplicity never exceeds the algebraic one, so a simple root (one that occurs once) always gives exactly one eigenvector. Only a repeated root can fall short, and a real symmetric matrix never does.

Example. Are these diagonalisable?

$$\mathbf{A} = \begin{pmatrix} 4 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 2 & 3 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 4 & 0 & 0 \\ 1 & 2 & 1 \\ 1 & 2 & 3 \end{pmatrix}.$$

Expand along row 1. The two matrices differ only in the $(3,1)$ entry, which the expansion never uses, so both have

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (4-\lambda)\big[(2-\lambda)(3-\lambda) - 2\big] = (4-\lambda)(\lambda-1)(\lambda-4).$$

So $\lambda = 4$ has algebraic multiplicity 2, and the simple root $\lambda = 1$ gives one eigenvector. For $\lambda = 4$,

$$\mathbf{A} - 4\mathbf{I} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & -2 & 1 \\ -1 & 2 & -1 \end{pmatrix}, \qquad \mathbf{B} - 4\mathbf{I} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & -2 & 1 \\ 1 & 2 & -1 \end{pmatrix}.$$

For $\mathbf{A}$, row 3 is -1 times row 2, so the one equation $v_1 - 2v_2 + v_3 = 0$ leaves two unknowns free, with eigenvectors $(1,0,-1)^{\mathsf{T}}$ and $(0,1,2)^{\mathsf{T}}$:

$$\mathbf{A}: \quad \boxed{\lambda = 4: \text{ alg. } 2, \text{ geom. } 2} \quad \text{diagonalisable.}$$

For $\mathbf{B}$, $R_3 - R_2 = (0,4,-2)$ gives $v_3 = 2v_2$, and then row 2 gives $v_1 = 0$. One unknown is free, and the only direction is $(0,1,2)^{\mathsf{T}}$:

$$\mathbf{B}: \quad \boxed{\lambda = 4: \text{ alg. } 2, \text{ geom. } 1} \quad \text{defective.}$$

Check. $\mathbf{A}(1,0,-1)^{\mathsf{T}} = (4,0,-4)^{\mathsf{T}}$ and $\mathbf{A}(0,1,2)^{\mathsf{T}} = (0,4,8)^{\mathsf{T}}$, both 4 times the vector. And $\operatorname{tr}\mathbf{A} = 9 = 4+4+1$, $\det\mathbf{A} = 16 = 4\cdot 4\cdot 1$. For $\mathbf{B}$, the first fails: $\mathbf{B}(1,0,-1)^{\mathsf{T}} = (4,0,-2)^{\mathsf{T}}$.

Pitfalls

- Calling every repeated eigenvalue a failure. $\mathbf{A}$ has $\lambda = 4$ twice and is diagonalisable; only the eigenspace decides.
- Counting rows instead of free unknowns. Row-reduce first: $\mathbf{B} - 4\mathbf{I}$ keeps two non-zero rows in three unknowns, so one unknown is free (geometric multiplicity 1); $\mathbf{A} - 4\mathbf{I}$ keeps one, so two are free.
- Working hard on the simple eigenvalue. A root of multiplicity 1 always has geometric multiplicity 1; check the repeated roots.

In the notes. Unit 6: Diagonalisation, for the criterion, and Unit 6: When diagonalisation fails, for the multiplicities.

Practise it. the Review set on this method

6.3 Diagonalising, and powers of a matrix

Problem. For a diagonalisable 2×2 matrix (Method 6.2), write $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$, and use it to find a closed form for $\mathbf{A}^k$.

Method

1. Find the eigenvalues and an eigenvector for each (Method 6.1).
2. Put the eigenvectors in the columns of $\mathbf{P}$, and the eigenvalues on the diagonal of $\mathbf{D}$, in the same order.
3. Compute $\det \mathbf{P}$, and $\mathbf{P}^{-1}$ by the 2×2 formula: swap the diagonal entries, negate the other two, divide by $\det \mathbf{P}$.
4. Then $\mathbf{A}^k = \mathbf{P}\mathbf{D}^k\mathbf{P}^{-1}$, with $\mathbf{D}^k = \operatorname{diag}(\lambda_1^k, \lambda_2^k)$.
5. Multiply $\mathbf{P}\mathbf{D}^k$ first, then multiply the result by $\mathbf{P}^{-1}$.
6. Check that $k = 1$ gives $\mathbf{A}$ back.

Example. Diagonalise $\mathbf{A} = \begin{pmatrix} -1 & 3 \\ -6 & 8 \end{pmatrix}$ and find a closed form for $\mathbf{A}^k$.

$\det(\mathbf{A} - \lambda\mathbf{I}) = (-1-\lambda)(8-\lambda)+18 = \lambda^2 - 7\lambda + 10 = (\lambda-2)(\lambda-5)$, so $\lambda_1 = 2$ and $\lambda_2 = 5$. For $\lambda_1 = 2$, $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} -3 & 3 \\ -6 & 6 \end{pmatrix}$ gives $v_1 = v_2$: $(1,1)^{\mathsf{T}}$. For $\lambda_2 = 5$, $\mathbf{A} - 5\mathbf{I} = \begin{pmatrix} -6 & 3 \\ -6 & 3 \end{pmatrix}$ gives $v_2 = 2v_1$: $(1,2)^{\mathsf{T}}$. With $\det \mathbf{P} = 2 - 1 = 1$,

$$\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}, \quad \mathbf{P}^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}.$$

Then $\mathbf{P}\mathbf{D}^k = \begin{pmatrix} 2^k & 5^k \\ 2^k & 2\cdot 5^k \end{pmatrix}$, and multiplying by $\mathbf{P}^{-1}$,

$$\boxed{\mathbf{A}^k = \begin{pmatrix} 2^{k+1} - 5^k & 5^k - 2^k \\ 2^{k+1} - 2\cdot 5^k & 2\cdot 5^k - 2^k \end{pmatrix}}$$

Check. At $k = 1$: $\begin{pmatrix} 4-5 & 5-2 \\ 4-10 & 10-2 \end{pmatrix} = \mathbf{A}$. At $k = 2$ the formula gives $\begin{pmatrix} -17 & 21 \\ -42 & 46 \end{pmatrix}$, which is $\mathbf{A}\mathbf{A}$ multiplied out.

Variants. For one eigenvector no $\mathbf{P}$ is needed: $\mathbf{A}^k\mathbf{v} = \lambda^k\mathbf{v}$. The same product with $\mathbf{D}^{-1} = \operatorname{diag}(1/\lambda_1, 1/\lambda_2)$ gives $\mathbf{A}^{-1}$, when no eigenvalue is 0.

Pitfalls

- Putting the eigenvalues in $\mathbf{D}$ in a different order from the columns of $\mathbf{P}$. Swapping one without the other gives a wrong $\mathbf{A}$.
- Multiplying in the wrong order: $\mathbf{P}^{-1}\mathbf{D}^k\mathbf{P}$ is not $\mathbf{A}^k$. Check $k = 1$ to catch it.
- Raising the entries of $\mathbf{A}$ to the power. Only a diagonal matrix powers entry by entry: $\mathbf{A}^2$ has $(1,1)$ entry -17, not $(-1)^2 = 1$.

In the notes. Unit 6: Diagonalisation, and Unit 6: Computing powers, with its 2×2 example.
Practise it. the Review set on this method

6.4 A recurrence, by decomposing the start vector

Problem. Find a closed form for a sequence given by a two-term linear recurrence, or for $\mathbf{x}_n$ when $\mathbf{x}_{n+1} = \mathbf{A}\mathbf{x}_n$, when the 2×2 step matrix has two distinct eigenvalues.

Method

1. Write the step as a matrix, $\mathbf{x}_{n+1} = \mathbf{A}\mathbf{x}_n$, so that $\mathbf{x}_n = \mathbf{A}^n\mathbf{x}_0$. For a two-term recurrence, stack two consecutive terms, the later one on top.
2. Find the eigenpairs $\lambda_1, \mathbf{v}_1$ and $\lambda_2, \mathbf{v}_2$ (Method 6.1).
3. Decompose the start: solve $\mathbf{x}_0 = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$ for c_1 and c_2.
4. Each eigenvector is scaled by its eigenvalue at every step: $\mathbf{x}_n = c_1\lambda_1^n\mathbf{v}_1 + c_2\lambda_2^n\mathbf{v}_2$.
5. Read off the component asked for.
6. Check a few small n against the recurrence. Only if the long run is asked for: when $|\lambda_1| > |\lambda_2|$ strictly and $c_1 \neq 0$, the λ_1 term dominates.

Example. A sequence has $x_{n+1} = x_n + 6x_{n-1}$, with $x_0 = 3$ and $x_1 = 4$. (a) Find a closed form for x_n. (b) What does x_{n+1}/x_n tend to?

(a) Stack two terms: the rows $x_{n+1} = x_n + 6x_{n-1}$ and $x_n = x_n$ give

$$\begin{pmatrix} x_{n+1} \\ x_n \end{pmatrix} = \begin{pmatrix} 1 & 6 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_n \\ x_{n-1} \end{pmatrix}, \qquad \begin{pmatrix} x_{n+1} \\ x_n \end{pmatrix} = \mathbf{A}^n \begin{pmatrix} 4 \\ 3 \end{pmatrix}.$$

$\det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^2 - \lambda - 6 = (\lambda - 3)(\lambda + 2)$. For $\lambda_1 = 3$, $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} -2 & 6 \\ 1 & -3 \end{pmatrix}$ gives $v_1 = 3v_2$: $(3, 1)^{\mathsf{T}}$. For $\lambda_2 = -2$, $\mathbf{A} + 2\mathbf{I} = \begin{pmatrix} 3 & 6 \\ 1 & 2 \end{pmatrix}$ gives $v_1 = -2v_2$: $(-2, 1)^{\mathsf{T}}$. Decompose the start:

$$\begin{pmatrix} 4 \\ 3 \end{pmatrix} = c_1 \begin{pmatrix} 3 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} -2 \\ 1 \end{pmatrix} : \quad 3c_1 - 2c_2 = 4, \quad c_1 + c_2 = 3,$$

so $c_1 = 2$ and $c_2 = 1$. Scaling each eigenvector by its eigenvalue,

$$\begin{pmatrix} x_{n+1} \\ x_n \end{pmatrix} = 2 \cdot 3^n \begin{pmatrix} 3 \\ 1 \end{pmatrix} + (-2)^n \begin{pmatrix} -2 \\ 1 \end{pmatrix},$$

and the second component is

$$\boxed{x_n = 2 \cdot 3^n + (-2)^n}$$

(b) $|3| > |-2|$ and $c_1 = 2 \neq 0$, so the first term wins: $x_n = 3^n\left(2 + (-\tfrac{2}{3})^n\right)$, and $\boxed{x_{n+1}/x_n \to 3}$.

Check. The recurrence gives $x_2 = 4 + 18 = 22$ and $x_3 = 22 + 24 = 46$; the formula gives $18 + 4 = 22$ and $54 - 8 = 46$.

Pitfalls

- Putting x_0 on top of the start vector. The top entry is the later term, so the start is $(x_1, x_0)^{\mathsf{T}} = (4, 3)^{\mathsf{T}}$.
- Reading off the wrong component: the top one is x_{n+1}, the bottom one x_n.
- Claiming a limit when the eigenvalues tie in modulus. With $\lambda = \pm 2$, $x_n = 2^n\left(c_1 + (-1)^n c_2\right)$, and unless c_1 or c_2 is 0 the ratio x_{n+1}/x_n never settles: it alternates between two values, or every other term is 0.

In the notes. Unit 6: Computing powers, with Fibonacci and Binet's formula, and Unit 6: Why this chain converges, for dominance.

Practise it. the Review set on this method

6.5 Symmetric and Hermitian matrices: orthogonal diagonalisation, PCA

Problem. For a real symmetric or a Hermitian matrix with distinct eigenvalues, find its real eigenvalues and orthogonal eigenvectors; for a real symmetric one, also an orthogonal $\mathbf{Q}$ with $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\mathsf{T}}$, and its principal axes.

Method

1. Check the structure: $\mathbf{A}^{\mathsf{T}} = \mathbf{A}$ for a real matrix; $\mathbf{H}^{\dagger} = \mathbf{H}$ for a complex one (a real diagonal, and conjugates across it).
2. Solve the characteristic equation. For a Hermitian matrix the off-diagonal product is $z\overline{z} = |z|^2$. The eigenvalues come out real.
3. Find an eigenvector for each as in Method 6.1: for a 2×2 from one non-zero row, choosing v_2 to clear any fraction; for a 3×3 by row-reducing $\mathbf{A} - \lambda\mathbf{I}$.
4. Check that the eigenvectors are orthogonal: each pair has dot product 0. For complex vectors use $\mathbf{u}^{\dagger}\mathbf{v} = \overline{u_1}v_1 + \overline{u_2}v_2$, not the plain product.
5. For a real symmetric matrix, divide each eigenvector by its length. These are the columns of $\mathbf{Q}$, with the eigenvalues in $\mathbf{D}$ in the same order. Then $\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}$ and $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\mathsf{T}}$.
6. PCA: for a covariance matrix, the eigenvector of the largest eigenvalue is the first principal axis. Each eigenvalue measures the variance along its axis, and they sum to $\operatorname{tr}\mathbf{A}$, the total variance, so that axis carries the share $\lambda/\operatorname{tr}\mathbf{A}$.

Example. (a) Orthogonally diagonalise the covariance matrix $\mathbf{A} = \begin{pmatrix} 3 & 2 \\ 2 & 6 \end{pmatrix}$, and give its first principal axis and the share of the variance it carries. (b) Find the eigenvalues and eigenvectors of $\mathbf{H} = \begin{pmatrix} 2 & 1+2\mathrm{i} \\ 1-2\mathrm{i} & -2 \end{pmatrix}$, and check that the eigenvectors are orthogonal.

(a) $\mathbf{A}^{\mathsf{T}} = \mathbf{A}$. $\det(\mathbf{A} - \lambda\mathbf{I}) = (3-\lambda)(6-\lambda) - 4 = \lambda^2 - 9\lambda + 14 = (\lambda-2)(\lambda-7)$, so $\lambda_1 = 2$, $\lambda_2 = 7$. Row 1 reads $v_1 + 2v_2 = 0$ for λ_1, giving $(2,-1)^{\mathsf{T}}$, and $-4v_1 + 2v_2 = 0$ for λ_2, giving $(1,2)^{\mathsf{T}}$. Their dot product is $2 - 2 = 0$, and both have length $\sqrt{5}$:

$$\boxed{\mathbf{Q} = \frac{1}{\sqrt{5}}\begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}}, \qquad \boxed{\mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 7 \end{pmatrix}}$$

The first principal axis is $\frac{1}{\sqrt{5}}(1,2)^{\mathsf{T}}$, the eigenvector of the larger eigenvalue 7. The total variance is $\operatorname{tr}\mathbf{A} = 9$, so it carries $\boxed{\frac{7}{9} \approx 77.8\%}$.

(b) Conjugating $\mathbf{H}$ and transposing returns $\mathbf{H}$, so $\mathbf{H}^{\dagger} = \mathbf{H}$. With $(1+2\mathrm{i})(1-2\mathrm{i}) = 5$,

$$\det(\mathbf{H} - \lambda\mathbf{I}) = (2-\lambda)(-2-\lambda) - 5 = \lambda^2 - 9,$$

so $\lambda = \pm 3$, both real. For $\lambda = 3$ row 1 reads $-v_1 + (1+2\mathrm{i})v_2 = 0$; take $v_2 = 1$. For $\lambda = -3$ it reads $5v_1 + (1+2\mathrm{i})v_2 = 0$; take $v_2 = 5$ to clear the fraction:

$$\boxed{\lambda = 3 : \begin{pmatrix} 1+2\mathrm{i} \\ 1 \end{pmatrix}}, \qquad \boxed{\lambda = -3 : \begin{pmatrix} -1-2\mathrm{i} \\ 5 \end{pmatrix}}$$

The conjugate product is $(1-2\mathrm{i})(-1-2\mathrm{i}) + 1\cdot 5 = -5 + 5 = 0$: orthogonal.

Check. $\mathbf{A}(1,2)^{\mathsf{T}} = (7,14)^{\mathsf{T}} = 7(1,2)^{\mathsf{T}}$, and $\mathbf{Q}\mathbf{D}\mathbf{Q}^{\mathsf{T}} = \frac{1}{5}\begin{pmatrix} 15 & 10 \\ 10 & 30 \end{pmatrix} = \mathbf{A}$. For $\mathbf{H}$: $\operatorname{tr}\mathbf{H} = 0 = 3 + (-3)$

and $\det \mathbf{H} = -4 - 5 = -9 = 3 \cdot (-3)$.

Pitfalls

- Using the plain product for complex vectors. In (b) it gives $(1+2\mathrm{i})(-1-2\mathrm{i}) + 5 = 8 - 4\mathrm{i}$, which says nothing about orthogonality.
- Forgetting to normalise. With the columns $(2,-1)^{\mathsf{T}}$ and $(1,2)^{\mathsf{T}}$ as they stand, $\mathbf{P}^{\mathsf{T}}\mathbf{P} = 5\mathbf{I}$, so $\mathbf{P}^{-1} \neq \mathbf{P}^{\mathsf{T}}$.
- Taking the principal axis from the largest diagonal entry. The 6 in $\mathbf{A}$ suggests $(0,1)^{\mathsf{T}}$, but the axis is $\frac{1}{\sqrt{5}}(1,2)^{\mathsf{T}}$.

In the notes. Unit 6: Symmetric and Hermitian matrices, with the note on the conjugate inner product, and Unit 6: Symmetric matrices and orthogonal diagonalisation.

Practise it. the Review set on this method

6.6 A Markov chain's long run

Problem. Set up the transition matrix of a Markov chain; find its steady state, a closed form for $\mathbf{x}_n$, and whether the chain settles.

Method

1. Build $\mathbf{P}$: column j lists the probabilities of moving *from* state j. Check that each column sums to one.
2. $\lambda_1 = 1$ always. Find λ_2 from the characteristic equation; for a 2×2, $\lambda_1 + \lambda_2 = \operatorname{tr}\mathbf{P}$ checks it, and $\lambda_2 = 1$ only for $\mathbf{P} = \mathbf{I}$, where $\mathbf{x}_n = \mathbf{x}_0$: stop.
3. Steady state: solve $(\mathbf{P} - \mathbf{I})\mathbf{x} = \mathbf{0}$, scale $\mathbf{x}$ so its entries sum to one, and check $\mathbf{P}\mathbf{x}_\infty = \mathbf{x}_\infty$.
4. Closed form, for two states: find the eigenvector $\mathbf{v}_2$ of λ_2, and write $\mathbf{x}_0 = c_1\mathbf{x}_\infty + c_2\mathbf{v}_2$. Since $\mathbf{v}_2$ sums to zero, $c_1 = 1$ for every probability start, and $\mathbf{x}_n = \mathbf{x}_\infty + c_2\lambda_2^n\mathbf{v}_2$.
5. Long run: the chain settles to $\mathbf{x}_\infty$ from every start when $\lambda = 1$ occurs just once and every other eigenvalue has $|\lambda| < 1$. A second eigenvalue of modulus 1 (such as -1) can keep it moving for ever.

Example. Each day a machine is either working or under repair. A working machine is still working the next day with probability 0.9; one under repair is working again the next day with probability 0.3. It starts under repair. Find the steady state, and the probability that it is working on day n.

With $\mathbf{x}_n = (\text{working},\ \text{under repair})$, the columns are the two from-states:

$$\mathbf{P} = \begin{pmatrix} 0.9 & 0.3 \\ 0.1 & 0.7 \end{pmatrix}, \qquad \mathbf{x}_0 = (0,\ 1).$$

Both columns sum to one. $\det(\mathbf{P} - \lambda\mathbf{I}) = \lambda^2 - 1.6\lambda + 0.6 = (\lambda - 1)(\lambda - 0.6)$, so $\lambda_1 = 1$ and $\lambda_2 = 0.6$, and $1 + 0.6 = 1.6 = \operatorname{tr}\mathbf{P}$. For $\lambda_1 = 1$, $(\mathbf{P} - \mathbf{I})\mathbf{x} = \mathbf{0}$ gives $-0.1x_1 + 0.3x_2 = 0$, so $x_1 = 3x_2$: the direction $(3,1)$. Scaled to sum one,

$$\boxed{\mathbf{x}_\infty = (0.75,\ 0.25)}$$

For $\lambda_2 = 0.6$, $\mathbf{P} - 0.6\mathbf{I} = \begin{pmatrix} 0.3 & 0.3 \\ 0.1 & 0.1 \end{pmatrix}$ gives $(1, -1)$. Then $\mathbf{x}_0 = (0, 1) = (0.75, 0.25) + c_2(1, -1)$ gives $c_2 = -0.75$, so

$$\mathbf{x}_n = (0.75,\ 0.25) - 0.75\,(0.6)^n\,(1, -1),$$

and the probability of working on day n is

$$\boxed{0.75 - 0.75\,(0.6)^n}$$

$\lambda = 1$ occurs once and $|0.6| < 1$, so this tends to 0.75 from any start.

Check. $\mathbf{P}(0.75, 0.25) = (0.675 + 0.075,\ 0.075 + 0.175) = (0.75, 0.25)$. One step from the start, $\mathbf{P}\mathbf{x}_0 = (0.3, 0.7)$, and the formula gives $0.75 - 0.45 = 0.3$.

Pitfalls

- Putting the from-states in the rows. That builds $\mathbf{P}^{\mathsf{T}}$, whose columns here sum to 1.2 and 0.8: the column check catches it.
- Giving the direction $(3, 1)$ as the steady state. A distribution sums to one: $(0.75, 0.25)$.
- Promising convergence without checking the other eigenvalues. The swap $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has eigenvalues ± 1 and oscillates for ever, and $\mathbf{I}$ has $\lambda = 1$ twice and never forgets its start.

In the notes. Unit 6: Markov chains, Unit 6: The steady state is an eigenvector, and Unit 6: Why this chain converges.

Practise it. the Review set on this method

Part III Differentiation

Units 7–8: limits, the rules of differentiation, and what a derivative is for, from tangents and the existence of roots to extreme values, sketches, optimisation and iterative methods. Most problems start by finding a derivative; the question decides what to do with it.

Unit 7 Differentiation I: limits and the rules

Seven methods: two for limits, one that finds a derivative as a limit, and four that differentiate. Which of the four depends on how the function is given. A tangent line, a velocity, a higher derivative and a sensitivity are each one step after the rules, and are variants of Method 7.4.

Which method?

What you see	What to do
one formula on both sides of a, and substituting gives a number	7.1: that number is the limit
$\frac{0}{0}$, and top and bottom share a factor	7.1: factor and cancel
$\frac{0}{0}$ with a square root, or with $1 - \cos x$	7.1: multiply by the conjugate
$\frac{0}{0}$ with the sine of something small	7.1: rewrite to $\frac{\sin u}{u}$
(a nonzero number)/0	7.1: one-sided limits; an asymptote
$x \to \infty$ with a rational function, or a difference with a root	7.1: conjugate first for a root; divide by the highest power
a join in a piecewise function	7.2: one-sided limits at the join
"from first principles"	7.3: the difference quotient
a product or quotient of two factors, a function of a function	7.4: name u and v, or outer and inner
y tangled with x in an equation	7.5: implicit differentiation
x and y both given in terms of t	7.6: parametric differentiation
the variable in both a base and an exponent; a product or quotient of three or more factors	7.7: take logarithms first

7.1 Evaluating a limit

Problem. Find $\lim_{x\to a} f(x)$, the one-sided limits at a, or $\lim_{x\to\infty} f(x)$, and the asymptotes they give.

Method

1. If f is one formula on both sides of a, substitute $x = a$ into it. If the result is a number, that is the limit. At the join of a piecewise function, find the two one-sided limits instead (Method 7.2, step 2).
2. If it is $\frac{0}{0}$, simplify for $x \neq a$: factor and cancel the common factor; multiply top and

bottom by the conjugate of a root (for $1-\cos x$, by $1+\cos x$, then $1-\cos^2 x=\sin^2 x$); or rewrite to $\dfrac{\sin u}{u}$ with $u\to 0$, using $\lim_{u\to 0}\dfrac{\sin u}{u}=1$.

3. Substitute into the simplified form, and state the limit.
4. If it is (a nonzero number)/0, find the limits from below and from above, $x\to a^-$ and $x\to a^+$, from the signs of the top and the bottom. An infinite one gives the vertical asymptote $x=a$. The limit exists only when the two one-sided limits exist and agree.
5. For $x\to\infty$, divide the top and the bottom by the highest power of x in the denominator. Each $1/x^k$ tends to 0: write the $\dfrac{(\cdot)-0}{(\cdot)+0}$ line. For a difference with a root, multiply and divide by its conjugate first, as in step 2.
6. If the limit is a number L, state it and the horizontal asymptote $y=L$. For a rational function the same L holds as $x\to-\infty$; for any other function, check that direction separately. If a power of x is left on the top, the limit is infinite and there is no horizontal asymptote.

Example. (a) For $f(x)=\dfrac{2x^2-5x+3}{x^2-x}$, find the limits of f as $x\to 1$, as $x\to 0^{\pm}$ and as $x\to\infty$, and the asymptotes. (b) Find $\lim\limits_{x\to 2}\dfrac{\sqrt{2x+5}-3}{x^2-4}$. (c) Find $\lim\limits_{x\to 0}\dfrac{\sin 3x}{x^2+2x}$.

(a) At $x=1$ substitution gives $\frac{0}{0}$. Factor and cancel, for $x\neq 1$:

$$f(x)=\frac{(2x-3)(x-1)}{x(x-1)}=\frac{2x-3}{x}\longrightarrow\frac{2-3}{1},$$

so $\boxed{\lim\limits_{x\to 1}f(x)=-1}$. At $x=0$ substitution gives $\frac{3}{0}$. Near 0 use the simplified form: its top tends to -3 and its bottom x has the sign of x, so $f\to-\infty$ as $x\to 0^+$ and $f\to+\infty$ as $x\to 0^-$. The vertical asymptote is $\boxed{x=0}$. As $x\to\infty$, divide by x^2:

$$f(x)=\frac{2-\frac{5}{x}+\frac{3}{x^2}}{1-\frac{1}{x}}\longrightarrow\frac{2-0+0}{1-0}=2.$$

The horizontal asymptote is $\boxed{y=2}$, approached as $x\to-\infty$ too.

(b) Substitution gives $\frac{0}{0}$. Multiply top and bottom by $\sqrt{2x+5}+3$; the top becomes $(2x+5)-9=2(x-2)$, and the bottom factors as $(x-2)(x+2)(\sqrt{2x+5}+3)$. Cancel $x-2$:

$$\frac{\sqrt{2x+5}-3}{x^2-4}=\frac{2}{(x+2)(\sqrt{2x+5}+3)}\longrightarrow\frac{2}{4\cdot 6}=\boxed{\frac{1}{12}}.$$

(c) Substitution gives $\frac{0}{0}$. Factor the bottom as $x(x+2)$ and supply the 3 that $\frac{\sin u}{u}$ needs, with $u=3x$:

$$\frac{\sin 3x}{x(x+2)}=\frac{\sin 3x}{3x}\cdot\frac{3}{x+2}\longrightarrow 1\cdot\frac{3}{2}=\boxed{\frac{3}{2}}.$$

Check. Substitute a value close to the target: in (b), $x=2.01$ gives 0.08308; in (c), $x=0.01$ gives 1.4923; in (a), $f(1.01)=-0.970$, $f(0.01)=-298$, $f(-0.01)=302$ and $f(1000)=1.997$.

Pitfalls

- Reading $\frac{0}{0}$ as "no limit". It only says that this form needs simplifying first.
- Treating $\frac{3}{0}$ as 0 or as ∞ without the signs. In (a) the two sides go to $-\infty$ and $+\infty$, so there is no limit at 0.
- Using $\frac{\sin 3x}{x} \to 1$. The standard limit needs the same u on the top and the bottom; here it gives 3.

In the notes. Unit 7: Limits at a point, and Unit 7: One-sided limits, infinite limits, and limits at infinity.

Practise it. the Review set on this method

7.2 Continuity and differentiability at a join

Problem. Decide whether a function, often defined piecewise, is continuous or differentiable at a point; choose constants that make it so.

Method

1. Find $f(a)$, from the piece that includes $x = a$.
2. Find $\lim_{x\to a} f(x)$. At a join, find the two one-sided limits, each from the piece on its own side.
3. f is continuous at a when the limit exists and equals $f(a)$. Equate them; with an unknown constant, this is an equation for it.
4. For differentiability, impose continuity first: a function differentiable at a is continuous there.
5. Find the one-sided limits of the difference quotient $\dfrac{f(a+\delta x)-f(a)}{\delta x}$ as $\delta x \to 0^-$ and as $\delta x \to 0^+$, each side from its own piece.
6. f is differentiable at a exactly when they are finite and agree, and $f'(a)$ is the common value. If they differ, the graph has a corner; if either is infinite, there is no derivative.

Example. (a) Find a and b so that $f(x) = \begin{cases} x^2+1, & x \le 1 \\ \dfrac{a}{x}+b, & x > 1 \end{cases}$ is differentiable at $x = 1$. (b) Show that $h(x) = \begin{cases} x^2, & x \le 0 \\ \sin x, & x > 0 \end{cases}$ is continuous but not differentiable at 0.

(a) $f(1) = 2$, from the left piece, and the one-sided limits are 2 from below and $a+b$ from above. Continuity needs $a+b=2$. With $b = 2-a$ the difference quotients are

$$\frac{(1+\delta x)^2+1-2}{\delta x} = 2+\delta x \xrightarrow[\delta x\to 0^-]{} 2,$$
$$\frac{1}{\delta x}\left(\frac{a}{1+\delta x}-a\right) = \frac{-a}{1+\delta x} \xrightarrow[\delta x\to 0^+]{} -a.$$

They agree only if $-a = 2$, and then $b = 2-a$: $\boxed{a=-2,\ b=4}$.

(b) $h(0) = 0$, and the one-sided limits are $0^2 = 0$ and $\sin 0 = 0$: h is continuous at 0. The difference quotients are $\dfrac{(\delta x)^2}{\delta x} = \delta x \to 0$ from below and $\dfrac{\sin \delta x}{\delta x} \to 1$ from above. They differ,

so h is *not differentiable* at 0: its graph has a corner there.

Check. In (a) the right piece is $4 - \frac{2}{x}$. As $x \to 1^+$ its value tends to 2 and its slope $\frac{2}{x^2}$ to 2, the same as the value and the slope $2x = 2$ of $x^2 + 1$ at 1. In (b) the slopes of the two pieces at 0 are $2x = 0$ and $\cos x = 1$: they differ, so there is a corner.

Pitfalls

- Matching the slopes and skipping continuity. With $a = -2$ and $b = 0$ the slopes match, but f jumps from 2 to -2, so it is not even continuous.
- Taking $f(a)$ from the wrong piece. Read $\leq$ and $<$ in the definition.
- Concluding from continuity that f is differentiable. The function h in (b) is continuous and has a corner.

In the notes. Unit 7: Continuity, and Unit 7: Notation, higher derivatives, and a function that fails.

Practise it. the Review set on this method

7.3 Differentiating from first principles

Problem. Find $f'(x)$ from the definition $f'(x) = \lim_{\delta x \to 0} \dfrac{f(x+\delta x) - f(x)}{\delta x}$, without the rules.

Method

1. Write the difference quotient $\dfrac{f(x+\delta x) - f(x)}{\delta x}$.
2. Simplify the numerator: expand a power; bring fractions to a common denominator; for a root, multiply top and bottom by the conjugate.
3. Every term of the numerator now carries a factor δx. Cancel it with the δx below.
4. Let $\delta x \to 0$ in what is left: "letting $\delta x \to 0$ leaves $f'(x) = \dots$".

Example. Differentiate from first principles (a) $f(x) = \dfrac{1}{x^2}$ and (b) $g(x) = \sqrt{x^2+4}$.

(a) Bring the two fractions to the denominator $x^2(x+\delta x)^2$ and expand the top:

$$\begin{aligned} \frac{f(x+\delta x) - f(x)}{\delta x} &= \frac{x^2 - (x+\delta x)^2}{\delta x\, x^2 (x+\delta x)^2} \\ &= \frac{-2x\,\delta x - (\delta x)^2}{\delta x\, x^2 (x+\delta x)^2} \\ &= \frac{-(2x+\delta x)}{x^2(x+\delta x)^2}. \end{aligned}$$

Letting $\delta x \to 0$ leaves $f'(x) = \dfrac{-2x}{x^4}$, so $\boxed{f'(x) = -\dfrac{2}{x^3}}$.

(b) Multiply top and bottom by $\sqrt{(x+\delta x)^2+4} + \sqrt{x^2+4}$. The top becomes $(x+\delta x)^2 + 4 - (x^2+4) = 2x\,\delta x + (\delta x)^2$, so

$$\frac{g(x+\delta x) - g(x)}{\delta x} = \frac{2x + \delta x}{\sqrt{(x+\delta x)^2+4} + \sqrt{x^2+4}}.$$

Letting $\delta x \to 0$ leaves $\dfrac{2x}{2\sqrt{x^2+4}}$, so $\boxed{g'(x) = \dfrac{x}{\sqrt{x^2+4}}}$.

Check. The rules give the same: $\frac{\mathrm{d}}{\mathrm{d}x}\,x^{-2} = -2x^{-3}$, and the chain rule gives $\frac{1}{2}(x^2+4)^{-1/2}\cdot 2x$.

Pitfalls

- Expanding $(x+\delta x)^2$ as $x^2 + (\delta x)^2$. The cross term $2x\,\delta x$ is the one that survives.
- Letting $\delta x \to 0$ before cancelling. That gives $\frac{0}{0}$ every time.
- Writing $f(x) + \delta x$ for $f(x+\delta x)$. In (b), $g(x+\delta x) = \sqrt{(x+\delta x)^2+4}$, not $\sqrt{x^2+4} + \delta x$.

In the notes. Unit 7: The derivative.

Practise it. the Review set on this method

7.4 Choosing between the product, quotient and chain rules

Problem. Differentiate a function built from standard functions by products, quotients and composition, and simplify the result.

Method

1. Find the last operation in the formula: a product uv, a quotient u/v, or a function of a function $f(g(x))$. It decides the rule.
2. Name u and v; or the outer function $f(u)$ and the inner $u = g(x)$.
3. Find u' and v' from the standard derivatives, using the chain rule inside when u or v is itself a function of a function.
4. Apply the rule: $(uv)' = u'v + uv'$; $\left(\frac{u}{v}\right)' = \dfrac{u'v - uv'}{v^2}$; $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u}\cdot\frac{\mathrm{d}u}{\mathrm{d}x}$.
5. Simplify: clear any fraction inside the numerator, and take out the common factor or power.

Variants. A *tangent line* at $x = a$: find $f(a)$ and $f'(a)$, write the point–slope form $y - f(a) = f'(a)(x-a)$, and rearrange to $y = mx + c$. *Motion* along a line: the velocity is $v = \dot{s}$ and the acceleration $a = \ddot{s}$, with units; the body is at rest where $v = 0$, found by factoring, and the sign of v gives the direction. *Higher derivatives*: differentiate again and again, listing f', f'', …; for a polynomial, stop at the first 0. A *sensitivity*: evaluate $\frac{\mathrm{d}L}{\mathrm{d}w}$ at the current w; a small step that reduces L moves w against the sign of $\frac{\mathrm{d}L}{\mathrm{d}w}$.

Example. (a) Differentiate $y = x\sqrt{5-x}$, and find the tangent line at $x = 4$. (b) Differentiate $y = \dfrac{\cos x}{1+\sin x}$.

(a) A product: $u = x$ and $v = (5-x)^{1/2}$. Then $u' = 1$, and by the chain rule, with inner $5-x$ of derivative -1, $v' = \frac{1}{2}(5-x)^{-1/2}\cdot(-1)$. So

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \sqrt{5-x} - \frac{x}{2\sqrt{5-x}}.$$

Put both terms over the common denominator $2\sqrt{5-x}$; the top is $2(5-x)-x$:

$$\boxed{\frac{dy}{dx} = \frac{10-3x}{2\sqrt{5-x}}}$$

At $x=4$: $y = 4\cdot 1 = 4$ and $\frac{dy}{dx} = \frac{-2}{2} = -1$. The point–slope form $y-4 = -(x-4)$ gives the tangent $\boxed{y = 8-x}$.

(b) A quotient: $u = \cos x$ and $v = 1+\sin x$, with $u' = -\sin x$ and $v' = \cos x$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{-\sin x\,(1+\sin x) - \cos^2 x}{(1+\sin x)^2} \\ &= \frac{-\sin x - (\sin^2 x + \cos^2 x)}{(1+\sin x)^2} \\ &= \frac{-(1+\sin x)}{(1+\sin x)^2},\end{aligned}$$

so $\boxed{\dfrac{dy}{dx} = -\dfrac{1}{1+\sin x}}$.

Check. The slope of a short secant centred on the point tests the derivative itself. In (a), $\dfrac{y(4.001)-y(3.999)}{0.002} \approx -1.000$, the slope -1 at $x=4$. In (b), $\dfrac{y(0.001)-y(-0.001)}{0.002} \approx -1.000$, and $-\frac{1}{1+0} = -1$.

Pitfalls

- Writing $(uv)' = u'v'$. In (a) it gives $-\frac{1}{2}$ at $x=4$ instead of -1.
- Dropping the inner derivative -1 of $5-x$. It gives $\frac{10-x}{2\sqrt{5-x}}$, which is 3 at $x=4$, and the wrong tangent.
- Turning the quotient rule round: $uv' - u'v$ over v^2 gives $+\frac{1}{1+\sin x}$ in (b), the right size with the wrong sign.

In the notes. Unit 7: The sum, product and quotient rules, Unit 7: The chain rule, and Unit 7: Three readings of the derivative, for the tangent line, the motion and the sensitivity.

Practise it. the Review set on this method

7.5 Implicit differentiation

Problem. A curve is given by an equation in x and y. Find the slope $y' = \frac{dy}{dx}$, its value at a point, and the second derivative y''.

Method

1. Differentiate both sides with respect to x, treating y as a function of x. By the chain rule $\frac{d}{dx}\,g(y) = g'(y)\,y'$; a product such as xy needs the product rule, $y + xy'$.
2. Collect the terms in y' on one side.
3. Solve for y', where its coefficient is not zero.
4. Evaluate at the point, having checked that the point lies on the curve.
5. For y'', differentiate the relation from step 1 once more, substitute the values of x, y and

y', and solve for y''.

Example. For the curve $e^y + xy = e$, find y', and find y' and y'' at the point $(0, 1)$.

The point lies on the curve: $e^1 + 0 = e$. Differentiate, with the chain rule on e^y and the product rule on xy:

$$e^y y' + y + xy' = 0 \quad \Longrightarrow \quad \boxed{y' = -\frac{y}{e^y + x}}$$

wherever $e^y + x \neq 0$. At $(0, 1)$ this gives $\boxed{y' = -\frac{1}{e}}$.

Differentiate the relation $e^y y' + y + xy' = 0$ again (product rule on $e^y y'$ and on xy'):

$$e^y (y')^2 + e^y y'' + y' + y' + xy'' = 0.$$

At $(0, 1)$ with $y' = -\frac{1}{e}$ this reads $\frac{1}{e} - \frac{2}{e} + e\,y'' = 0$, so $\boxed{y'' = \frac{1}{e^2}}$.

Check. This equation can be solved for x instead: $x = \dfrac{e - e^y}{y}$, with $\frac{dx}{dy} = \dfrac{-ye^y - (e - e^y)}{y^2}$, which is $-e$ at $y = 1$. The slope is $\frac{dy}{dx} = 1 \big/ \frac{dx}{dy} = -\frac{1}{e}$.

Pitfalls

- Writing $\frac{d}{dx} e^y = e^y$. The chain rule adds the factor y'.
- Differentiating xy as xy' alone. The product rule gives $y + xy'$.
- Substituting the point before differentiating. With $x = 0$ put in first, the xy term has vanished, and $e^y y' = 0$ gives the false slope 0.

In the notes. Unit 7: Implicit differentiation.

Practise it. the Review set on this method

7.6 Parametric differentiation

Problem. A curve is given by $x = x(t)$ and $y = y(t)$. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$, and the points with a horizontal or a vertical tangent.

Method

1. Find dx/dt and dy/dt. (Dots are for derivatives with respect to time only.)
2. State where $dx/dt \neq 0$: the slope formula holds only there.
3. Divide, $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$, and simplify.
4. For $\frac{d^2y}{dx^2}$, differentiate $\frac{dy}{dx}$ with respect to t and divide by dx/dt again.
5. The tangent is horizontal where $dy/dt = 0$ and $dx/dt \neq 0$, and vertical where $dx/dt = 0$ and $dy/dt \neq 0$. Give the points (x, y), not just the values of t.

Example. The figure-of-eight curve $x = 2\cos t$, $y = \sin 2t$ is traced once for $0 \leq t < 2\pi$. (a) Find $\frac{dy}{dx}$. (b) Find the points with a horizontal or a vertical tangent. (c) Find $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$.

(a) $\dfrac{dx}{dt} = -2\sin t$ and $\dfrac{dy}{dt} = 2\cos 2t$, so for $\sin t \neq 0$,

$$\boxed{\frac{dy}{dx} = \frac{2\cos 2t}{-2\sin t} = -\frac{\cos 2t}{\sin t}}$$

(b) *Horizontal*: $\cos 2t = 0$ at $t = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$, where $\sin t \neq 0$. They give $\boxed{(\pm\sqrt{2},\ \pm 1)}$, all four sign choices. *Vertical*: $\sin t = 0$ at $t = 0$ and $t = \pi$, where $\dfrac{dy}{dt} = 2 \neq 0$. They give $\boxed{(\pm 2,\ 0)}$.

(c) By the quotient rule,

$$\frac{d}{dt}\left(-\frac{\cos 2t}{\sin t}\right) = \frac{2\sin 2t\,\sin t + \cos 2t\,\cos t}{\sin^2 t}.$$

At $t = \frac{\pi}{4}$, where $\sin 2t = 1$, $\cos 2t = 0$ and $\sin t = \frac{\sqrt{2}}{2}$, this is $\dfrac{\sqrt{2}}{1/2} = 2\sqrt{2}$. Divide by $\dfrac{dx}{dt} = -\sqrt{2}$: $\boxed{\dfrac{d^2y}{dx^2} = -2}$. The curve is concave down at $(\sqrt{2}, 1)$.

Check. At $t = \frac{\pi}{4}$, $y = \sin 2t$ takes its largest value 1, so a horizontal tangent there, with the curve bending down, is expected. Eliminating t on $0 < t < \pi$, where $\sin t > 0$, gives $y = \frac{1}{2}x\sqrt{4 - x^2}$; its derivative at $x = \sqrt{2}$ is 0 and its second derivative there is -2.

Pitfalls

- Dividing the second derivatives, $\dfrac{d^2y/dt^2}{d^2x/dt^2}$. At $t = \frac{\pi}{4}$ that gives $\frac{-4}{-\sqrt{2}} = 2\sqrt{2}$, the wrong size and the wrong sign.
- Dividing the wrong way round, $(dx/dt)/(dy/dt)$. That is $\frac{dx}{dy}$.
- Calling a point horizontal without checking $dx/dt \neq 0$ there. Where both vanish, the formula says nothing.

In the notes. Unit 7: Parametric differentiation.
Practise it. the Review set on this method

7.7 Logarithmic differentiation

Problem. Differentiate a function with the variable in both a base and an exponent, or a product or quotient of many factors.

Method

1. Take $\ln|\cdot|$ of both sides. Where $y > 0$ throughout, as for x^x with $x > 0$, plain ln will do; say so.
2. Simplify with the log laws: a product becomes a sum, a quotient a difference, and a power a multiplier.
3. Differentiate both sides with respect to x. The left side gives $\dfrac{1}{y}\dfrac{dy}{dx}$ by the chain rule; the right side goes term by term.
4. Multiply back by y, written in terms of x, and state where the result holds: away from the zeros of the factors. At a zero of a power factor, such as $x = 0$ for x^3, y is still defined:

find y' there by the product rule instead.

Example. (a) Differentiate $y = x^{\sqrt{x}}$ for $x > 0$, and evaluate $\frac{dy}{dx}$ at $x = 4$. (b) Differentiate $y = \dfrac{x^3\sqrt{x+1}}{(x-2)^2}$, and evaluate $\frac{dy}{dx}$ at $x = 3$.

(a) For $x > 0$, $y > 0$, so $\ln y = \sqrt{x}\,\ln x$. Differentiate, by the product rule on the right:

$$\frac{1}{y}\frac{dy}{dx} = \frac{\ln x}{2\sqrt{x}} + \sqrt{x}\cdot\frac{1}{x} = \frac{\ln x + 2}{2\sqrt{x}}.$$

Multiply back by y:

$$\boxed{\frac{dy}{dx} = x^{\sqrt{x}}\,\frac{\ln x + 2}{2\sqrt{x}}}$$

At $x = 4$: $y = 4^2 = 16$, so $\frac{dy}{dx} = 16\cdot\dfrac{\ln 4 + 2}{4} = \boxed{8(1+\ln 2)}$.

(b) For $x > -1$, $x \neq 0$, $x \neq 2$: $\ln|y| = 3\ln|x| + \frac{1}{2}\ln(x+1) - 2\ln|x-2|$. Differentiate:

$$\frac{1}{y}\frac{dy}{dx} = \frac{3}{x} + \frac{1}{2(x+1)} - \frac{2}{x-2}.$$

Multiply back by y:

$$\frac{dy}{dx} = \frac{x^3\sqrt{x+1}}{(x-2)^2} \times \left(\frac{3}{x} + \frac{1}{2(x+1)} - \frac{2}{x-2}\right).$$

At $x = 3$, $y = \frac{27\cdot 2}{1} = 54$ and the bracket is $1 + \frac{1}{8} - 2 = -\frac{7}{8}$, so

$$\left.\frac{dy}{dx}\right|_{x=3} = 54\cdot\left(-\frac{7}{8}\right) = \boxed{-\frac{189}{4}}.$$

At $x = 0$, where $y = 0$, the product rule on $x^3\cdot\dfrac{\sqrt{x+1}}{(x-2)^2}$ gives two terms, each with a factor x^2, so $y'(0) = 0$.

Check. The slope of a short secant centred on the point. In (a), $\dfrac{y(4.001) - y(3.999)}{0.002} \approx 13.545$, and $8(1+\ln 2) \approx 13.545$. In (b), $\dfrac{y(3.001) - y(2.999)}{0.002} \approx -47.25$, and $-\frac{189}{4} = -47.25$.

Pitfalls

- Using the power rule on $x^{\sqrt{x}}$ as if $\sqrt{x}$ were a constant: $\sqrt{x}\,x^{\sqrt{x}-1}$ is 8 at $x = 4$, not 13.5.
- Stopping at $\dfrac{y'}{y} = \ldots$. The derivative needs the factor y multiplied back.
- Taking plain ln in (b). For $-1 < x < 0$ the factor x^3 is negative, and $\ln x^3$ is not defined there; $\ln|x^3| = 3\ln|x|$ is.

In the notes. Unit 7: Logarithmic differentiation.
Practise it. the Review set on this method

Unit 8 Differentiation II: applications

Seven methods. Nearly every problem starts by computing f', so what decides the method is what the question asks for. The derivative of an inverse function is a variant of Method 8.2.

Which method?

The question asks for	Method
the tangent or the normal at a point	8.1
"show that the equation has a root in $[a, b]$"	8.2, a sign change (IVT)
"… and it has exactly one root"	8.2, add the sign of f'
a value c with $f'(c)$ equal to a secant slope; a bound on $\lvert f(b) - f(a)\rvert$	8.2, the mean value theorem
the slope of f^{-1} at a point	8.2, variant: $1/f'(a)$
a limit of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$	8.3, l'Hôpital's rule
a limit of the form $0 \cdot \infty$ or $\infty - \infty$	8.3, rewrite as a quotient first
a local maximum or minimum; "classify"	8.4, the derivative tests
the largest or smallest value on $[a, b]$	8.4, the closed-interval method
a sketch of $y = f(x)$	8.5, the checklist
the shape of f from a graph of f'	8.5, variant: read the signs
the largest or smallest value, from a description or on an open interval	8.6, reduce to one variable
a root to six decimal places	8.7, Newton's method
a minimum, by iterating with a step size η	8.7, gradient descent

8.1 Tangent and normal

Problem. Find the tangent line and the normal line to the curve $y = f(x)$ at the point where $x = a$, or where the tangent has a given slope.

Method

1. Compute $f(a)$ and $f'(a)$: differentiate first, then put $x = a$. If the point is given by its slope instead (a horizontal tangent has slope 0), first solve $f'(a) =$ that slope for a, keeping solutions in the domain.
2. Tangent: $y = f(a) + f'(a)\,(x - a)$. Simplify it to the form $y = mx + c$.
3. Normal: its slope is $-1/f'(a)$, so $y = f(a) - \dfrac{1}{f'(a)}\,(x - a)$. If $f'(a) = 0$, the tangent is the horizontal line $y = f(a)$ and the normal is the vertical line $x = a$.
4. Check that the two slopes multiply to -1 (when $f'(a) \neq 0$), and that both lines pass through $\big(a, f(a)\big)$.

Example. For the curve $y = x\sqrt{x+3}$, find (a) the tangent and the normal at the point where $x = 1$, and (b) the tangent and the normal at the point where the tangent is horizontal.

By the product rule,

$$f'(x) = \sqrt{x+3} + \frac{x}{2\sqrt{x+3}} = \frac{3x+6}{2\sqrt{x+3}}.$$

(a) $f(1) = 1 \cdot \sqrt{4} = 2$ and $f'(1) = 2 + \frac{1}{4} = \frac{9}{4}$. The tangent is

$$y = 2 + \tfrac{9}{4}(x-1) \quad \Longrightarrow \quad \boxed{y = \tfrac{9}{4}x - \tfrac{1}{4}}$$

The normal has slope $-1/\frac{9}{4} = -\frac{4}{9}$:

$$y = 2 - \tfrac{4}{9}(x-1) \quad \Longrightarrow \quad \boxed{y = -\tfrac{4}{9}x + \tfrac{22}{9}}$$

(b) $f'(x) = 0$ where $3x + 6 = 0$, that is at $x = -2$, which is in the domain $x \geq -3$. There $f(-2) = -2\sqrt{1} = -2$. The tangent is horizontal, so the normal is vertical:

$$\boxed{\text{tangent } y = -2, \quad \text{normal } x = -2}$$

Check. (a) The slopes multiply to $\frac{9}{4} \cdot \left(-\frac{4}{9}\right) = -1$, and both lines pass through $(1, 2)$: $\frac{9}{4} - \frac{1}{4} = 2$ and $-\frac{4}{9} + \frac{22}{9} = 2$. The slope itself: $\big(f(1.001) - f(1)\big)/0.001 \approx 2.2502$, close to $\frac{9}{4}$. (b) $y = -2$ and $x = -2$ both pass through $(-2, -2) = \big(-2, f(-2)\big)$.

Pitfalls

- Leaving x in the slope. The tangent's slope is the number $f'(1) = \frac{9}{4}$, not the function $f'(x)$.
- Taking the normal's slope as $\frac{4}{9}$ or as $-\frac{9}{4}$. It is minus the reciprocal, $-1/f'(a)$.
- Writing the normal at a horizontal tangent as $y = \ldots$. Its slope $-1/0$ does not exist; the normal is the line $x = a$.

In the notes. Unit 8: Tangent and normal lines.
Practise it. the Review set on this method

8.2 A root exists and is unique (IVT and monotonicity); the mean value theorem

Problem. Show that an equation has a root in a given interval, and that it has no other; find the point c that the mean value theorem promises, or use the theorem to bound a difference.

Method

1. For a root: write the equation as $f(x) = 0$, with everything on one side.
2. State that f is continuous on $[a, b]$: polynomials, e^x, $\ln x$, $\sin x$, $\cos x$ and their sums, products, quotients and compositions are continuous wherever they are defined, so check that f is defined at every point of $[a, b]$.
3. Evaluate $f(a)$ and $f(b)$, and show that they have opposite signs.
4. Conclude by the intermediate value theorem: $f(c) = 0$ for some $c \in (a, b)$.
5. Uniqueness: show that $f' > 0$ (or $f' < 0$) on the whole domain of f, which must be an interval, usually $\mathbb{R}$. By the monotonicity corollary, which holds on an interval, f is then strictly monotone, so it takes the value 0 at most once.
6. Mean value theorem (start here for a secant slope or a bound): check that f is continuous

on $[a, b]$ and differentiable on (a, b). Compute the secant slope $\dfrac{f(b) - f(a)}{b - a}$, solve $f'(c) =$ slope, and keep only the solutions with $c \in (a, b)$.

Variants. A *bound*: apply the mean value theorem between the two points, $f(b) - f(a) = f'(c)\,(b - a)$, and replace $|f'(c)|$ by the largest value $|f'|$ takes between the two points. The *derivative of an inverse*: once $f' > 0$ (or $f' < 0$) shows that f is one-to-one, $(f^{-1})'(b) = 1/f'(a)$, where a solves $f(a) = b$ (usually by inspection) and $f'(a) \neq 0$. No formula for f^{-1} is needed.

Example. (a) Show that $e^x + x = 2$ has exactly one real solution, and that it lies in $(0, 1)$. (b) Find the value of c that the mean value theorem gives for $f(x) = \dfrac{1}{x}$ on $[1, 4]$.

(a) Write the equation as $g(x) = e^x + x - 2 = 0$. The function g, an exponential plus a polynomial, is continuous on $\mathbb{R}$, and

$$g(0) = 1 + 0 - 2 = -1 < 0, \qquad g(1) = e - 1 > 0.$$

By the intermediate value theorem, $g(c) = 0$ for some $c \in (0, 1)$. And $g'(x) = e^x + 1 > 0$ for every x, so g is strictly increasing and has at most one zero. So there is $\boxed{\text{exactly one root, in } (0, 1)}$.

(b) $f(x) = 1/x$ is continuous on $[1, 4]$ and differentiable on $(1, 4)$, since 0 is not in the interval. The secant slope is

$$\frac{f(4) - f(1)}{4 - 1} = \frac{\frac{1}{4} - 1}{3} = -\frac{1}{4}.$$

Solve $f'(c) = -\dfrac{1}{c^2} = -\dfrac{1}{4}$: $c^2 = 4$, so $c = \pm 2$. Only one of them lies in $(1, 4)$: $\boxed{c = 2}$.

Check. (a) One more sign: $g(\frac{1}{2}) = \sqrt{e} - \frac{3}{2} \approx 0.15 > 0$, so the root is in fact in $(0, \frac{1}{2})$, and Method 8.7 finds it there. (b) $f'(2) = -\frac{1}{4}$, the slope of the secant.

Pitfalls

- Stopping at the sign change when "exactly one" is asked. The intermediate value theorem gives at least one root, never uniqueness.
- Skipping the continuity check. On $[-1, 1]$ the function $1/x$ has $f(-1) < 0 < f(1)$ but no root. Its secant slope is 1 while $f' < 0$ everywhere, so there is no mean-value point either: $1/x$ is not defined at 0, so it is not continuous on $[-1, 1]$.
- Keeping $c = -2$. The theorem promises a point strictly inside (a, b).

In the notes. Unit 8: Two theorems for continuous functions, Unit 8: The Mean Value Theorem, and Unit 8: The derivative of an inverse function.

Practise it. the Review set on this method For the derivative of an inverse, the Review set on this variant

8.3 l'Hôpital's rule

Problem. Evaluate a limit of an indeterminate form: $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0 \cdot \infty$ or $\infty - \infty$.

Method

1. Substitute the limit point and name the form. If it is not indeterminate, the rule does not apply: a finite value is the answer, and a non-zero number over 0 means an infinite limit,

or none if the two sides differ (Unit 7).

2. Rewrite $0 \cdot \infty$ as a quotient, $fg = \dfrac{f}{1/g}$, sending down a factor g whose reciprocal is simple (for $g = x$, $1/x$); if the new quotient is no simpler, send down the other factor. Rewrite a difference of fractions over a common denominator; with no fractions, factor out the larger term: $x - \ln x = x\left(1 - \frac{\ln x}{x}\right)$. Each leaves a $\frac{0}{0}$ or $\frac{\infty}{\infty}$ quotient.
3. Apply the rule: differentiate the numerator and the denominator separately, and mark the step $\overset{\text{H}}{=}$.
4. Simplify, and check the form again before every repeat. Stop using the rule as soon as the form is no longer indeterminate.
5. Evaluate, as in step 1.

Example. Evaluate (a) $\lim_{x\to\infty} x\left(e^{2/x} - 1\right)$ and (b) $\lim_{x\to 0}\left(\dfrac{1}{x} - \dfrac{1}{e^x - 1}\right)$.

(a) As $x \to \infty$, $e^{2/x} \to 1$, so the form is $\infty \cdot 0$. Rewrite it as $\dfrac{e^{2/x} - 1}{1/x}$, a $\frac{0}{0}$ form. By the chain rule the top differentiates to $-\frac{2}{x^2}e^{2/x}$ and the bottom to $-\frac{1}{x^2}$, so

$$\lim_{x\to\infty} \frac{e^{2/x} - 1}{1/x} \overset{\text{H}}{=} \lim_{x\to\infty} 2e^{2/x} = \boxed{2}.$$

(b) The form is $\infty - \infty$. Over the common denominator $x(e^x - 1)$,

$$\frac{1}{x} - \frac{1}{e^x - 1} = \frac{e^x - 1 - x}{x(e^x - 1)},$$

a $\frac{0}{0}$ form. Then

$$\lim_{x\to 0} \frac{e^x - 1 - x}{x(e^x - 1)} \overset{\text{H}}{=} \lim_{x\to 0} \frac{e^x - 1}{e^x - 1 + xe^x},$$

which is still $\frac{0}{0}$, so apply the rule again:

$$\lim_{x\to 0} \frac{e^x - 1}{e^x - 1 + xe^x} \overset{\text{H}}{=} \lim_{x\to 0} \frac{e^x}{2e^x + xe^x} = \boxed{\tfrac{1}{2}}.$$

Check. Numerically: at $x = 1000$ the expression in (a) is 2.0020, and at $x = 0.01$ the one in (b) is 0.4992.

Pitfalls

- Using the rule once too often. In (b) the last quotient tends to $\frac{1}{2}$, not $\frac{0}{0}$; one more application would give $\frac{1}{3}$.
- Applying the rule to $0 \cdot \infty$ or $\infty - \infty$ before rewriting it as a single quotient.
- Differentiating f/g by the quotient rule. The rule differentiates the top and the bottom separately.

In the notes. Unit 8: l'Hôpital's rule.

Practise it. the Review set on this method

8.4 Classifying critical points; the closed-interval method

Problem. Find the critical points of f and classify each as a local maximum, a local minimum or neither; find the absolute maximum and minimum of a continuous f on a closed interval $[a, b]$.

Method

1. Find f' and factor it. The critical points are the solutions of $f'(x) = 0$ in the domain, together with any point of the domain where f' does not exist.
2. Classify each stationary point by the second-derivative test where it is quick: $f''(c) > 0$ gives a local minimum, $f''(c) < 0$ a local maximum.
3. If $f''(c) = 0$, or $f'(c)$ does not exist, use the first-derivative test instead: the sign of f' on each side of c. From + to − is a maximum, from − to + a minimum, and the same strict sign on both sides is neither. If $f' = 0$ on one side, as for $(\mathrm{ReLU}(x))^2$ at 0, no pattern applies: compare $f(x)$ with $f(c)$ on both sides.
4. State each result as a point: "local maximum at $(c, f(c))$", or neither.
5. On a closed interval: keep the critical points inside $[a, b]$, and evaluate f there and at both endpoints.
6. The largest of these values is the absolute maximum, and the smallest is the absolute minimum.

Example. Let $f(x) = x^4 - 4x^3$. (a) Find and classify its critical points. (b) Find its absolute maximum and minimum on $[-1, 4]$.

(a) $f'(x) = 4x^3 - 12x^2 = 4x^2(x-3)$, which is zero at $x = 0$ and $x = 3$ and exists everywhere. With $f''(x) = 12x^2 - 24x$,

$$f''(3) = 108 - 72 = 36 > 0 \;\Rightarrow\; \text{local min at } (3, -27).$$

At $x = 0$, $f''(0) = 0$ and the test is inconclusive. The factor $4x^2$ is positive on both sides of 0 and $x - 3$ is negative near 0, so $f' < 0$ on both sides: f keeps decreasing through $x = 0$, and $f''(x) = 12x(x-2)$ changes sign there. So

$$\boxed{\text{local min at } (3, -27)}, \qquad \boxed{\text{neither at } (0, 0)},$$

the second a horizontal point of inflection.

(b) Both critical points lie in $[-1, 4]$. The values there and at the endpoints are

$$f(-1) = 5, \quad f(0) = 0, \quad f(3) = -27, \quad f(4) = 0.$$

So $\boxed{\text{abs. max 5 at } x = -1}$ and $\boxed{\text{abs. min } -27 \text{ at } x = 3}$.

Check. $f(x) + 27 = (x-3)^2(x^2 + 2x + 3)$, and $x^2 + 2x + 3 = (x+1)^2 + 2 > 0$, so $f(x) \geq -27$ for every x, with equality only at $x = 3$. And the signs used at 0: $f'(-1) = -16$ and $f'(1) = -8$.

Pitfalls

- Reading $f''(0) = 0$ as "an inflection" or "neither" without looking further: x^4 also has $f''(0) = 0$, at a minimum. Fall back on the sign of f'.
- Skipping the endpoints. Here the absolute maximum, 5 at $x = -1$, is at an endpoint, not at a critical point.
- Dividing $4x^2(x-3) = 0$ by x^2, which loses the critical point $x = 0$.

In the notes. Unit 8: Critical points and the first-derivative test, Unit 8: Concavity and the second-derivative test, and Unit 8: Two theorems for continuous functions, for the extreme value theorem.

Practise it. the Review set on this method

8.5 Sketching a curve by the checklist, or from a graph of f'

Problem. Sketch $y = f(x)$ for a polynomial or a rational function, showing its intercepts, asymptotes, turning points and inflections; or describe f from a graph of f' alone.

Method

1. *Domain and symmetry*: where f is defined; whether it is even or odd.
2. *Intercepts*: $f(0)$, if 0 is in the domain, and the solutions of $f(x) = 0$.
3. *Asymptotes*, for a rational function: cancel any common factor of top and bottom first (a zero that cancels completely leaves a hole); then vertical where the denominator vanishes and the numerator does not; horizontal from the limits as $x \to \pm\infty$.
4. *Increase and decrease*: the sign of f' on each interval between its zeros and the gaps in the domain. Classify the stationary points (Method 8.4).
5. *Concavity and inflections*: the sign of f''. An inflection is a sign change of f'' at a point of the domain.
6. Describe the shape in a sentence, then draw it, labelling the points you found.

Variants. From a graph of f': f increases where $f' > 0$ and decreases where $f' < 0$. A zero where f' changes from + to − is a local maximum of f; one where it changes from − to +, a local minimum. The slope of the graph of f' is f'': where that graph rises, f is concave up, where it falls, concave down, and each turning point of f' is an inflection of f.

Example. Find the features needed to sketch $f(x) = \dfrac{x-1}{x^2}$, and describe the curve.

- *Domain and symmetry.* $\mathbb{R} \setminus \{0\}$. Neither even nor odd: $f(-1) = -2$ but $f(1) = 0$.
- *Intercepts.* None on the y-axis, since 0 is not in the domain. $f(x) = 0$ only at $x = 1$.
- *Asymptotes.* At $x = 0$ the denominator vanishes and the numerator is -1, so $x = 0$ is a vertical asymptote, with $f \to -\infty$ on both sides since $x^2 > 0$. And $f(x) = \frac{1}{x} - \frac{1}{x^2} \to 0$ as $x \to \pm\infty$, so the asymptotes are $\boxed{x = 0 \text{ and } y = 0}$.
- *Increase/decrease.* By the quotient rule, $f'(x) = \dfrac{2-x}{x^3}$. Its sign is −, +, − on $(-\infty, 0)$, $(0, 2)$ and $(2, \infty)$: f decreases, increases, then decreases.
- *Classification.* The only stationary point is $x = 2$, where f' changes from + to −: a local maximum at $\boxed{(2, \frac{1}{4})}$.
- *Concavity.* $f''(x) = \dfrac{2x-6}{x^4}$ has the sign of $x - 3$: concave down on $(-\infty, 0)$ and on $(0, 3)$, concave up on $(3, \infty)$, with an inflection at $\boxed{(3, \frac{2}{9})}$.

The left branch starts just below $y = 0$ far to the left and falls to $-\infty$ beside $x = 0$. The right branch rises from $-\infty$ beside $x = 0$, crosses the axis at $(1, 0)$, peaks at $(2, \frac{1}{4})$, changes its bend

at $(3, \frac{2}{9})$ and falls back towards $y = 0$, staying above it.

Check. $\frac{1}{4} - f(x) = \dfrac{(x-2)^2}{4x^2} \geq 0$, so $(2, \frac{1}{4})$ is the highest point of the whole graph, as the description says. The sign of f agrees too: negative for $x < 1$, positive for $x > 1$.

Pitfalls

- Reading a sign change across an asymptote as a turning point. Here f' changes from $-$ to $+$ across $x = 0$, but $x = 0$ is not in the domain, so it is no minimum.
- Giving a y-intercept when 0 is not in the domain.
- Putting a branch on the wrong side of the horizontal asymptote $y = L$. The sign of $f(x) - L$ for large $|x|$ decides it. Here $L = 0$, so it is the sign of f: below $y = 0$ on the left, above on the right.

In the notes. Unit 8: Curve sketching, with its two full sketches, and Unit 8: Concavity and the second-derivative test.

Practise it. the Review set on this method

8.6 An optimisation word problem

Problem. A quantity described in words is to be made as large or as small as possible, subject to a constraint.

Method

1. Name the quantity Q to be optimised and the variables it depends on. A labelled sketch helps.
2. Use the constraint to write Q as a function of a single variable, and state the domain: the physically meaningful interval.
3. Find the critical points: solve $Q' = 0$, and add any interior point of the domain where Q' does not exist.
4. Identify the optimum by the second-derivative test or the sign of Q'. Compare it with the endpoints of a closed interval, or with the behaviour at the ends of an open one.
5. Answer the question asked: every dimension and the optimal value, with units.

Example. An open-topped box with a square base must hold $500\,\text{cm}^3$. What dimensions use the least card, and how much card is that?

The quantity to minimise is the area A of card. Let the base have side x and the box height h. The base and the four sides give $A = x^2 + 4xh$. The volume $x^2h = 500$ fixes $h = 500/x^2$, so

$$A(x) = x^2 + 4x \cdot \frac{500}{x^2} = x^2 + \frac{2000}{x}, \qquad x > 0.$$

Then

$$A'(x) = 2x - \frac{2000}{x^2} = 0 \implies x^3 = 1000, \quad x = 10.$$

$A''(x) = 2 + \dfrac{4000}{x^3} > 0$ for all $x > 0$, so this is a minimum; and $A \to \infty$ both as $x \to 0^+$ and as

$x \to \infty$, so it is the global minimum. The height is $h = 500/100 = 5$:

$$\boxed{10 \times 10 \times 5 \text{ cm}}, \qquad \boxed{A_{\min} = 300 \text{ cm}^2}.$$

Check. $10^2 \cdot 5 = 500$. Neighbouring widths cost more: $A(9) \approx 303.2$ and $A(11) \approx 302.8$. Exactly, $A(x) - 300 = \dfrac{(x-10)^2(x+20)}{x} \geq 0$ for $x > 0$.

Pitfalls

- Counting a lid: $2x^2 + 4xh$ is the card for a closed box.
- Stopping at the second-derivative test on an open domain. It shows a local minimum; the behaviour at the ends shows that it is the global one.
- Stopping at $x = 10$. The question asks for every dimension and for the amount of card.

In the notes. Unit 8: Solving optimisation problems, with the most economical can.
Practise it. the Review set on this method

8.7 Newton's method and gradient descent

Problem. Find a root of $f(x) = 0$ to a given number of decimal places by Newton's method; minimise f numerically by gradient descent with a learning rate η.

Method

1. Newton: write f, f' and the iteration $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$ explicitly, with f and f' substituted.
2. Start from x_0 near the root: a point of the interval that the intermediate value theorem gives (an endpoint will do), with $f'(x_0) \neq 0$. Tabulate x_n and $f(x_n)$ to six decimal places. If the iterates cycle or leave the interval, halve it by the sign at its midpoint and start again there.
3. Stop when two iterates agree to the places asked for, and confirm the rounding by a sign change: for six places, f changes sign between $x - 0.0000005$ and $x + 0.0000005$, where x is the answer. Comment on the speed: near a simple root the number of correct digits roughly doubles each step.
4. Gradient descent: find f', write the update $x_{n+1} = x_n - \eta f'(x_n)$ in simplified form, and iterate from x_0.
5. For a quadratic with its minimum at x^* and $k = f''$, the error $e_n = x_n - x^*$ obeys $e_{n+1} = (1 - \eta k)\, e_n$. The iterates converge exactly when $0 < \eta < 2/k$.

Why it works. x_{n+1} is where the tangent at x_n meets the axis: put $y = 0$ in $y = f(x_n) + f'(x_n)(x - x_n)$ (Method 8.1).

Example. (a) Method 8.2 showed that $\mathrm{e}^x + x - 2 = 0$ has exactly one root, in $(0, 1)$. Find it to six decimal places by Newton's method from $x_0 = 0$. (b) Minimise $f(x) = 3x^2 - 6x + 2$ by gradient descent with $\eta = 0.25$ from $x_0 = 3$: compute x_1, x_2, x_3, and find the values of η for which the iteration converges.

(a) With $f(x) = \mathrm{e}^x + x - 2$ and $f'(x) = \mathrm{e}^x + 1$,

$$x_{n+1} = x_n - \frac{\mathrm{e}^{x_n} + x_n - 2}{\mathrm{e}^{x_n} + 1}.$$

The first step is $x_1 = 0 - \frac{-1}{2} = 0.5$, and then

n	x_n	$f(x_n)$
0	0.000000	−1.000000
1	0.500000	0.148721
2	0.443852	0.002551
3	0.442855	0.000001
4	0.442854	0.000000
5	0.442854	0.000000

The last two iterates, x_4 and x_5, agree to six places: $\boxed{x \approx 0.442854}$. The errors run 0.44, 0.057, 0.0010, 3×10^{-7}: the number of correct digits roughly doubles each step.

(b) $f'(x) = 6x - 6$, so $x_{n+1} = x_n - 0.25(6x_n - 6) = -0.5\,x_n + 1.5$, and

$$\boxed{x_1 = 0,\ x_2 = 1.5,\ x_3 = 0.75}$$

The minimum is at $x^* = 1$, where $f' = 0$. Here $k = f'' = 6$, so $e_{n+1} = (1 - 0.25 \cdot 6)\, e_n = -0.5\, e_n$: the error halves and changes sign each step, 2, −1, 0.5, −0.25. The iteration converges exactly when $0 < \eta < 2/k$, that is $\boxed{0 < \eta < \frac{1}{3}}$.

Check. (a) $f(0.4428535) < 0 < f(0.4428545)$, so the root rounds to 0.442854. (b) With $\eta = 0.4$, outside the range, the update is $x_{n+1} = -1.4\,x_n + 2.4$, and $3 \to -1.8 \to 4.92 \to -4.488$ runs away from 1.

Pitfalls

- Rounding x_n to two or three places between steps. The rounding error then swamps what the method gains; keep six places or more.
- Stepping uphill. The update subtracts $\eta f'(x_n)$; adding it climbs away from the minimum.
- Taking k to be the coefficient of x^2. It is f'': here 6, not 3, so the bound is $\eta < \frac{1}{3}$, not $\eta < \frac{2}{3}$.

In the notes. Unit 8: Newton's method, and Unit 8: Gradient descent.

Practise it. the Review set on this method

Part IV Integration

Units 9–10: the definite integral, the fundamental theorem and substitution; then the techniques for the integrands substitution cannot reach, and the integrals that model an area, a volume, work, a mass or a probability. Try a substitution first. Unit 10's table says what to try next.

Unit 9 Integration I: the definite integral and substitution

Six methods. The table of antiderivatives comes straight after Riemann sums, because FTC 2 needs an antiderivative; the rest follow the order of the notes. The properties of the definite integral (splitting the interval, reversing the limits, linearity) are single moves inside them. The last method is for the integrals the others cannot reach: a continuous function such as e^{-x^2} has an antiderivative, but not an elementary one, and its definite integral is bracketed by Riemann sums.

9.1 Riemann sums and their limit

Problem. Estimate $\int_a^b f(x)\,dx$ by a Riemann sum with n strips, and say whether it is an over- or an under-estimate; for a polynomial f of degree at most two, find the right sum for general n in closed form and take its limit.

Method

1. Cut $[a,b]$ into n strips of width $\Delta x = \dfrac{b-a}{n}$. The division points are $x_k = a + k\,\Delta x$: right endpoints for $k = 1,\dots,n$, left endpoints for $k = 0,\dots,n-1$.
2. Add the rectangles, $\sum f(x_k)\,\Delta x$, listing the heights $f(x_k)$.
3. If f increases on the whole of $[a,b]$, the right sum over-estimates and the left sum under-estimates; if it decreases, it is the other way round. Either way the exact value is trapped between the two. If f turns inside $[a,b]$, the two sums need not trap it.
4. For general n, take the right sum, $S_n = \sum_{k=1}^{n} f(a+k\,\Delta x)\,\Delta x$. Write it as a sum in k, expand, and pull the powers of n out of the sums.
5. Quote $\sum_{k=1}^{n} k = \frac{1}{2}n(n+1)$ and $\sum_{k=1}^{n} k^2 = \frac{1}{6}n(n+1)(2n+1)$, and simplify to a closed form in n.
6. Let $n \to \infty$, and compare the limit with the value from FTC 2 (Method 9.3).

Example. Let $f(x) = 6 - x - x^2$ on $[0,2]$. (a) Estimate $\int_0^2 f(x)\,dx$ with $n = 4$ strips, by the right and the left sums. Which is the over-estimate? (b) Find the right sum S_n with n strips in closed form, and its limit.

(a) $\Delta x = \frac{2-0}{4} = 0.5$. The heights at $x = 0,\ 0.5,\ 1,\ 1.5,\ 2$ are $6,\ 5.25,\ 4,\ 2.25,\ 0$, so

$$\text{right} = 0.5\,(5.25 + 4 + 2.25 + 0) = 5.75,$$

$$\text{left} = 0.5\,(6 + 5.25 + 4 + 2.25) = 8.75.$$

f is decreasing on $[0,2]$, since x and x^2 both increase there. So the left end is the tallest point

of each strip: the left sum is the over-estimate, and

$$\boxed{5.75 < \int_0^2 f(x)\,\mathrm{d}x < 8.75}$$

(b) Now $\Delta x = \frac{2}{n}$ and $x_k = \frac{2k}{n}$, so $f(x_k) = 6 - \frac{2k}{n} - \frac{4k^2}{n^2}$ and

$$\begin{aligned} S_n &= \sum_{k=1}^{n}\left(6 - \frac{2k}{n} - \frac{4k^2}{n^2}\right)\frac{2}{n} \\ &= 12 - \frac{4}{n^2}\sum_{k=1}^{n} k - \frac{8}{n^3}\sum_{k=1}^{n} k^2 \\ &= 12 - \frac{2(n+1)}{n} - \frac{4(n+1)(2n+1)}{3n^2}. \end{aligned}$$

Since $\frac{2(n+1)}{n} = 2 + \frac{2}{n}$ and $\frac{4(n+1)(2n+1)}{3n^2} = \frac{8n^2+12n+4}{3n^2} = \frac{8}{3} + \frac{4}{n} + \frac{4}{3n^2}$,

$$\boxed{S_n = \frac{22}{3} - \frac{6}{n} - \frac{4}{3n^2}}$$

As $n \to \infty$ the last two terms vanish, and

$$\int_0^2 f(x)\,\mathrm{d}x = \lim_{n\to\infty} S_n = \boxed{\tfrac{22}{3}}.$$

Check. $n = 4$ gives $S_4 = \frac{22}{3} - \frac{3}{2} - \frac{1}{12} = \frac{23}{4} = 5.75$, the right sum of (a). FTC 2 gives

$$\left[6x - \tfrac{1}{2}x^2 - \tfrac{1}{3}x^3\right]_0^2 = 12 - 2 - \tfrac{8}{3} = \tfrac{22}{3} \approx 7.33,$$

inside the bracket. And $S_n < \frac{22}{3}$ for every n, as an under-estimate must be.

Pitfalls

- Calling the right sum an over-estimate out of habit. That holds only on an increasing function; here f decreases and the right sum is the under-estimate.
- Summing over $k = 0, \ldots, n$: that is $n + 1$ rectangles, one of them outside $[a, b]$.
- Writing $x_k = \frac{k}{n}$ on an interval that is not $[0, 1]$. On $[0, 2]$ the points are $x_k = \frac{2k}{n}$, and the width is $\frac{2}{n}$.

In the notes. Unit 9: The area under a curve, and Unit 9: Computing an integral from the definition.

Practise it. the Review set on this method

9.2 Antiderivatives from the table: rewrite, then read off a

Problem. Find an antiderivative of a sum of powers, roots, reciprocals and the table's entries e^{ax}, $\sin ax$, $\cos ax$, $\sec^2 ax$, $\frac{1}{a^2+x^2}$ and $\frac{1}{\sqrt{a^2-x^2}}$, as an indefinite integral or as the F that FTC 2 needs.

Method

1. Rewrite the integrand as a sum of powers and table entries. Write roots and reciprocals as powers ($\sqrt{x} = x^{1/2}$, $\frac{1}{x^2} = x^{-2}$), split a quotient by a one-term denominator term by term (a sum in the denominator does not split), and take constant multiples out.
2. Integrate each power by the power rule, $x^n \to \dfrac{x^{n+1}}{n+1}$ for $n \neq -1$. The exception, $\frac{1}{x}$, gives $\ln|x|$.
3. Match every other term to a table row and read off $a > 0$. In e^{ax}, $\sin ax$, $\cos ax$ and $\sec^2 ax$, a is the coefficient of x, and the answer carries $\frac{1}{a}$ (with a minus sign for sin).
4. In $\frac{1}{a^2+x^2}$ and $\frac{1}{\sqrt{a^2-x^2}}$, the constant is a^2, so a is its square root. The arctan row carries $\frac{1}{a}$; the arcsin row has none.
5. Add a single $+C$. For a definite integral, leave out the $+C$ and go on to Method 9.3.
6. Check by differentiating the answer term by term.

Example. Find (a) $\displaystyle\int \frac{x^3 - 2\sqrt{x} + 5}{x}\,dx$ and (b) $\displaystyle\int \left(3\sin\frac{x}{2} + \frac{6}{3+x^2}\right)dx$.

(a) Divide term by term: the integrand is $x^2 - 2x^{-1/2} + \frac{5}{x}$. The powers give $\frac{1}{3}x^3$ and $-2 \cdot \frac{x^{1/2}}{1/2} = -4x^{1/2}$, and $\frac{5}{x}$ gives $5\ln|x|$:

$$\boxed{\tfrac{1}{3}x^3 - 4\sqrt{x} + 5\ln|x| + C}$$

(b) In $\sin\frac{x}{2}$, $a = \frac{1}{2}$, so $\frac{1}{a} = 2$ and $\int 3\sin\frac{x}{2}\,dx = -6\cos\frac{x}{2}$. In $\frac{6}{3+x^2}$, $a^2 = 3$, so $a = \sqrt{3}$, and the term gives $6 \cdot \frac{1}{\sqrt{3}} \arctan\frac{x}{\sqrt{3}} = 2\sqrt{3}\arctan\frac{x}{\sqrt{3}}$. Hence

$$\boxed{-6\cos\tfrac{x}{2} + 2\sqrt{3}\arctan\tfrac{x}{\sqrt{3}} + C}$$

Check. (a) Differentiating gives $x^2 - 2x^{-1/2} + \frac{5}{x}$, the integrand again. The $\ln|x|$ matters, because the integrand is defined for $x < 0$ too, and $\frac{d}{dx}\ln(-x) = \frac{1}{x}$ there. (b) Differentiating gives

$$3\sin\frac{x}{2} + 2\sqrt{3} \cdot \frac{1/\sqrt{3}}{1 + x^2/3}$$
$$= 3\sin\frac{x}{2} + \frac{6}{3+x^2}.$$

Pitfalls

- Reading a off a^2. Taking $a = 3$ in $3 + x^2$ gives $2\arctan\frac{x}{3}$, whose derivative is $\frac{6}{9+x^2}$, not $\frac{6}{3+x^2}$.
- Multiplying by a instead of dividing. $\int 3\sin\frac{x}{2}\,dx$ is not $-\frac{3}{2}\cos\frac{x}{2}$, whose derivative is $\frac{3}{4}\sin\frac{x}{2}$: with $a = \frac{1}{2}$, dividing by a doubles. Dropping the minus sign of the sin row is the other common slip: $6\cos\frac{x}{2}$ differentiates to $-3\sin\frac{x}{2}$.
- Writing $5\ln x$. It is undefined for $x < 0$, where $\frac{5}{x}$ is perfectly well defined. And the power rule cannot do $\frac{1}{x}$ at all: it would divide by $n + 1 = 0$.

In the notes. Unit 9: The antiderivative and the indefinite integral, for the power rule and $\ln|x|$, and Unit 9: A table of standard integrals.

Practise it. the Review set on this method

9.3 A definite integral by FTC 2; signed against total area

Problem. Evaluate $\int_a^b f(x)\,dx$ exactly; find the net signed area and the total area between a graph and the x-axis.

Method

1. Find an antiderivative F of f (Method 9.2). No $+C$ is needed.
2. Check that f is continuous on the whole of $[a,b]$ (a denominator that is 0 inside it rules this out). If it is not, FTC 2 does not apply: stop. Otherwise write $\big[F(x)\big]_a^b$, then $F(b)-F(a)$ with both values visible, and simplify.
3. For the total area, find where f changes sign in $[a,b]$.
4. Cut $[a,b]$ at the crossings and apply step 2 to each piece. The net signed area is the sum of the pieces; the total area is the sum of their magnitudes. State both.

Variants. A *net change*: the integral of a rate over $[a,b]$ is the change in the quantity, $\int_a^b Q'(t)\,dt = Q(b)-Q(a)$. Use step 2, or steps 3–4 applied to v for a total such as the distance travelled, $\int_a^b |v|\,dt$, and give the answer its unit.

Example. Find the net signed area and the total area between the graph of $f(x)=\dfrac{x-4}{\sqrt{x}}$ and the x-axis on $[1,9]$.

Split the quotient: $f(x)=\dfrac{x}{\sqrt{x}}-\dfrac{4}{\sqrt{x}}=x^{1/2}-4x^{-1/2}$. Term by term, $F(x)=\frac{2}{3}x^{3/2}-8x^{1/2}$, so

$$F(1)=\tfrac{2}{3}-8=-\tfrac{22}{3},\quad F(4)=\tfrac{16}{3}-16=-\tfrac{32}{3},\quad F(9)=18-24=-6.$$

The net signed area is

$$\big[F(x)\big]_1^9=-6-\left(-\frac{22}{3}\right)=\boxed{\tfrac{4}{3}}.$$

f changes sign only at $x=4$: it is negative on $[1,4)$ and positive on $(4,9]$. The two pieces are

$$\int_1^4 f\,dx=-\tfrac{32}{3}+\tfrac{22}{3}=-\tfrac{10}{3},\qquad \int_4^9 f\,dx=-6+\tfrac{32}{3}=\tfrac{14}{3},$$

and the total area is $\left|-\frac{10}{3}\right|+\frac{14}{3}=\boxed{8}$.

Check. Differentiate: $F'(x)=\frac{2}{3}\cdot\frac{3}{2}x^{1/2}-8\cdot\frac{1}{2}x^{-1/2}=x^{1/2}-4x^{-1/2}=f(x)$. The pieces add up to the net value, $-\frac{10}{3}+\frac{14}{3}=\frac{4}{3}$, and the first is negative, as the sign of f on $[1,4)$ says it must be.

Pitfalls

- Giving the net value $\frac{4}{3}$ as the area. The integral counts the part below the axis as negative; the area is 8.
- Taking $F(a)$ to be 0. It is not, unless the lower limit makes it so: here $F(1)=-\frac{22}{3}$, and leaving it out gives -6.
- Using FTC 2 where f is not continuous on $[a,b]$. For $\frac{1}{x}$ on $[-1,1]$ the bracket $\big[\ln|x|\big]_{-1}^{1}=0$ is a number, but the theorem does not apply, so it is not the integral.

In the notes. Unit 9: The antiderivative and the indefinite integral, Unit 9: The Fundamental Theorem of Calculus, and Unit 9: Properties of the definite integral, with its signed-area example.

Practise it. the Review set on this method

9.4 Differentiating an integral (FTC 1)

Problem. Differentiate, with respect to x, an integral $\int f(t)\,\mathrm{d}t$ whose limits depend on x but whose integrand does not, without evaluating it.

Method

1. Find which limit depends on x. If only the upper one does, go to step 3.
2. If the lower limit varies, reverse the limits, which flips the sign: $\int_{g(x)}^{b} f = -\int_{b}^{g(x)} f$. If both vary, split at a constant c: $\int_{g(x)}^{h(x)} f = \int_{c}^{h(x)} f - \int_{c}^{g(x)} f$.
3. In each piece $\int_{c}^{g(x)} f(t)\,\mathrm{d}t$, substitute the upper limit into the integrand: $f\big(g(x)\big)$.
4. Multiply by the derivative of that limit, $g'(x)$ (the chain rule). For the limit x itself, the factor is 1.
5. Combine the pieces with their signs. No antiderivative is needed, so this works even when there is no elementary one.

Why it works. With $F(u) = \int_{c}^{u} f(t)\,\mathrm{d}t$, FTC 1 says $F' = f$. The integral up to $g(x)$ is the composite $F\big(g(x)\big)$, and the chain rule gives $F'\big(g(x)\big)\,g'(x) = f\big(g(x)\big)\,g'(x)$.

Example. Find (a) $\dfrac{\mathrm{d}}{\mathrm{d}x}\displaystyle\int_{x}^{1} \frac{\sin t}{t}\,\mathrm{d}t$ for $x > 0$, and (b) $\dfrac{\mathrm{d}}{\mathrm{d}x}\displaystyle\int_{x}^{2x} \frac{\mathrm{d}t}{1+t^2}$.

(a) The variable is the lower limit. Reverse the limits: $\int_{x}^{1} \frac{\sin t}{t}\,\mathrm{d}t = -\int_{1}^{x} \frac{\sin t}{t}\,\mathrm{d}t$. Substitute x into the integrand; the factor is 1. So

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_{x}^{1} \frac{\sin t}{t}\,\mathrm{d}t = \boxed{-\frac{\sin x}{x}}.$$

(b) Both limits vary. Split at 0: $\int_{x}^{2x} = \int_{0}^{2x} - \int_{0}^{x}$. In the first piece substitute $2x$ and multiply by $(2x)' = 2$; the second gives $\frac{1}{1+x^2}$. So

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_{x}^{2x} \frac{\mathrm{d}t}{1+t^2} = \boxed{\frac{2}{1+4x^2} - \frac{1}{1+x^2}}.$$

Check. In (b) the integral can be evaluated first: $\big[\arctan t\big]_{x}^{2x} = \arctan 2x - \arctan x$, whose derivative is $\frac{2}{1+4x^2} - \frac{1}{1+x^2}$. In (a) it cannot, since $\frac{\sin t}{t}$ has no elementary antiderivative, but the sign can be checked: for $0 < x < \pi$ the integrand is positive, and moving the lower limit up takes a strip of height $\frac{\sin x}{x}$ away, so the derivative is negative.

Pitfalls

- Forgetting the chain-rule factor: the first piece of (b) is $\frac{2}{1+4x^2}$, not $\frac{1}{1+4x^2}$.
- Forgetting the sign when the variable is the lower limit: (a) is $-\frac{\sin x}{x}$, not $\frac{\sin x}{x}$.
- Letting the constant limit contribute: $\frac{\mathrm{d}}{\mathrm{d}x}\int_{1}^{x} f(t)\,\mathrm{d}t$ is $f(x)$, not $f(x) - f(1)$.

In the notes. Unit 9: The Fundamental Theorem of Calculus, from Part 1 to the example with a variable limit that is a function.

Practise it. the Review set on this method

9.5 Substitution, indefinite and definite

Problem. Integrate a function that is not in the table but contains an inside function together with its derivative; evaluate such a definite integral.

Method

1. Choose $u = g(x)$, the inside function whose derivative $g'(x)$ is also present, up to a constant factor.
2. Write $\mathrm{d}u = g'(x)\,\mathrm{d}x$, and solve it for the piece the integrand contains (for instance $x\,\mathrm{d}x = \tfrac{1}{2}\,\mathrm{d}u$).
3. For a definite integral, convert the limits: $x = a \Rightarrow u = g(a)$, $x = b \Rightarrow u = g(b)$.
4. Rewrite the whole integral in u. No x may be left. A constant may be moved across the integral sign; a function may not.
5. Integrate in u, from the table.
6. Indefinite: put $u = g(x)$ back and add $+C$. Definite: evaluate with the u-limits, and never return to x.
7. Check an indefinite answer by differentiating it.

Variants. *The f'/f pattern*: when the numerator is a constant c times the derivative of the denominator, $u =$ denominator gives $c\int \frac{\mathrm{d}u}{u}$, so $\int \frac{c\,g'(x)}{g(x)}\,\mathrm{d}x = c\ln|g(x)| + C$. Drop the bars only when $g > 0$ is known. *A stray x*: if $u = x + k$ leaves a factor x behind, write $x = u - k$ and expand into powers of u. *Completing the square*: a quadratic with no real roots in a denominator becomes $A\big[(x+h)^2 + a^2\big]$, with A its x^2 coefficient; take the $\frac{1}{A}$ out, put $u = x + h$ and use the table entry $\int \frac{\mathrm{d}u}{a^2+u^2} = \frac{1}{a}\arctan\frac{u}{a}$. *Route 1* for a definite integral: integrate, return to x, then apply the x-limits (Method 9.3). Both routes give the same number.

Example. Evaluate (a) $\displaystyle\int_0^{\pi/3} \frac{\sin x}{\cos^3 x}\,\mathrm{d}x$, and find (b) $\displaystyle\int \frac{\mathrm{d}x}{2x^2 + 12x + 36}$.

(a) The inside function is $u = \cos x$, and $\mathrm{d}u = -\sin x\,\mathrm{d}x$, so $\sin x\,\mathrm{d}x = -\mathrm{d}u$. The limits:

$$x = 0 \;\Rightarrow\; u = 1, \qquad x = \tfrac{\pi}{3} \;\Rightarrow\; u = \tfrac{1}{2}.$$

Hence

$$\begin{aligned}\int_0^{\pi/3} \frac{\sin x}{\cos^3 x}\,\mathrm{d}x &= -\int_1^{1/2} u^{-3}\,\mathrm{d}u \\ &= \int_{1/2}^{1} u^{-3}\,\mathrm{d}u = \Big[-\frac{1}{2u^2}\Big]_{1/2}^{1} \\ &= -\frac{1}{2} - (-2) = \boxed{\tfrac{3}{2}}.\end{aligned}$$

(b) The denominator has no real roots. Take out the 2 and complete the square: $2x^2+12x+36 = 2\big[(x+3)^2+9\big]$. Put $u = x + 3$, so $\mathrm{d}u = \mathrm{d}x$, and use the table with $a = 3$:

$$\tfrac{1}{2}\int \frac{\mathrm{d}u}{9+u^2} = \tfrac{1}{6}\arctan\frac{u}{3} + C = \boxed{\tfrac{1}{6}\arctan\frac{x+3}{3} + C}.$$

Check. (a) by Route 1: $-\int u^{-3}\,du = \frac{1}{2u^2}$, so an antiderivative is $\frac{1}{2\cos^2 x}$. Its derivative is $\frac{\sin x}{\cos^3 x}$, the integrand; with the minus sign lost it would be minus the integrand. And

$$\left[\frac{1}{2\cos^2 x}\right]_0^{\pi/3} = 2 - \tfrac{1}{2} = \tfrac{3}{2}.$$

(b) Differentiate:

$$\tfrac{1}{6}\cdot\frac{1/3}{1+(x+3)^2/9} = \frac{1}{18+2(x+3)^2} = \frac{1}{2x^2+12x+36}.$$

Pitfalls

- Keeping the x-limits after changing the variable, as in $\int_0^{\pi/3} u^{-3}\,du$. The limits belong to the variable of integration.
- Losing the minus sign of $du = -\sin x\,dx$. It gives $-\frac{3}{2}$, a negative value for an integrand that is positive on $(0, \frac{\pi}{3})$.
- Losing a factor. Dropping the $\frac{1}{a}$ of the table entry gives $\frac{1}{2}\arctan\frac{x+3}{3}$; dropping the $\frac{1}{2}$ taken out gives $\frac{1}{3}\arctan\frac{x+3}{3}$. They differentiate to $\dfrac{3}{2x^2+12x+36}$ and $\dfrac{2}{2x^2+12x+36}$.

In the notes. Unit 9: Integration by substitution, Unit 9: Substitution in a definite integral, and Unit 9: A table of standard integrals.

Practise it. the Review set on this method

9.6 No elementary antiderivative: decide, then bracket

Problem. Decide whether FTC 2 can evaluate an integral with an elementary antiderivative. When it cannot, say why, say what antiderivative there is, and estimate the definite integral from Riemann sums.

Method

1. Look for an inside function whose derivative is also present, up to a constant. If there is one, the integral is elementary: substitute (Method 9.5) and evaluate by FTC 2.
2. If the integrand is one the notes name, e^{-x^2}, $\frac{\sin x}{x}$, $\sin(x^2)$, $\frac{e^x}{x}$ or $\frac{1}{\ln x}$, it has no elementary antiderivative (Liouville). Quote this. It still has an antiderivative, its area function $F(x) = \int_a^x f(t)\,dt$ (FTC 1). What it lacks is a formula, so FTC 2 has nothing to substitute into.
3. Estimate the definite integral by the left and right sums with n strips (Method 9.1, steps 1–2).
4. Decide the direction of each sum from monotonicity (Method 9.1, step 3), and state the bracket, rounding outwards. More strips tighten it.

Example. Of $P = \displaystyle\int_2^4 \frac{dx}{\ln x}$ and $Q = \displaystyle\int_2^4 \frac{dx}{x\ln x}$, one can be evaluated by FTC 2. Evaluate it. For the other, say why FTC 2 cannot, and bracket it with $n = 4$ strips.

Q: the inside function is $u = \ln x$, and its derivative $\frac{1}{x}$ is present. So $du = \frac{dx}{x}$, and the

limits are

$$x = 2 \;\Rightarrow\; u = \ln 2, \qquad x = 4 \;\Rightarrow\; u = \ln 4 = 2\ln 2.$$

Hence

$$\begin{aligned} Q &= \int_{\ln 2}^{2\ln 2} \frac{\mathrm{d}u}{u} \\ &= \ln(2\ln 2) - \ln(\ln 2) \\ &= \boxed{\ln 2} \approx 0.693. \end{aligned}$$

P: $\frac{1}{\ln x}$ is on the notes' list, and no factor $\frac{1}{x}$ is present. (Forcing $u = \ln x$ gives $\mathrm{d}x = \mathrm{e}^u\,\mathrm{d}u$ and turns P into $\int \frac{\mathrm{e}^u}{u}\,\mathrm{d}u$, another entry on the list.) So there is no elementary antiderivative, only the area function $F(x) = \int_2^x \frac{\mathrm{d}t}{\ln t}$. With $\Delta x = 0.5$, the heights at $x = 2,\ 2.5,\ 3,\ 3.5,\ 4$ are $1.4427,\ 1.0914,\ 0.9102,\ 0.7982,\ 0.7213$. The first four add to 4.2425 and the last four to 3.5211, so

$$\text{left} = 0.5 \cdot 4.2425 \approx 2.1213,$$

$$\text{right} = 0.5 \cdot 3.5211 \approx 1.7606.$$

$\ln x$ increases and is positive on $[2, 4]$, so $\frac{1}{\ln x}$ decreases: the left sum is the over-estimate. Rounding outwards,

$$\boxed{1.760 < P < 2.122}$$

Check. Q: $\frac{\mathrm{d}}{\mathrm{d}x}\ln(\ln x) = \frac{1}{\ln x}\cdot\frac{1}{x}$, the integrand, and $\ln 2 > 0$, as a positive integrand requires. P: the two sums share the three middle heights, so they differ by $0.5\,(f(2) - f(4)) = 0.5\,(1.4427 - 0.7213) \approx 0.3607$, and they do $(2.1213 - 1.7606)$. A computer gives $P \approx 1.9224$, inside the bracket.

Pitfalls

- Reading "no elementary antiderivative" as "no antiderivative". P has one, $F(x) = \int_2^x \frac{\mathrm{d}t}{\ln t}$ with $F'(x) = \frac{1}{\ln x}$. What it lacks is a formula.
- Dividing by the derivative of the inside. $\int \mathrm{e}^{-x^2}\,\mathrm{d}x$ is not $\frac{\mathrm{e}^{-x^2}}{-2x}$: its derivative is $\mathrm{e}^{-x^2}\left(1 + \frac{1}{2x^2}\right)$. Substitution needs the derivative present as a factor; you cannot divide by it.
- Saying "no" because a substitution fails. Only the functions the notes list may be quoted as non-elementary. $x\,\mathrm{e}^x$ offers no substitution, yet it integrates, by integration by parts, in Unit 10.

In the notes. Unit 9: When there is no elementary antiderivative, Unit 9: The Fundamental Theorem of Calculus (Part 1), and Unit 9: The area under a curve, for the bracket.

Practise it. the Review set on this method

Unit 10 Integration II: techniques and applications

Seven methods: four techniques of integration, then three uses of the integral. On this unit choosing the technique is most of the work. Look for a substitution first (Method 9.5); when there is none, the shape of the integrand says what to try next.

Which method?

The integrand, or the question, has	Method
an inside function times its derivative; an f'/f	9.5, substitution
a polynomial times e^{ax}, $\sin ax$, $\cos ax$ or $\sec^2 ax$	10.1, by parts, u = the polynomial
a polynomial times $\ln x$ or $\arctan x$	10.1, by parts, $u = \ln x$ or $\arctan x$
a lone $\ln x$ or $\arctan x$	10.1, by parts with $v' = 1$
e^{ax} times $\sin bx$ or $\cos bx$	10.1, parts twice, then solve for the integral
a polynomial over a polynomial, with no repeated irreducible quadratic factor	10.2: divide if improper, then factor
1 over a quadratic with no real roots	10.2: complete the square, then arctan
an odd power of $\sin x$ or of $\cos x$	10.3: split off one factor
only even powers of $\sin x$ and $\cos x$	10.3: the half-angle identities
$\tan^m x \sec^n x$ with $n \geq 2$ even or m odd; a power of $\tan x$ alone	10.3: peel off $\sec^2 x$ if you can, else $\tan^2 x$
$\sqrt{a^2 - x^2}$	10.4, $x = a \sin\theta$
$\sqrt{a^2 + x^2}$, or a power of $a^2 + x^2$, such as $1/(a^2 + x^2)^2$	10.4, $x = a \tan\theta$
$\sqrt{x^2 - a^2}$	10.4, $x = a \sec\theta$
a limit of integration ∞ or $-\infty$	10.5, improper integral
a finite interval with a point where the integrand is unbounded	10.5, a limit at the blow-up; split at an interior one
an area, a volume, work, a mass or an average value, in words	10.6, slice and sum; an average: its Variants
a density: its constant, a probability, a mean	10.7, density

10.1 Integration by parts

Problem. Integrate a product of two different kinds of function, such as a polynomial times an exponential, or a lone logarithm or inverse trigonometric function, such as $\ln x$.

Method

1. Name the choice by LIATE: u is the factor that comes first in Logarithm, Inverse trig, Algebraic, Trigonometric, Exponential, and the rest is v'. For a lone function such as $\ln x$ or $\arctan x$, take $v' = 1$.
2. Write down u' and v.
3. Apply $\int uv'\,dx = uv - \int u'v\,dx$. For a definite integral the first term is $\left[uv\right]_a^b$.
4. Evaluate the new integral. If it needs parts again, repeat with the same kind of choice.

5. If the integral you started with comes back, call it I and solve for I algebraically. Do not keep going.
6. Add $+C$ to an indefinite result.

Example. Evaluate (a) $\displaystyle\int_0^{\pi/4} x\sec^2 x\,\mathrm{d}x$ and (b) $\displaystyle\int \sin(\ln x)\,\mathrm{d}x$ for $x>0$.

(a) LIATE: x (algebraic) before $\sec^2 x$ (trigonometric), so $u=x$, $v'=\sec^2 x$, giving $u'=1$ and $v=\tan x$. Call the integral J. Then

$$\begin{aligned} J &= \big[x\tan x\big]_0^{\pi/4} - \int_0^{\pi/4} \tan x\,\mathrm{d}x \\ &= \tfrac{\pi}{4} - \big[\ln|\sec x|\big]_0^{\pi/4} = \tfrac{\pi}{4} - \ln\sqrt{2}, \end{aligned}$$

that is, $\boxed{\tfrac{\pi}{4}-\tfrac{1}{2}\ln 2}\approx 0.439$.

(b) A lone function, so try $v'=1$, as for $\ln x$: take $u=\sin(\ln x)$, $v'=1$: then $u'=\cos(\ln x)/x$ and $v=x$, and the x cancels:

$$I=\int \sin(\ln x)\,\mathrm{d}x = x\sin(\ln x) - \int \cos(\ln x)\,\mathrm{d}x.$$

Parts again, the same way ($u=\cos(\ln x)$, $v'=1$, $u'=-\sin(\ln x)/x$):

$$\int \cos(\ln x)\,\mathrm{d}x = x\cos(\ln x) + I.$$

The original integral has come back. Substitute and solve for I:

$$I = x\sin(\ln x) - x\cos(\ln x) - I,$$

so $2I = x\big(\sin(\ln x)-\cos(\ln x)\big)$ and

$$I=\boxed{\tfrac{x}{2}\big(\sin(\ln x)-\cos(\ln x)\big)+C}$$

Check. Differentiate. In (a), $\frac{\mathrm{d}}{\mathrm{d}x}\big(x\tan x-\ln|\sec x|\big)=\tan x+x\sec^2 x-\tan x=x\sec^2 x$. In (b), the product rule gives $\frac{1}{2}\big(\sin(\ln x)-\cos(\ln x)\big)+\frac{1}{2}\big(\cos(\ln x)+\sin(\ln x)\big)=\sin(\ln x)$.

Pitfalls

- Choosing the other way round. In (a), $u=\sec^2 x$ and $v'=x$ give $\int \frac{1}{2}x^2\cdot 2\sec^2 x\tan x\,\mathrm{d}x$, worse than the integral you started with.
- Applying parts a third time in (b). Once I reappears, the next round only undoes the last one; solve for I instead.
- Dropping the minus sign in front of $\int u'v\,\mathrm{d}x$, or in $u'=-\sin(\ln x)/x$. In (b) that turns $-I$ into $+I$, and I cancels from the equation.

In the notes. Unit 10: Integration by parts, with LIATE and its three examples.

Practise it. the Review set on this method

10.2 Partial fractions, branched by the denominator

Problem. Integrate a rational function $\dfrac{P(x)}{Q(x)}$, a polynomial over a polynomial. A power of an irreducible quadratic, such as $1/(x^2+1)^2$, is not split here: use Method 10.4.

Method

1. Is it proper? If the degree of P is at least that of Q, divide first: a polynomial plus a proper remainder.
2. Factor Q into linear factors and irreducible quadratics (negative discriminant). If Q is a single irreducible quadratic and P is a constant, there is nothing to split: complete the square, $(x+p)^2+q$, and put $u=x+p$.
3. Write the template: $\dfrac{A}{ax+b}$ for each linear factor; one term for each power of a repeated linear factor, up to the highest; and $\dfrac{Ax+B}{ax^2+bx+c}$ for an irreducible quadratic.
4. Clear the denominators. Cover-up: put x equal to the root of each linear factor, which wipes out every term but one. Match a coefficient (or substitute another x) for each constant cover-up cannot reach.
5. Integrate term by term: $\dfrac{A}{ax+b}$ gives $\frac{A}{a}\ln|ax+b|$, and $\dfrac{1}{(ax+b)^2}$ gives $-\dfrac{1}{a(ax+b)}$. Over an irreducible quadratic, split the numerator into the f'/f part (a ln) and the rest (an arctan, after completing the square).

Example. Evaluate (a) $\displaystyle\int \frac{x^2+2}{x^2+x-2}\,\mathrm{d}x$ and (b) $\displaystyle\int \frac{2x^2+x+2}{(x-1)(x^2+4)}\,\mathrm{d}x$.

(a) Both degrees are 2, so divide first:

$$\frac{x^2+2}{x^2+x-2} = 1 + \frac{4-x}{x^2+x-2}.$$

The denominator factors as $(x-1)(x+2)$, two distinct linear factors, so

$$\frac{4-x}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2},$$

and $4-x = A(x+2)+B(x-1)$. Cover-up: $x=1$ gives $3=3A$, so $A=1$; $x=-2$ gives $6=-3B$, so $B=-2$. Integrating term by term,

$$\boxed{x + \ln|x-1| - 2\ln|x+2| + C}$$

(b) Proper, with a linear factor and the quadratic x^2+4, which is irreducible (discriminant -16). The template is

$$\frac{2x^2+x+2}{(x-1)(x^2+4)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+4},$$

so $2x^2+x+2 = A(x^2+4)+(Bx+C)(x-1)$. Cover-up at $x=1$: $5=5A$, so $A=1$. Cover-up cannot reach B and C. Match coefficients instead: the x^2 terms give $2=A+B$, so $B=1$; the

constants give $2 = 4A - C$, so $C = 2$. Split the quadratic term into its f'/f part and the rest:

$$\frac{x+2}{x^2+4} = \frac{1}{2}\cdot\frac{2x}{x^2+4} + \frac{2}{x^2+4}.$$

With $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan\frac{x}{a}$ and $a = 2$,

$$\boxed{\ln|x-1| + \tfrac{1}{2}\ln(x^2+4) + \arctan\tfrac{x}{2} + C}$$

Check. Recombine at a convenient x. At $x = 0$, the integrand of (a) is $\frac{2}{-2} = -1$ and the split form gives $1 + \frac{1}{-1} - \frac{2}{2} = -1$; the integrand of (b) is $\frac{2}{(-1)\cdot 4} = -\frac{1}{2}$ and the split form gives $\frac{1}{-1} + \frac{2}{4} = -\frac{1}{2}$. But $x = 0$ wipes out the Bx term, so test (b) at $x = 2$ too: $\frac{12}{1\cdot 8} = \frac{3}{2}$ and $\frac{1}{1} + \frac{4}{8} = \frac{3}{2}$.

Pitfalls

- Splitting before dividing. In (a), the template $\frac{A}{x-1} + \frac{B}{x+2}$ can never produce the 1 that division gives, and the equations for A and B are inconsistent.
- Putting a constant over an irreducible quadratic. In (b), the numerator over x^2+4 must be $Bx + C$, with two constants.
- Reaching for partial fractions when the numerator is the derivative of the denominator: $\int \frac{2x+1}{x^2+x-2}\,dx$ is $\ln|x^2+x-2| + C$ at once, by substitution.

In the notes. Unit 10: Partial fractions, with the recipe and its four examples.

Practise it. the Review set on this method

10.3 Trigonometric integrals: odd or even powers, tan and sec

Problem. Integrate a product of powers of $\sin x$ and $\cos x$, or of $\tan x$ and $\sec x$.

Method

1. Classify. Powers of sin and cos: if one power is odd, split off one factor of it; if both are even, use the half-angle identities. Powers $\tan^m x \sec^n x$: if $n \geq 2$ is even, peel off $\sec^2 x$; otherwise, if m is odd or there is no $\sec x$, peel off $\tan^2 x = \sec^2 x - 1$.
2. Rewrite the rest with an identity: $\sin^2 x + \cos^2 x = 1$ for an odd power; $\cos^2 x = \frac{1}{2}(1+\cos 2x)$ and $\sin^2 x = \frac{1}{2}(1-\cos 2x)$ for even powers; $1 + \tan^2 x = \sec^2 x$ for tangents and secants.
3. Substitute: $u = \cos x$ if you split off $\sin x$ ($du = -\sin x\,dx$), $u = \sin x$ if you split off $\cos x$, $u = \tan x$ for each piece $\tan^k x \sec^2 x$, $u = \sec x$ for a piece $\sec^k x \sec x \tan x$. For a definite integral, map the limits, and keep the sign.
4. A leftover $\int \tan x\,dx$ is $\ln|\sec x| + C$.
5. Write the result in terms of x, and add $+C$ to an indefinite one.

Example. Evaluate (a) $\displaystyle\int_0^{\pi/2} \sin^5 x \cos^3 x\,dx$ and (b) $\displaystyle\int \tan^3 x \sec^4 x\,dx$.

(a) Both powers are odd, so either can be split. Split off one $\cos x$ and write the rest in sines: $\sin^5 x \cos^3 x = \sin^5 x\,(1 - \sin^2 x)\cos x$. Put $u = \sin x$, $du = \cos x\,dx$, with $x = 0 \mapsto u = 0$

and $x = \frac{\pi}{2} \mapsto u = 1$:

$$\int_0^{\pi/2} \sin^5 x \cos^3 x \, dx = \int_0^1 u^5(1-u^2)\, du$$
$$= \tfrac{1}{6} - \tfrac{1}{8} = \boxed{\tfrac{1}{24}}.$$

(b) The power of sec is even, so peel off $\sec^2 x$ (m is odd too, but n comes first) and convert the rest, $\tan^3 x \sec^4 x = \tan^3 x\,(1+\tan^2 x)\sec^2 x$. With $u = \tan x$, $du = \sec^2 x \, dx$,

$$\int \tan^3 x \sec^4 x \, dx = \int u^3(1+u^2)\, du = \tfrac{1}{4}u^4 + \tfrac{1}{6}u^6 + C,$$

that is, $\boxed{\tfrac{1}{4}\tan^4 x + \tfrac{1}{6}\tan^6 x + C}$.

Check. In (a), split off $\sin x$ instead: $u = \cos x$ maps $0 \mapsto 1$ and $\frac{\pi}{2} \mapsto 0$, and the minus sign from du turns the limits round, giving $\int_0^1 (1-u^2)^2 u^3 \, du = \frac{1}{4} - \frac{1}{3} + \frac{1}{8} = \frac{1}{24}$ again. In (b), differentiate: $\tan^3 x \sec^2 x + \tan^5 x \sec^2 x = \tan^3 x \sec^2 x\,(1+\tan^2 x) = \tan^3 x \sec^4 x$.

Pitfalls

- Using the half-angle identity on an odd power. It works, but slowly; split off one factor instead.
- Losing the minus sign of $du = -\sin x \, dx$. With $u = \cos x$ in (a) it gives $-\frac{1}{24}$: a negative answer for an integrand that is positive on $(0, \frac{\pi}{2})$.
- Converting every power of sec. In (b), $\sec^4 x = (1+\tan^2 x)^2$ leaves no $\sec^2 x$ to make du.

In the notes. Unit 10: Trigonometric integrals.
Practise it. the Review set on this method

10.4 Trigonometric substitution

Problem. Integrate an expression containing $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$ or $\sqrt{x^2 - a^2}$ (or a power of $a^2 + x^2$) that no plain substitution clears.

Method

1. Read off the form and $a > 0$, and substitute: $\sqrt{a^2 - x^2}$: $x = a\sin\theta$, $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$; $\sqrt{a^2 + x^2}$: $x = a\tan\theta$, $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$; $\sqrt{x^2 - a^2}$ with $x \geq a$: $x = a\sec\theta$, $\theta \in [0, \frac{\pi}{2})$.
2. Write dx, and simplify the root by a Pythagorean identity. Use the range of θ to drop the modulus: $\sqrt{a^2 - a^2\sin^2\theta} = a\,|\cos\theta| = a\cos\theta$.
3. Integrate in θ, often a trigonometric integral (Method 10.3). For a definite integral, change the limits to θ and finish there.
4. For an indefinite integral, convert back to x: through the substitution and a right triangle, or algebraically (for instance $\cos\theta = \sqrt{1 - \sin^2\theta}$ on the range).
5. Add $+C$ to an indefinite result.

Example. Evaluate (a) $\displaystyle\int \frac{dx}{(x^2+1)^2}$ and (b) $\displaystyle\int_0^1 \frac{x^2}{\sqrt{4-x^2}}\,dx$.

(a) A power of a^2+x^2 with $a=1$: put $x=\tan\theta$, $\theta\in(-\tfrac{\pi}{2},\tfrac{\pi}{2})$, so $dx=\sec^2\theta\,d\theta$ and $(x^2+1)^2=\sec^4\theta$. Then, by the half-angle identity,

$$\int \frac{\sec^2\theta}{\sec^4\theta}\,d\theta=\int \cos^2\theta\,d\theta=\tfrac{1}{2}\theta+\tfrac{1}{4}\sin 2\theta+C.$$

Convert back: $\theta=\arctan x$, and $\sin 2\theta=2\sin\theta\cos\theta=\dfrac{2\tan\theta}{\sec^2\theta}=\dfrac{2x}{1+x^2}$. So the integral is

$$\boxed{\tfrac{1}{2}\arctan x+\frac{x}{2(1+x^2)}+C}$$

(b) The form $\sqrt{a^2-x^2}$ with $a=2$: put $x=2\sin\theta$, $\theta\in[-\tfrac{\pi}{2},\tfrac{\pi}{2}]$, so $dx=2\cos\theta\,d\theta$ and $\sqrt{4-x^2}=2\cos\theta$. The limits: $x=0\mapsto\theta=0$, and $x=1\mapsto\sin\theta=\tfrac{1}{2}$, so $\theta=\tfrac{\pi}{6}$. Then

$$\begin{aligned}\int_0^1 \frac{x^2\,dx}{\sqrt{4-x^2}}&=\int_0^{\pi/6}\frac{4\sin^2\theta\cdot 2\cos\theta}{2\cos\theta}\,d\theta\\&=\int_0^{\pi/6}4\sin^2\theta\,d\theta\\&=\int_0^{\pi/6}2(1-\cos 2\theta)\,d\theta\\&=\big[2\theta-\sin 2\theta\big]_0^{\pi/6},\end{aligned}$$

which is $\boxed{\tfrac{\pi}{3}-\tfrac{\sqrt{3}}{2}}\approx 0.181$.

Check. In (a), differentiate, using the quotient rule on the second term:

$$\frac{1}{2(1+x^2)}+\frac{1-x^2}{2(1+x^2)^2}=\frac{1}{(1+x^2)^2}.$$

In (b), the integrand increases from 0 to $\tfrac{1}{\sqrt{3}}\approx 0.577$ on $[0,1]$, so the value must lie between 0 and 0.577; it does.

Pitfalls

- Choosing by habit: $\sqrt{4-x^2}$ takes sin, $\sqrt{x^2-4}$ takes sec. With the wrong one no identity clears the root: $x=2\sin\theta$ makes $x^2-4=-4\cos^2\theta$, which is negative.
- Forgetting to change the limits in (b), or changing them and then converting back to x as well.
- Using a trigonometric substitution where a plain one works: $\int x\sqrt{4-x^2}\,dx$ needs only $u=4-x^2$.

In the notes. Unit 10: Trigonometric substitution, with its sine and secant examples and the right triangle.

Practise it. the Review set on this method

10.5 Improper integrals

Problem. Evaluate an integral over an unbounded interval, or of a function that is unbounded on a finite interval, or show that it diverges.

Method

1. Replace ∞ by t (and $-\infty$ by s). Split a doubly infinite integral at any convenient point c into two pieces.
2. Integrate on the finite interval, $[a, t]$, as an ordinary definite integral.
3. Take the limit $t \to \infty$. Justify each term: a term such as $t\,\mathrm{e}^{-t}$ tends to 0 by l'Hôpital's rule (Unit 8).
4. State the verdict: the integral converges to the limit, or it diverges. A split integral converges only if each piece does.

Example. Evaluate, or show divergent: (a) $\displaystyle\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{x^2-6x+13}$ and (b) $\displaystyle\int_{2}^{\infty} \frac{x}{x^2+1}\,\mathrm{d}x$.

(a) Complete the square: $x^2-6x+13 = (x-3)^2+4$, so an antiderivative is $\frac{1}{2}\arctan\frac{x-3}{2}$. Split at $c=3$:

$$\int_3^t \frac{\mathrm{d}x}{x^2-6x+13} = \tfrac{1}{2}\arctan\tfrac{t-3}{2} \to \tfrac{\pi}{4},$$
$$\int_s^3 \frac{\mathrm{d}x}{x^2-6x+13} = -\tfrac{1}{2}\arctan\tfrac{s-3}{2} \to \tfrac{\pi}{4},$$

as $t \to \infty$ and $s \to -\infty$. Both pieces converge, so the integral converges to $\frac{\pi}{4}+\frac{\pi}{4} = \boxed{\frac{\pi}{2}}$.

(b) The numerator is half the derivative of the denominator:

$$\int_2^t \frac{x}{x^2+1}\,\mathrm{d}x = \left[\tfrac{1}{2}\ln(x^2+1)\right]_2^t = \tfrac{1}{2}\ln(t^2+1) - \tfrac{1}{2}\ln 5.$$

This grows without bound as $t \to \infty$, so the integral $\boxed{\text{diverges}}$.

Variants. An integrand that is unbounded at a point of a finite interval makes the integral improper too. At an endpoint a, integrate from s and let $s \to a^+$; at an interior point c, split at c and require both halves to converge, since FTC 2 applied straight across the blow-up can return a finite but wrong answer. For instance,

$$\int_{-1}^{8} \frac{\mathrm{d}x}{\sqrt[3]{x^2}} = \lim_{t\to 0^-}\left[3\sqrt[3]{x}\right]_{-1}^{t} + \lim_{s\to 0^+}\left[3\sqrt[3]{x}\right]_{s}^{8} = 3+6 = 9.$$

Check. In (a), differentiate the antiderivative: the chain rule gives $\frac{1}{2}\cdot\frac{1}{2}\cdot 1/\left(1+\frac{1}{4}(x-3)^2\right) = 1/\left((x-3)^2+4\right)$, the integrand; a missing $\frac{1}{2}$ would double it. In (b), $\frac{x}{x^2+1} \geq \frac{1}{2x}$ for $x \geq 1$ (it rearranges to $x^2 \geq 1$), and the tail of $\frac{1}{x}$ has infinite area, so divergence is no surprise.

Pitfalls

- Putting ∞ into the formula as if it were a number. $\left[\cdot\right]_2^{\infty}$ is only shorthand for the limit as $t \to \infty$, taken after the integral is evaluated, and it has to be justified.
- Deciding from the integrand alone. The integrand of (b) tends to 0, yet the integral diverges.
- Evaluating a doubly infinite integral as one symmetric limit, $\lim_{t\to\infty} \int_{-t}^{t}$. That can be finite when the two halves diverge; take the pieces separately.

In the notes. Unit 10: Improper integrals, probability and expectation, its definition and first example.

Practise it. the Review set on this method

10.6 Setting up an integral by slice and sum

Problem. Turn a description into an integral: the area between two curves, the volume of a solid of revolution or of known cross-section, the work done lifting something, or the mass of a rod.

Method

1. Say what x measures, from where, and the range it covers. For an area, find where the curves meet; sketch if asked.
2. Write the contribution of one slice of width $\mathrm{d}x$ as (something) $\times$ $\mathrm{d}x$. A strip of height (top − bottom), with the top curve checked by a test point. A disc of area $\pi f(x)^2$. A slab of area $A(x)$, by similar triangles if need be. A mass $\mathrm{d}m = \rho(x)\,\mathrm{d}x$. Work $\mathrm{d}W$ = force $\times$ distance.
3. Integrate over the range. For an area, split at every crossing, so that each piece has the right curve on top.
4. Evaluate, and give the units.
5. Check against a known formula or a simple bound.

Variants. The *average value* of f on $[a, b]$ is $\bar{f} = \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x$, in the units of f: the height of the rectangle with the same area.

Example. (a) Find the area enclosed between $y = x^3$ and $y = 4x$. (b) The region under $y = \dfrac{1}{\sqrt{1+x^2}}$, $0 \le x \le 1$, is revolved about the x-axis. Find the volume. (c) A cylindrical tank of radius 1 m and height 2 m is full of water, of density $1000\,\mathrm{kg\,m^{-3}}$. Find the work done pumping all of it out over the top. Take $g = 9.8\,\mathrm{m\,s^{-2}}$.

(a) The curves meet where $x^3 = 4x$, that is $x(x-2)(x+2) = 0$: at $x = -2, 0, 2$. A test point in each interval: at $x = -1$, $x^3 = -1 > -4$, so x^3 is on top on $[-2, 0]$; at $x = 1$, $4x = 4 > 1$, so $4x$ is on top on $[0, 2]$. Split at 0:

$$\begin{aligned} A &= \int_{-2}^{0} (x^3 - 4x)\,\mathrm{d}x + \int_{0}^{2} (4x - x^3)\,\mathrm{d}x \\ &= \left[\tfrac{1}{4}x^4 - 2x^2\right]_{-2}^{0} + \left[2x^2 - \tfrac{1}{4}x^4\right]_{0}^{2} \\ &= 4 + 4, \end{aligned}$$

so $\boxed{A = 8}$.

(b) The disc at x has radius $f(x) = \dfrac{1}{\sqrt{1+x^2}}$ and area $\pi f(x)^2 = \dfrac{\pi}{1+x^2}$, so

$$V = \pi \int_0^1 \frac{\mathrm{d}x}{1+x^2} = \pi \left[\arctan x\right]_0^1 = \boxed{\tfrac{\pi^2}{4}} \approx 2.47.$$

(c) Let x (m) be the depth below the top, $0 \le x \le 2$. The slice at depth x has volume $\pi \cdot 1^2 \, \mathrm{d}x$, so mass $1000\pi \, \mathrm{d}x$ and weight $9800\pi \, \mathrm{d}x$ (N), and it is raised a distance x: $\mathrm{d}W = 9800\pi x \, \mathrm{d}x$. Then

$$W = \int_0^2 9800\pi x \, \mathrm{d}x = 9800\pi \left[\tfrac{1}{2}x^2\right]_0^2,$$

so $\boxed{W = 19600\pi \text{ J}} \approx 6.16 \times 10^4$ J (joules: 1 J is 1 N m).

Check. (a) On $[0, 2]$ the piece is the triangle under $y = 4x$, of area $\frac{1}{2} \cdot 2 \cdot 8 = 8$, less the area $\int_0^2 x^3 \, \mathrm{d}x = 4$ under $y = x^3$: 4. Both curves are odd, so a half-turn about the origin carries it onto the other piece: 4 each. (b) One disc at the midpoint, with $f(\frac{1}{2})^2 = \frac{4}{5}$ and width 1, estimates $V \approx \frac{4\pi}{5} \approx 2.51$, close to 2.47; forgetting to square gives $\pi \ln(1 + \sqrt{2}) \approx 2.77$, far off. (c) The water weighs $2000\pi \cdot 9.8 = 19600\pi$ N and its centre is 1 m below the top: weight times that depth is 19600π J.

Pitfalls

- Integrating across a crossing. In (a), $\int_{-2}^{2}(4x - x^3) \, \mathrm{d}x = 0$: the two pieces cancel, but the area is 8.
- Forgetting to square the radius: $\pi \int_0^1 \frac{\mathrm{d}x}{\sqrt{1+x^2}}$ is not the volume in (b).
- Measuring x from the bottom of the tank in (c) but keeping the distance x: a slice at height x above the bottom is raised $2 - x$.

In the notes. Unit 10: Area between two curves, Unit 10: Volumes of revolution, Unit 10: The average value of a function (the Variant), and Unit 10: Setting up integrals from a description.

Practise it. the Review set on this method For the average value, the Review set on this variant

10.7 Probability density: normalise, probability, expectation

Problem. A random quantity X has a density f, perhaps with an unknown constant. Fix the constant, then find a probability and the expectation.

Method

1. Check that f is a density: $f(x) \ge 0$, and $\int f = 1$ over its range. If f contains an unknown constant k, impose $\int f = 1$ to find it.
2. A probability is an area: $P(a \le X \le b) = \int_a^b f(x) \, \mathrm{d}x$. Over an infinite range the integral is improper (Method 10.5).
3. The expectation is $\mathbb{E}[X] = \int x \, f(x) \, \mathrm{d}x$ over the whole range. Over an infinite range it is improper too, and it must converge.
4. Check: the probabilities of an event and of its complement add to 1, and $\mathbb{E}[X]$ lies between the smallest and largest values X can take.

Example. X has density $f(x) = \dfrac{k}{(1+x)^3}$ on $[0,\infty)$, and 0 elsewhere. Find k, $P(X>1)$ and $\mathbb{E}[X]$.

Normalise. An antiderivative of $(1+x)^{-3}$ is $-\frac{1}{2}(1+x)^{-2}$, so

$$\int_0^t \frac{k\,dx}{(1+x)^3} = \frac{k}{2}\Big(1 - \frac{1}{(1+t)^2}\Big),$$

which tends to $\frac{k}{2}$ as $t\to\infty$. Setting $\frac{k}{2}=1$ gives $\boxed{k=2}$, and then $f\ge 0$.

The probability is an improper integral:

$$P(X>1) = \lim_{t\to\infty}\Big[-\frac{1}{(1+x)^2}\Big]_1^t = 0 + \frac{1}{4} = \boxed{\tfrac{1}{4}}.$$

For the mean, write $x = (1+x)-1$ in the numerator:

$$\frac{2x}{(1+x)^3} = \frac{2}{(1+x)^2} - \frac{2}{(1+x)^3}.$$

Then

$$\begin{aligned}\int_0^t \frac{2x\,dx}{(1+x)^3} &= \Big[-\frac{2}{1+x} + \frac{1}{(1+x)^2}\Big]_0^t \\ &= -\frac{2}{1+t} + \frac{1}{(1+t)^2} + 1,\end{aligned}$$

which tends to 1 as $t\to\infty$: $\boxed{\mathbb{E}[X]=1}$.

Check. $P(X\le 1) = \big[-\frac{1}{(1+x)^2}\big]_0^1 = 1-\frac{1}{4} = \frac{3}{4}$, and $\frac{3}{4}+\frac{1}{4}=1$. For the mean, $\mathbb{E}[X]\ge 0$ is a weak test, and 1 is also the value of $\int f$, the first Pitfall's wrong answer, so differentiate instead: $\frac{d}{dx}\big(-\frac{2}{1+x}+\frac{1}{(1+x)^2}\big) = \frac{2}{(1+x)^2} - \frac{2}{(1+x)^3}$, the split integrand.

Pitfalls

- Finding $\mathbb{E}[X]$ from $\int f$ instead of $\int xf$. The first is 1 for every density.
- Taking $P(X>1)$ over $[0,1]$. Draw the interval the event describes first.
- Using k before fixing it, or fixing it from $\int_0^1 f = 1$ when the range is $[0,\infty)$.

In the notes. Unit 10: Improper integrals, probability and expectation, from "Probability densities and expectation" on.

Practise it. the Review set on this method